

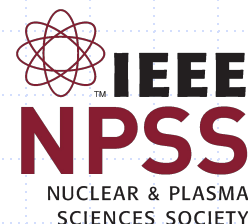


SESTRA

User Training Session

Michael Holik, Vladimir Vicha, Stanislav Pospisil

*Institute of Experimental and Applied Physics
Czech Technical University in Prague*



2024/06/26



Training Outline

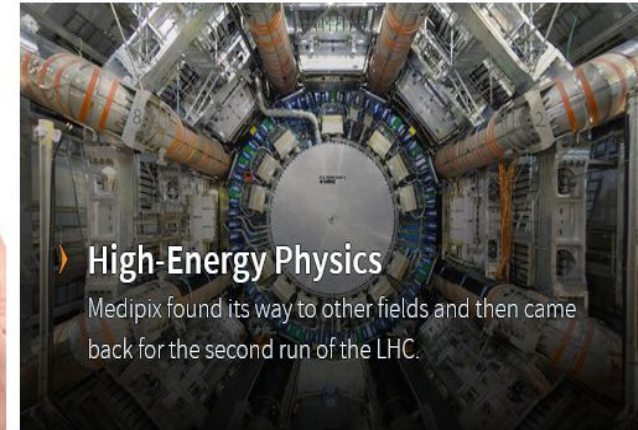
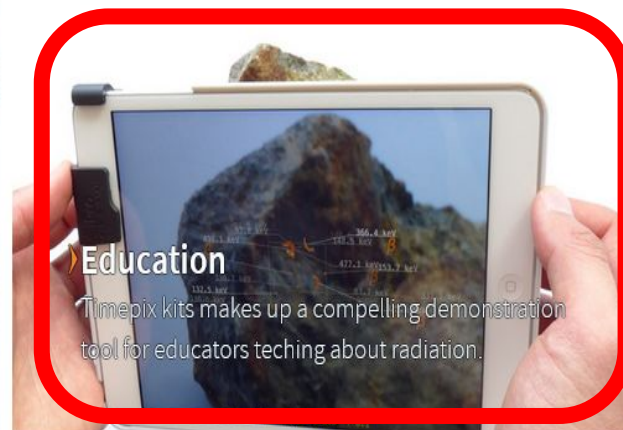
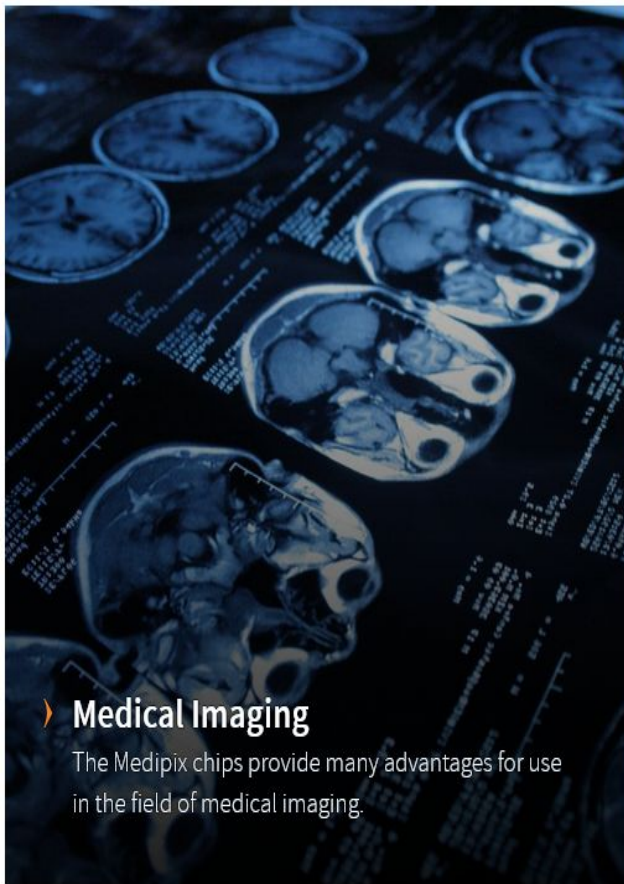
- **Session 1**
Introduction into Timepix detector, particle camera working principle, content of SESTRA kit, Pixelman simple preview software and its functionality
- **Session 2**
Uranium glass experiment, Potassium experiment, Thoriated electrode experiment, Study of particle properties, Using of online visualisation tools, americium isotope radiation source, beam collimation, study of alpha radiation properties, start of long term background measurement
- **Session 3**
Evaluation of long term background measurement, Radiation at on Earth ground vs orbit, observation of radon decay, Cosmic radiation, utilization of virtual particle camera, utilization of advanced particle type analysis
- **Session 4**
alpha radiography, absorption of gammas in metals, gamma radiography, Own radiation sample study
- **Discussion**



CERN Medipix Collaboration



- ◆ Development of pixel detectors
- ◆ Technology transfer initiative
- ◆ **Application of pixel detectors in the field of education**

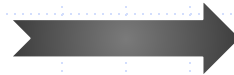




Particle Camera = Modern Instrument for Teaching of Physics

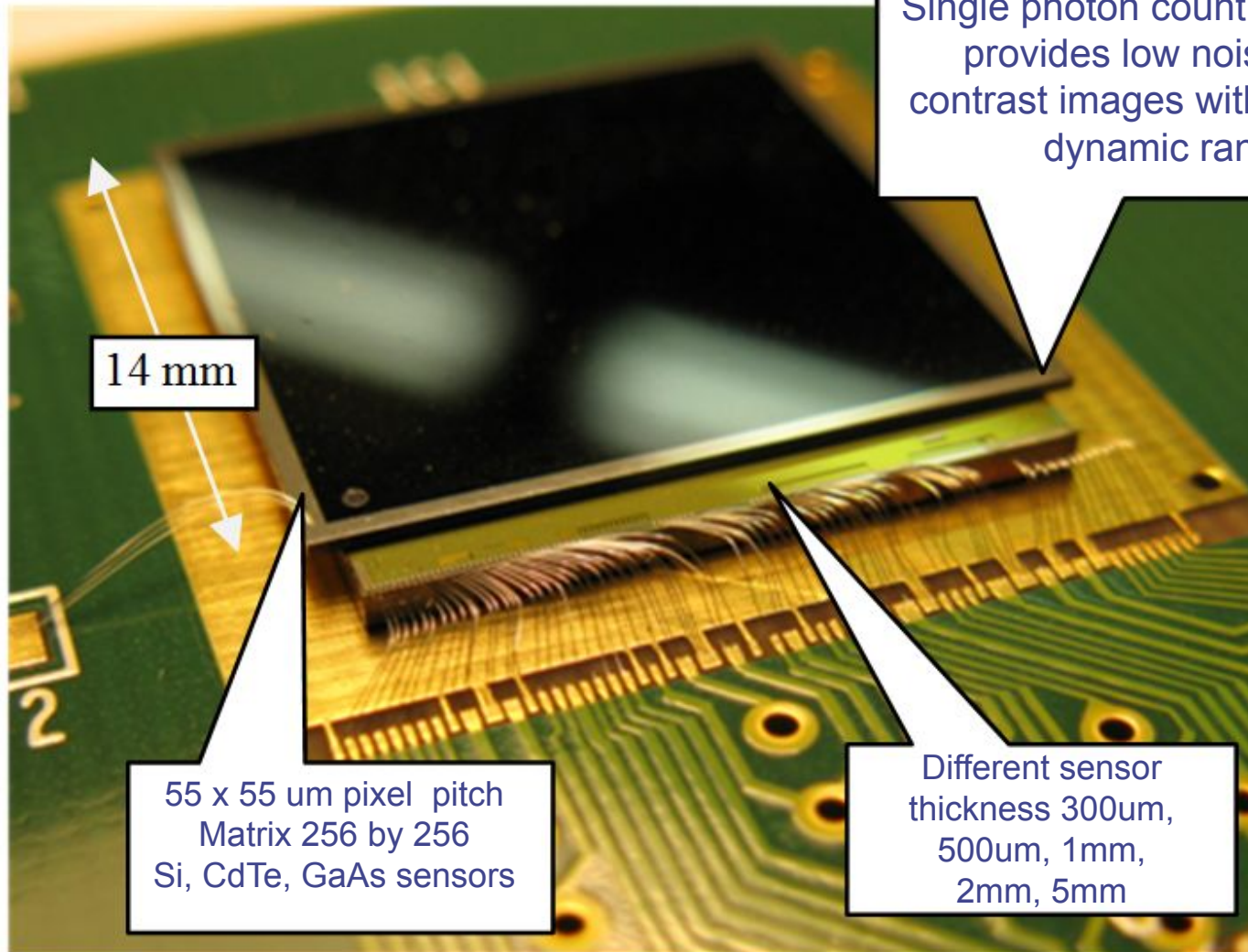


Geiger-Muller counter vs Timepix based Particle camera





Timepix - Hybrid Pixel Detector with Single Particle Detecting Capability



14 mm

Single photon counting readout provides low noise, high contrast images with very high dynamic range

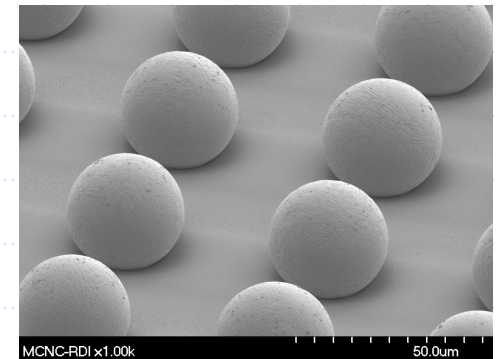
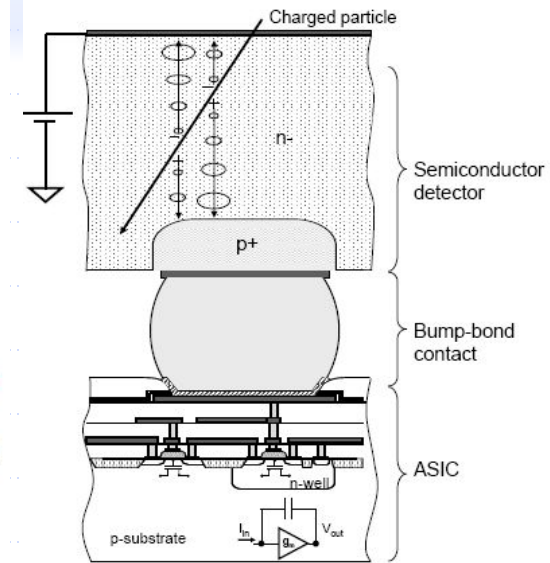
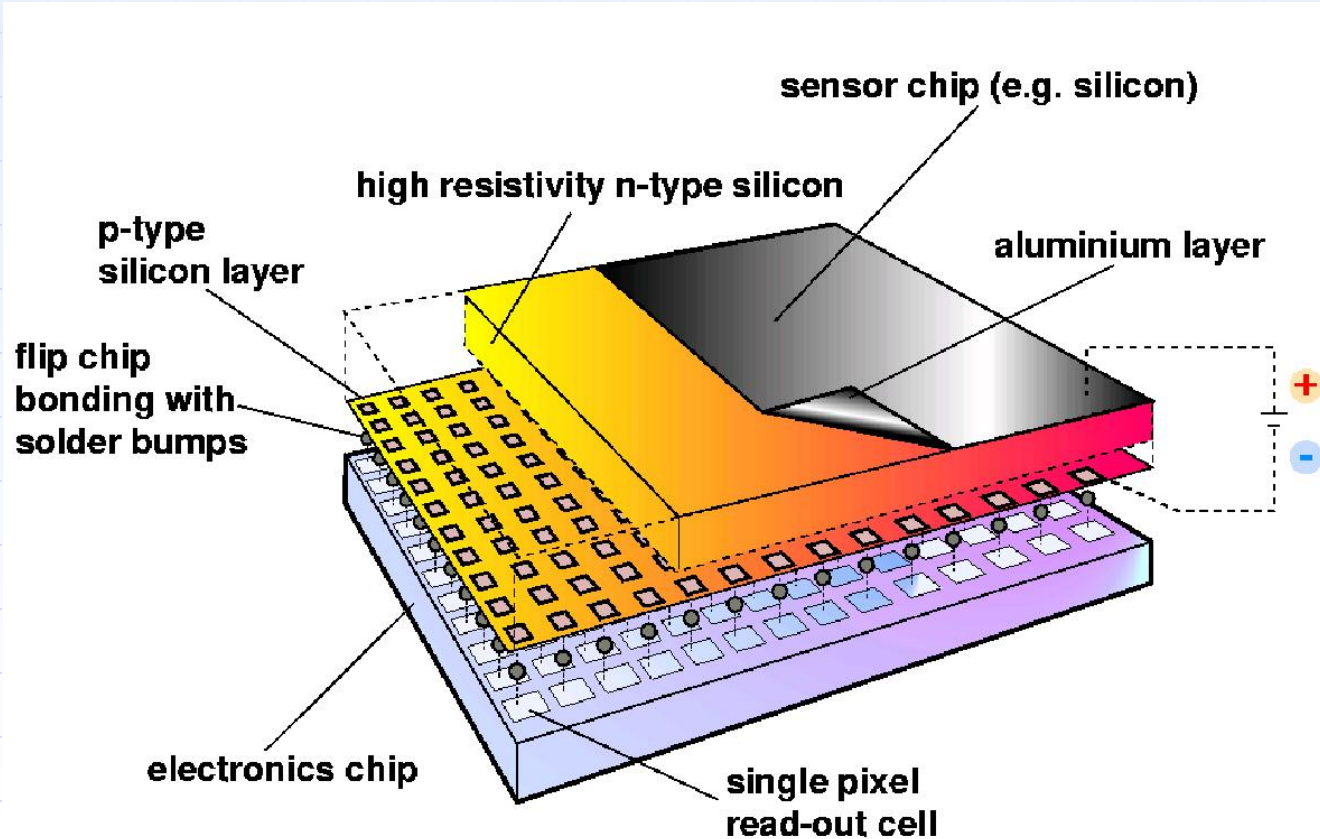
55 x 55 um pixel pitch
Matrix 256 by 256
Si, CdTe, GaAs sensors

Different sensor thickness 300um, 500um, 1mm, 2mm, 5mm

Timepix



Timepix - Hybrid Pixel Detector with Single Particle Detection Capability

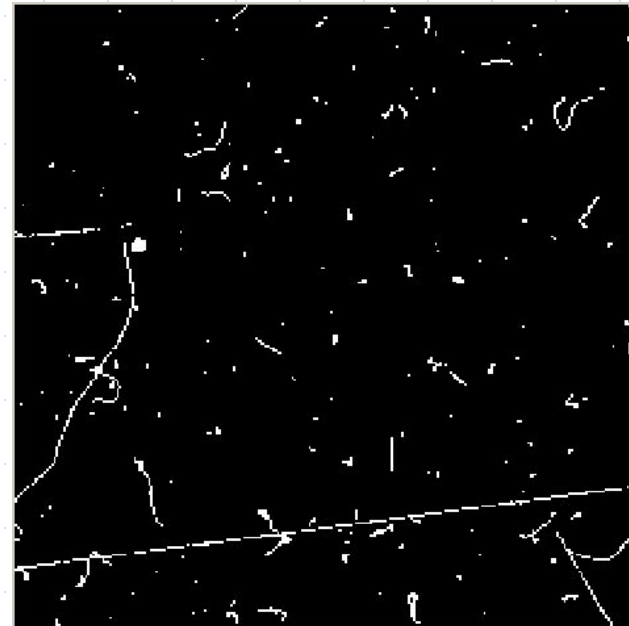
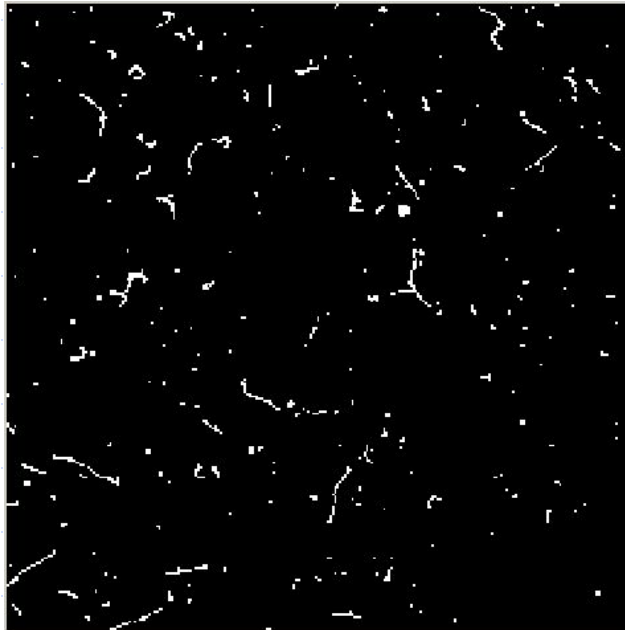


**Sensor part and readout electronics part are optimized separately
(sensor materials – Si, CdTe, GaAs)**



Pixel detector as "Active emulsion"

- ◆ Some particles can be identified from shape of their tracks as in formerly used nuclear emulsions.
- ◆ Example: Cosmic rays and natural background

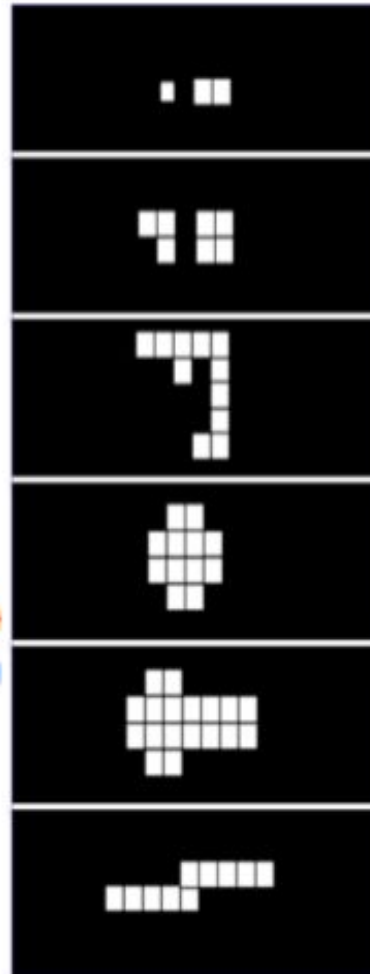
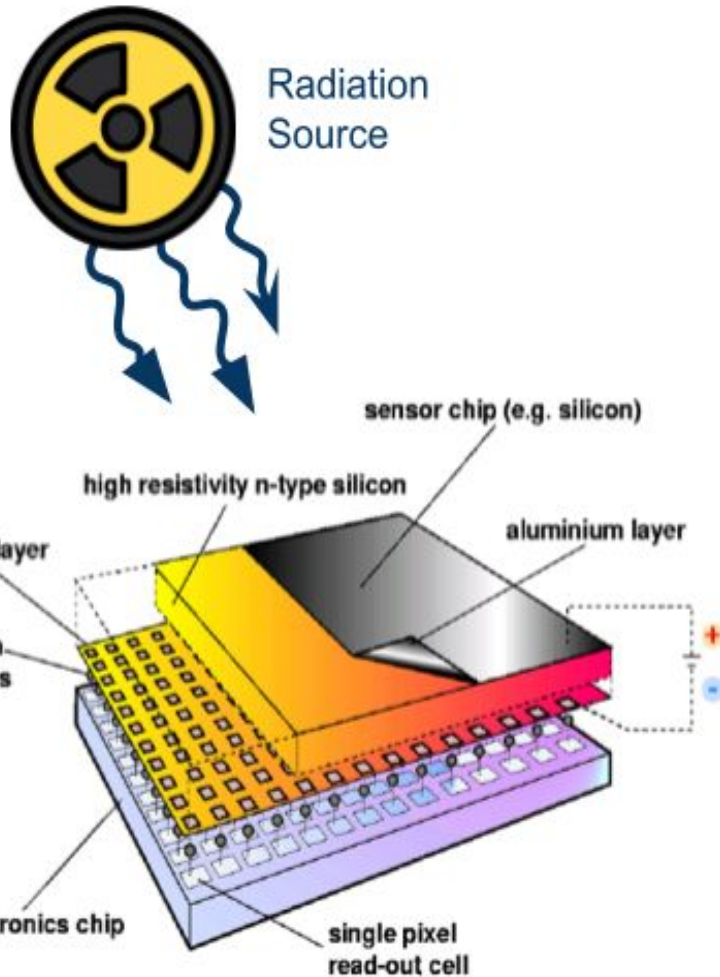


Are particles photogenic?

Yes of course, they are when observing them with Timepix detector.



Particle Camera Working Principle



(1) Dot

Photons and electrons (10keV)

(2) Small blob

Photons and electrons

(3) Curly track

Electrons (MeV range)

(4) Heavy blob

Heavy ionizing particles with low range (alpha particles,...)

(5) Heavy track

Heavy ionizing particles (protons,...)

(6) Straight track

Energetic light charged particles (MIP, Muons,...)

....Different kind of radiation produces tracks of different shape



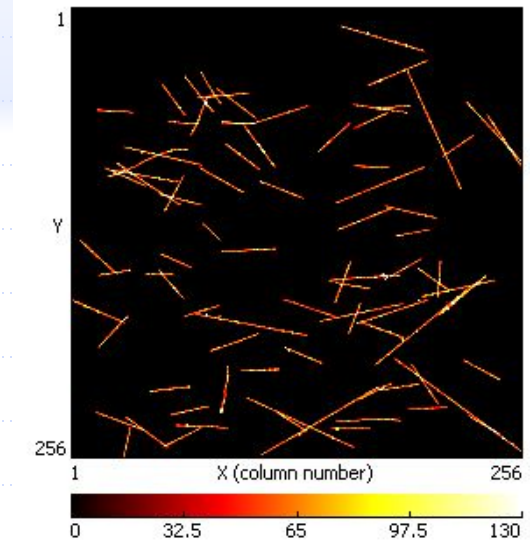
Particle Tracking and Recognition

Muons

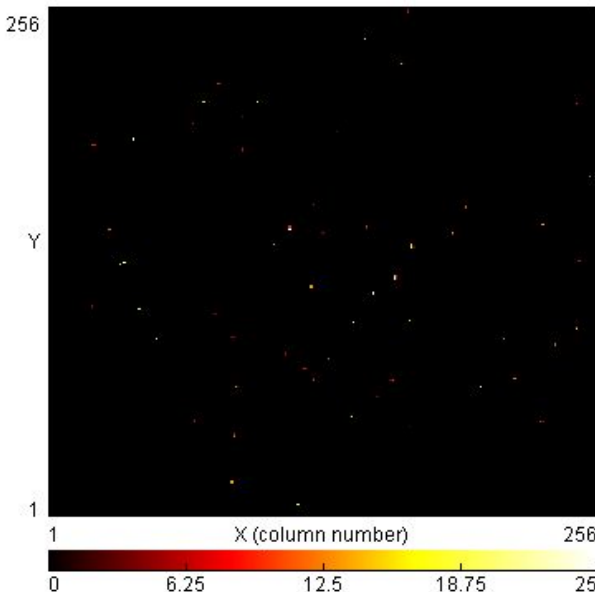


Type of interacting particle can be recognized by its specific track left in the pixel matrix

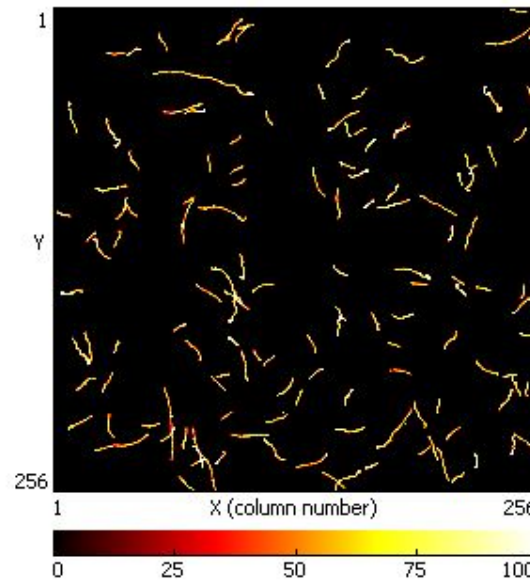
Track shape depends on interacting particle energy, mass, sensor material, bias voltage, angle,...



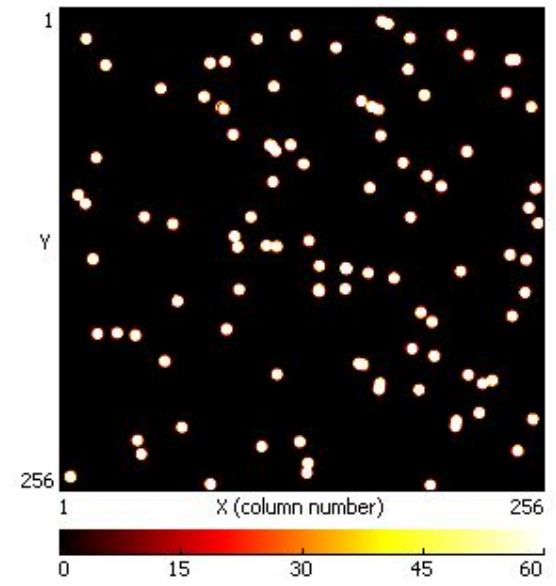
Gammas/X-Rays



Electrons

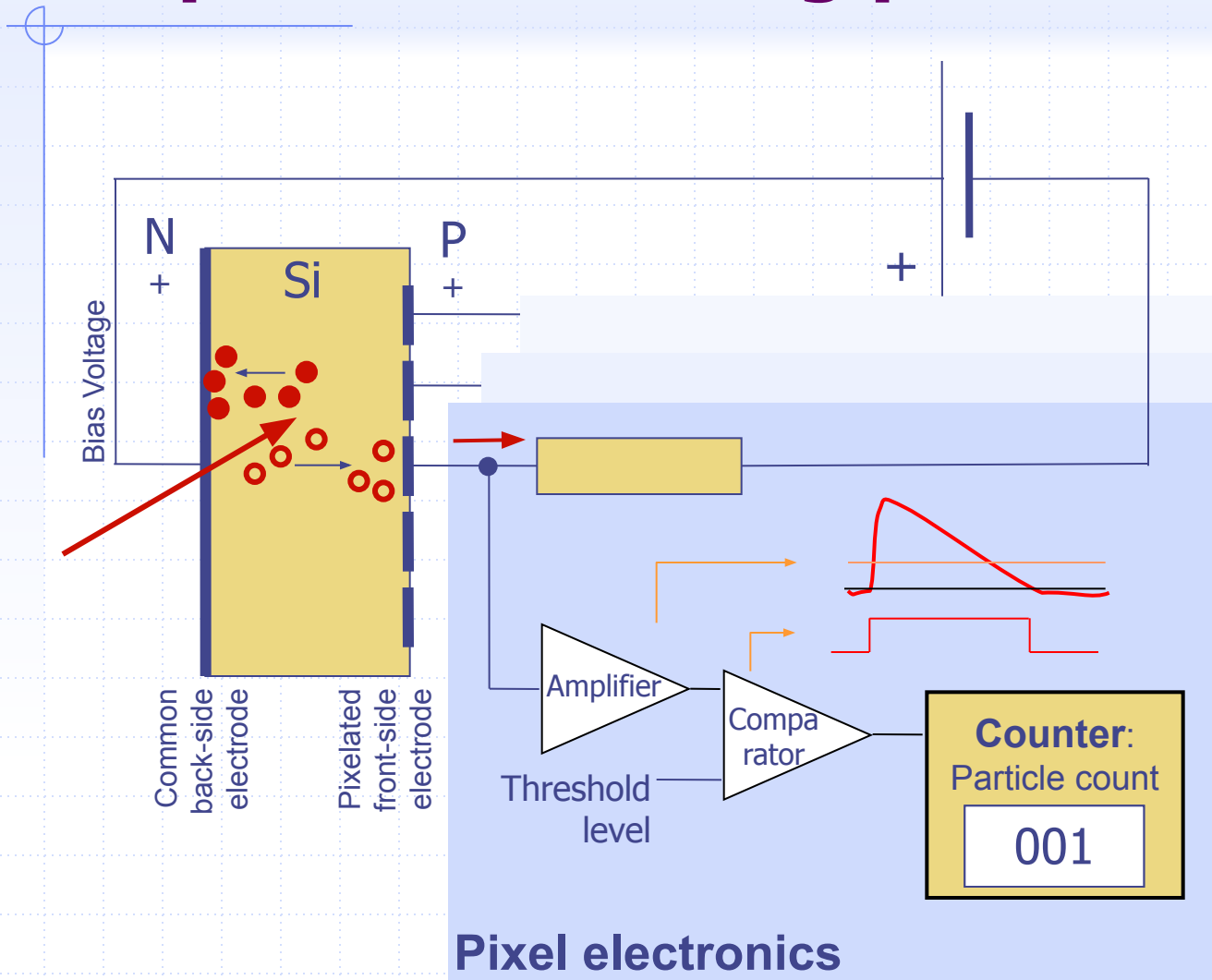


Alphas





Principle of semiconductor single particle counting pixel detector



Threshold level above
electronic noise
⇒ No false counting.

Digital integration
(counting)
⇒ No dark current.



Unlimited dynamic
range and exposure
time.

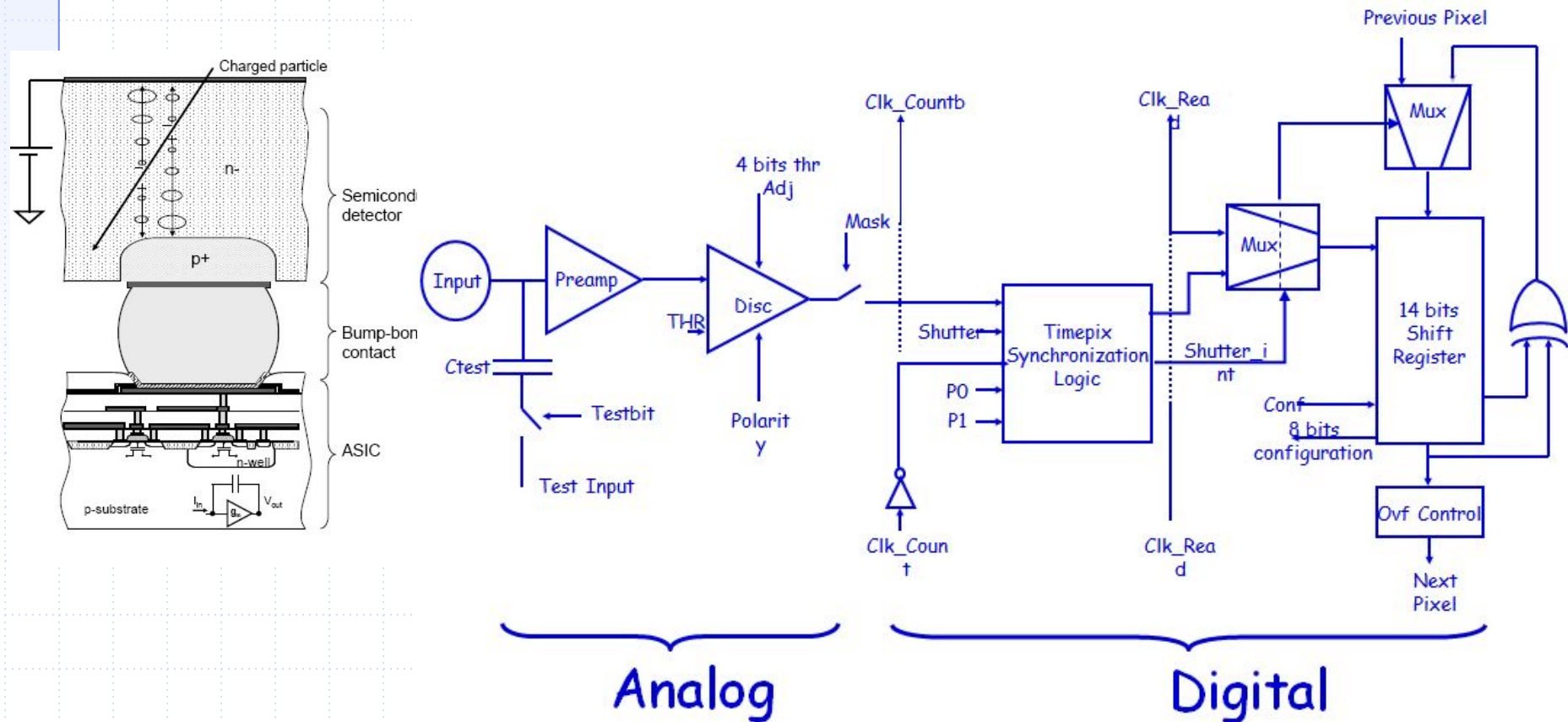
Detected count obeys
poissonian distribution



Timepix read-out chip

Single pixel schematics

- Independent signal processing is performed in each individual pixel
- Very high level of miniaturization and functionality implementation
- All 65 536 detector channels (pixels) are running in parallel





Timepix readout chip layout

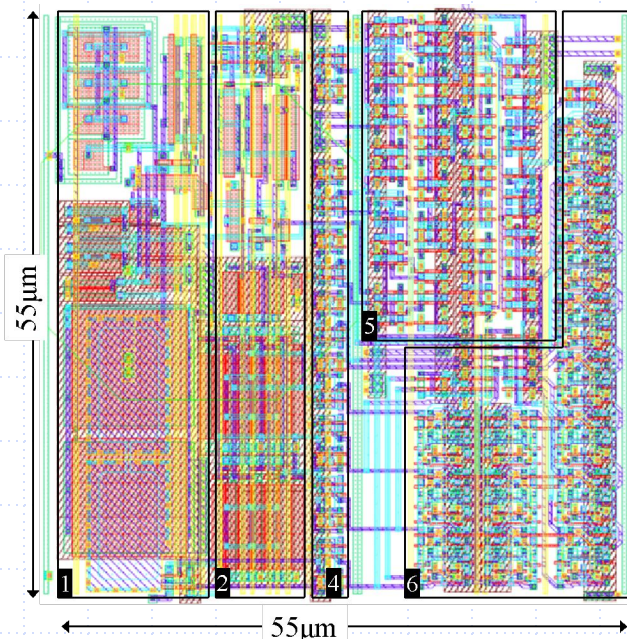


256x256 55 μ m square pixels

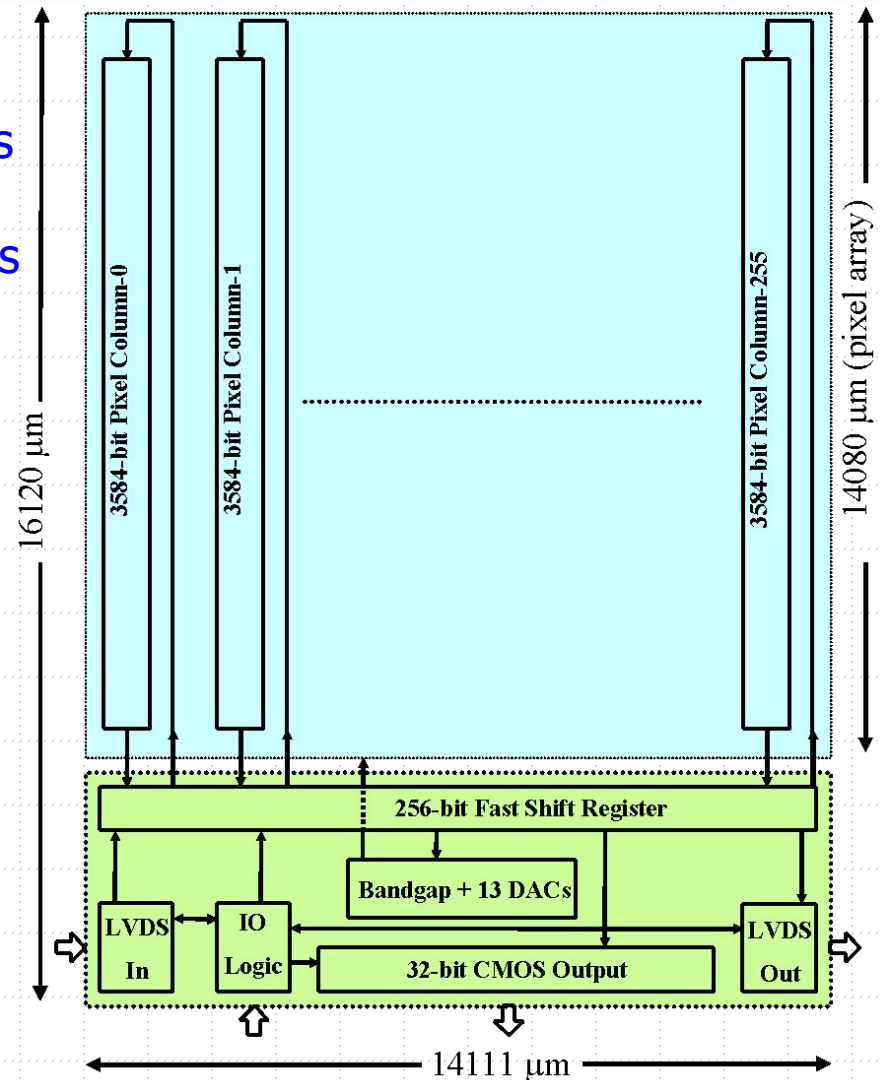
36M Transistors

Serial readout (@100MHz) -> 9.17 ms
(~100 frames per second)

Parallel readout (@100MHz) -> 287 μ s
(~3000 frames per second)



Single pixel floor plan





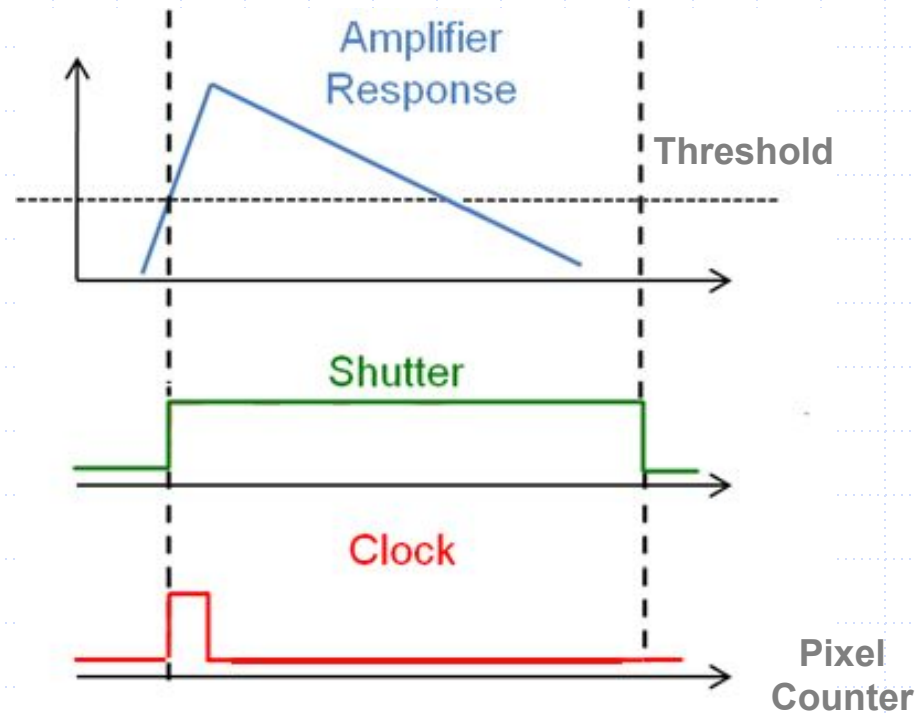
Pixel Operation Modes

- ◆ Each pixel can be configured to operate in different mode independently to other pixels
- ◆ **Medipix mode**
Counter is incremented if incoming particle is registered
- ◆ **Time of Arrival mode**
Counter is continuously incremented from the time when first particle is registered
- ◆ **Time Over Threshold mode**
Counter is incremented as long as a signal is over threshold
- ◆ **Masked mode**
pixel is disabled



Medipix / Event Counting Mode

- Each event registered is counted as 1
- Counter value equals to a number of interacted particles
- Event rate is limited by run up time and time of integration





High contrast imaging:

Unlimited dynamic range

Example 1:

X-ray transmission
Image of mouse head
(dose 0.9 mGy)

Single photon counting device can count unlimited number of photons, noise in images is given just by poissonian statistics:

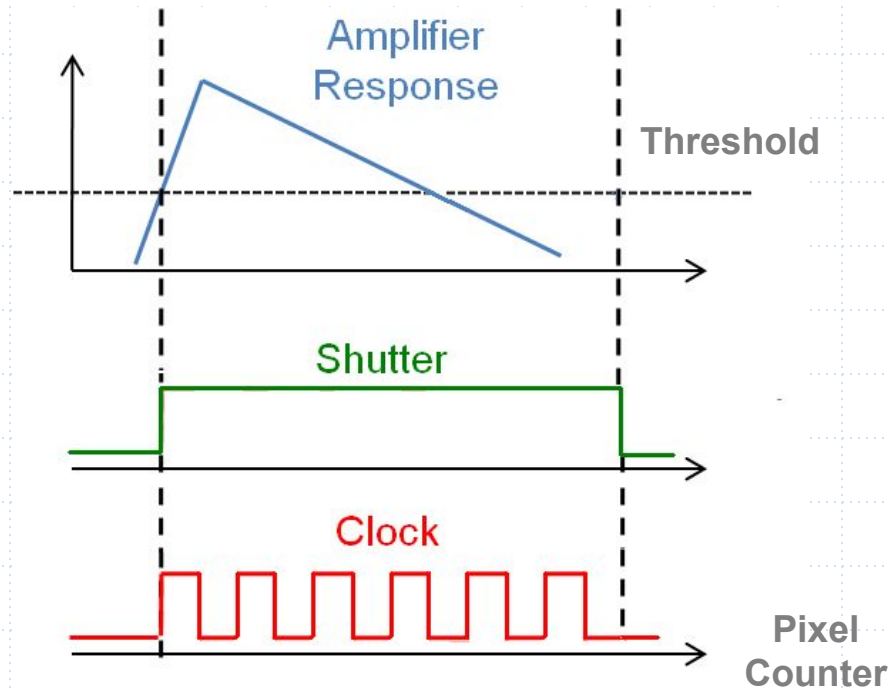
=> Unlimited number of gray levels in images depending just on the beam intensity and exposure time.

=> Almost unlimited contrast



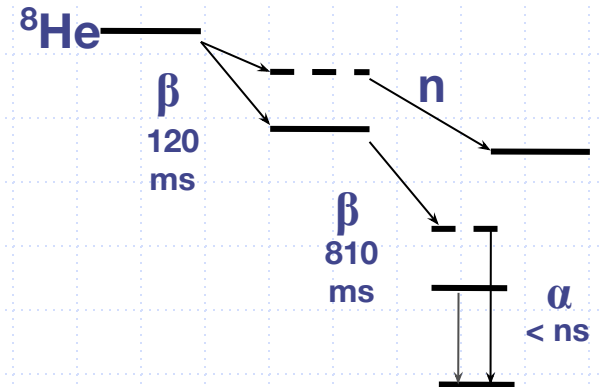
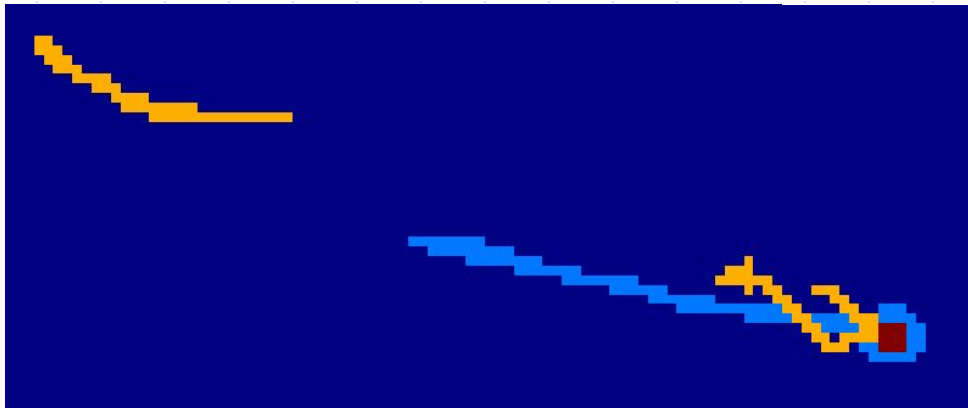
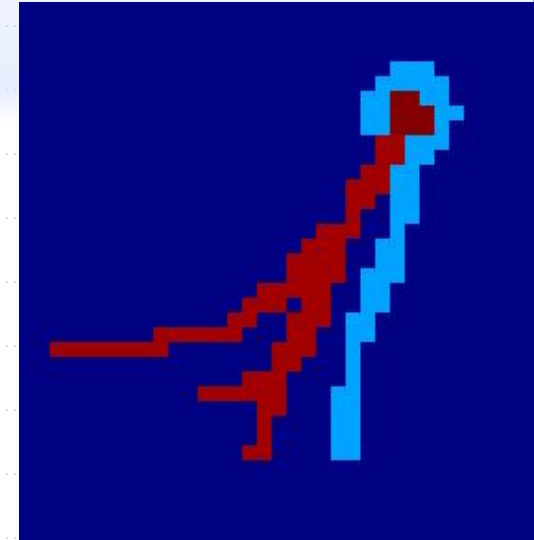
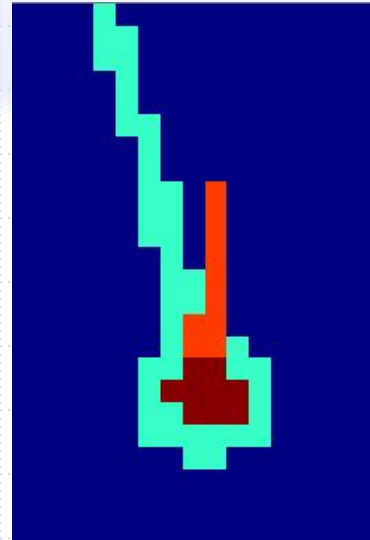
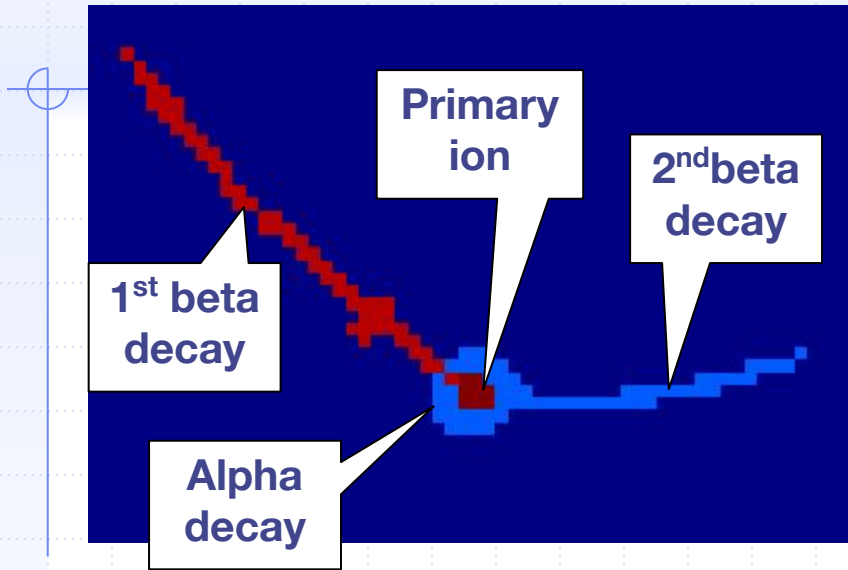
Timepix / Time Of Arrival Mode

- Counts from passing threshold to closing shutter
- Allows accurate timing of hits in individual pixels





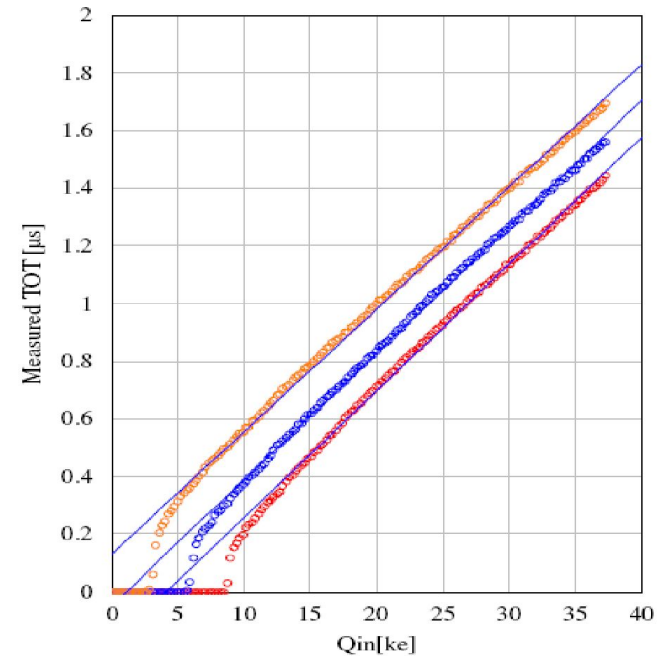
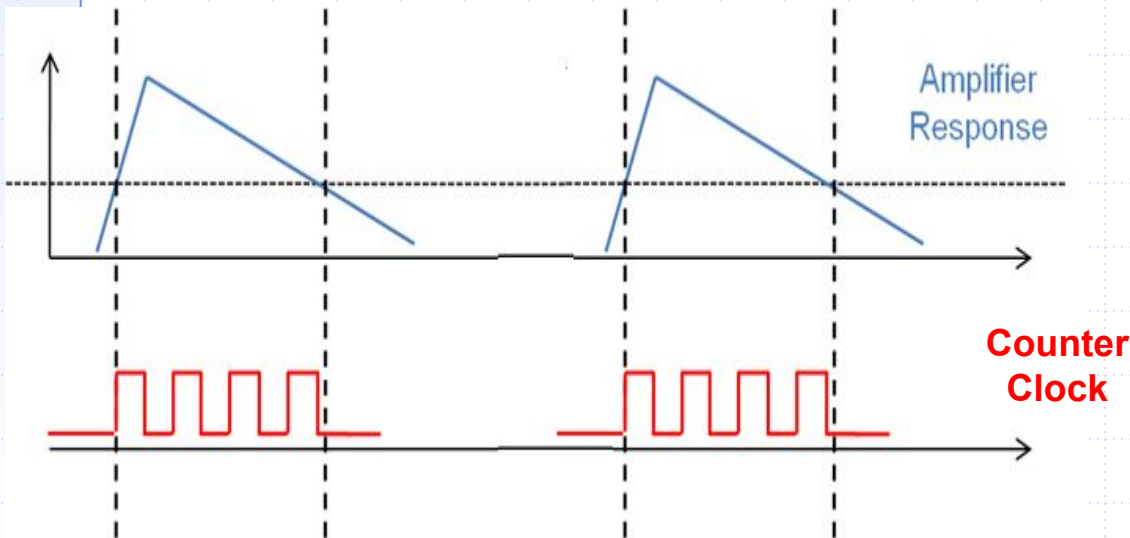
Measured events: ^8He decays (TOA Mode)





Time Over Threshold Mode Energy measurement

Counter is incremented as long as a signal is over threshold



E.g. 20ke- $\sim 1\mu s = 40 \times 25ns$ (i.e. base clock 40 MHz)

Linearity of one pixel measured at
three different threshold values

Residual errors come from non linearity with small
charge signals and a slow return to baseline

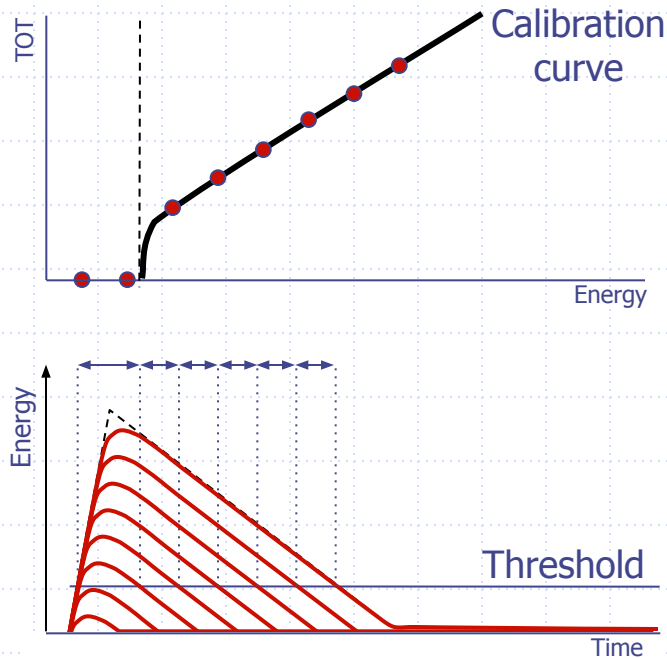
Preamp has fast rise (90ns) but slow (500ns -2500ns)
constant current return to zero



per pixel energy measurement: Time Over Threshold mode

Counter in each pixel can be used as

- Wilkinson type **ADC** to measure energy of each particle detected.



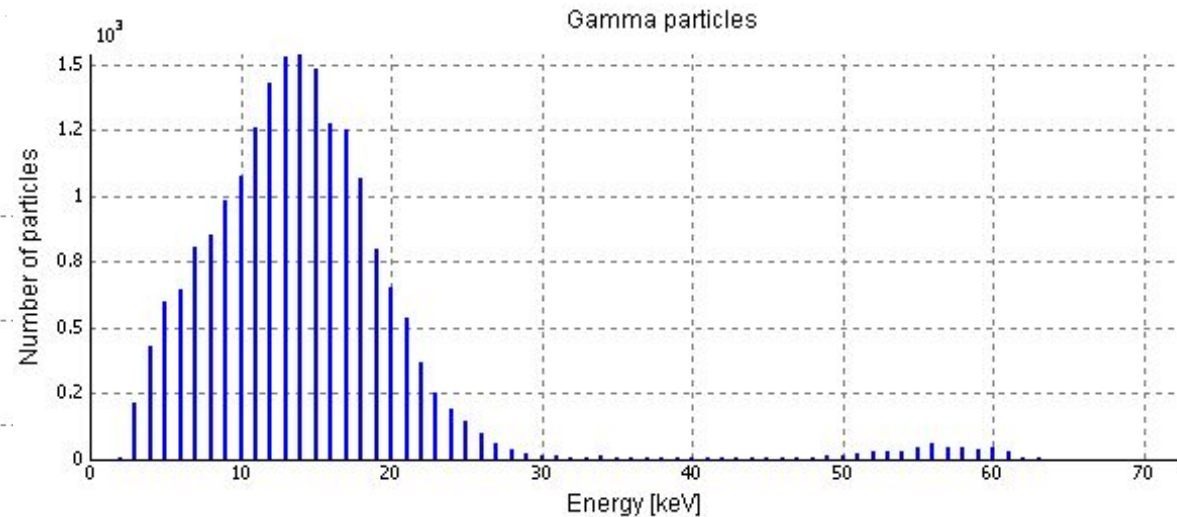
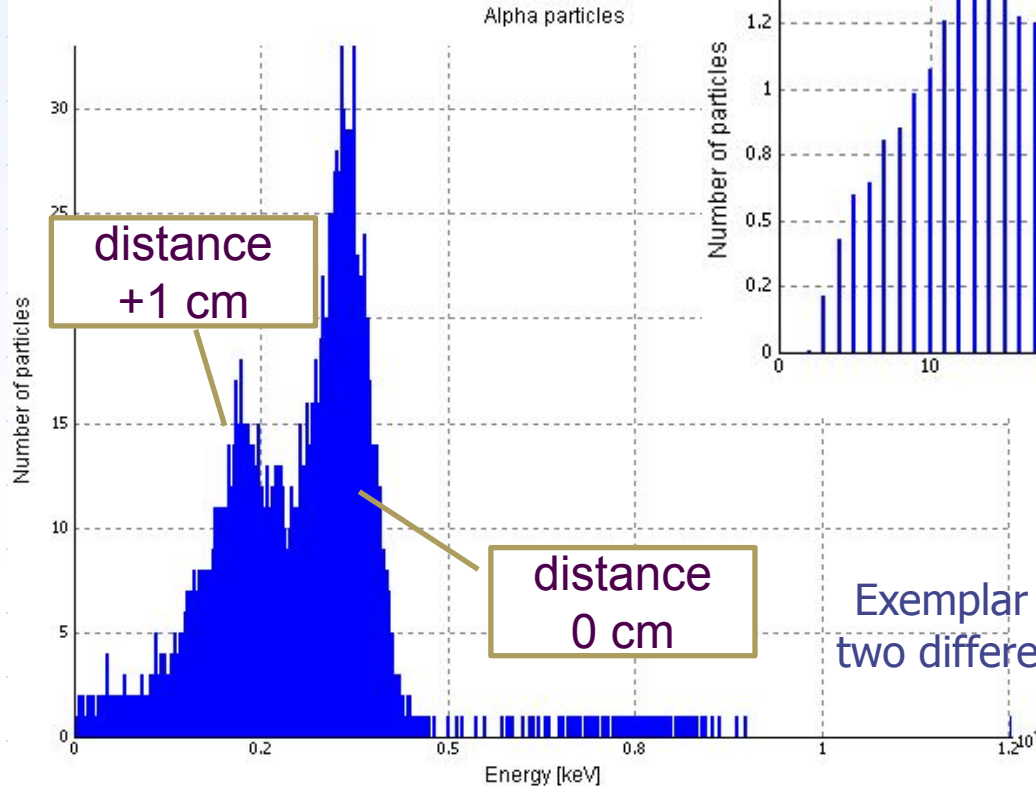
- If the pulse shape is triangular then Time over Threshold is proportional to collected charge i.e. to energy.
- Due to limited bandwidth the pulse can be NEVER perfectly triangular.
- Non-linear TOT to energy dependence



And when per-pixel calibration is applied on measured TOT values...



Exemplar ^{241}Am spectra



Exemplar ^{241}Am alpha spectrum measured in air at two different distances between the radiation source and the Timepix detector



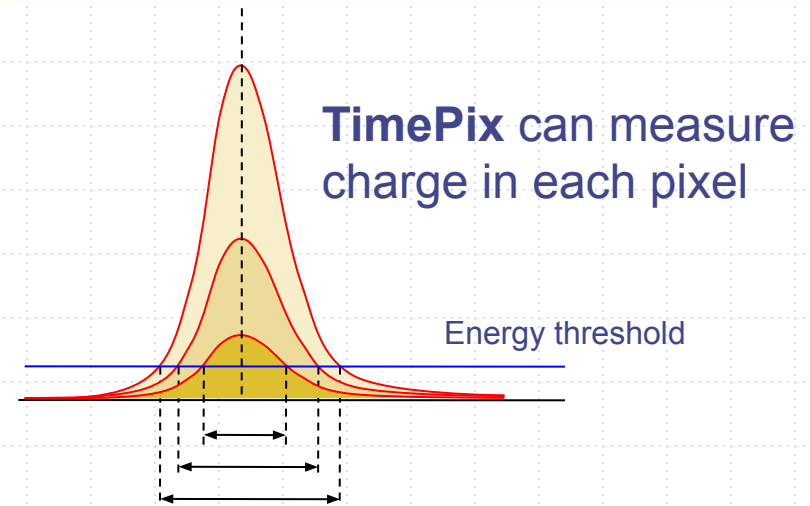
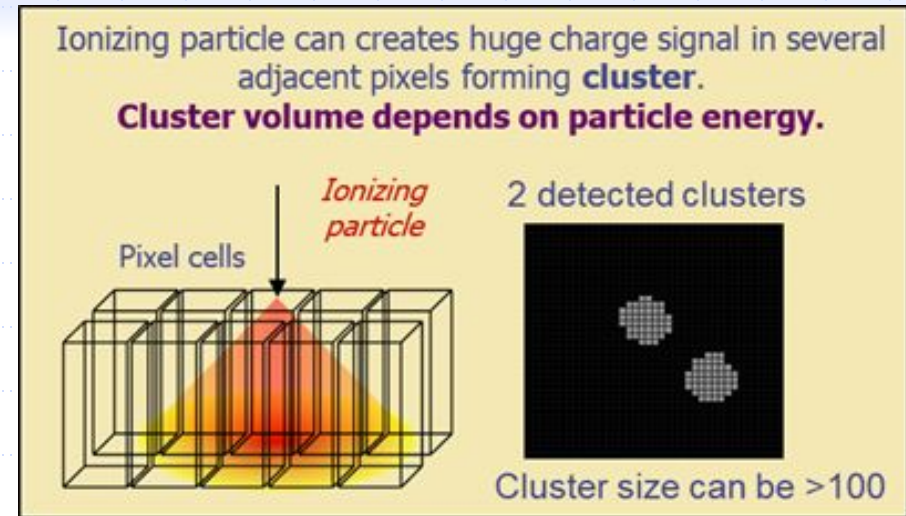
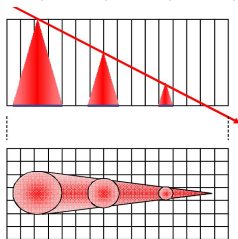
Charge Sharing Effect

creation of clusters

- Ionizing particle creates a charge in the sensor.
- The charge is collected by external electric field => the process takes some time
- Due to charge diffusion the charge cloud expands
- The charge cloud can overlap several adjacent pixels => **CLUSTER**
- Pixels in a cluster will detect the charge if it is higher than certain threshold

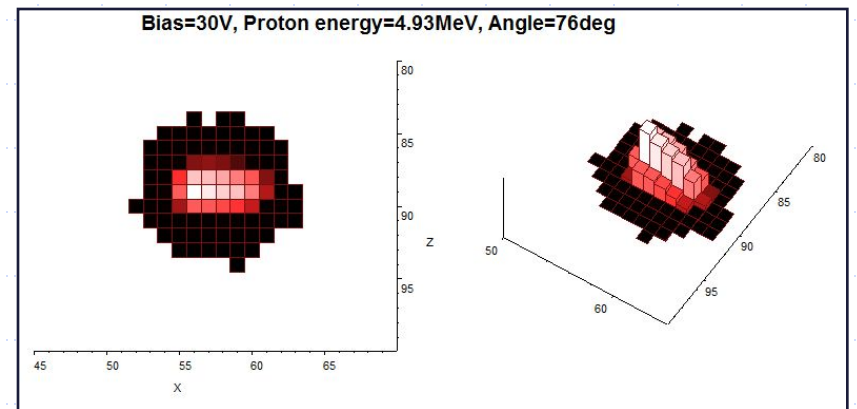
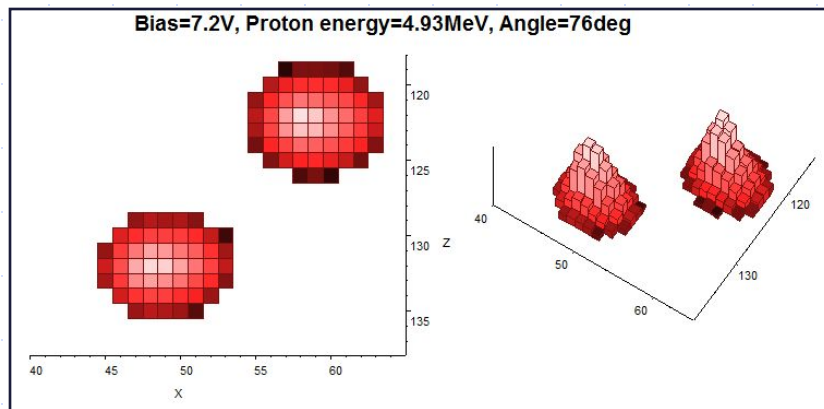
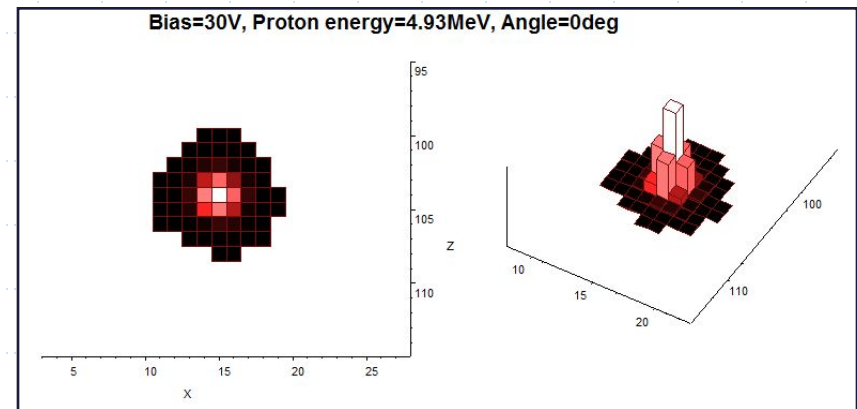
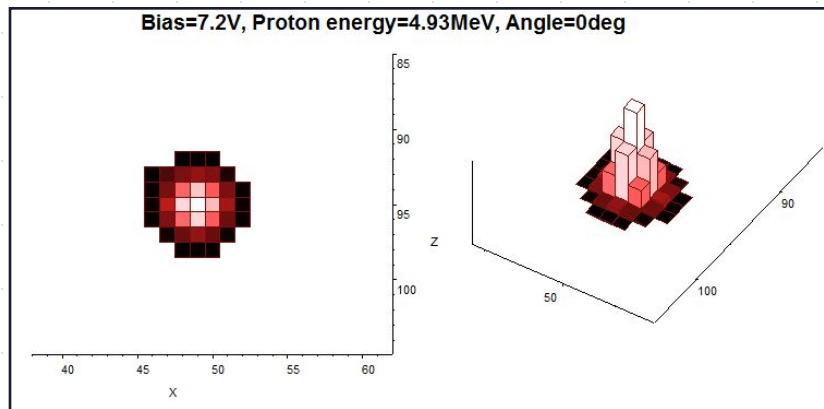
The Cluster size depends on:

- Particle energy and range
- Depth of interaction
- Detector Bias Voltage
- Local CCE (e.g. due to a material inhomogeneities and radiation damage)





4.93 MeV proton tracks in silicon pixel detector recorded by Timepix device. Exposition under different angles and different applied detector biases.

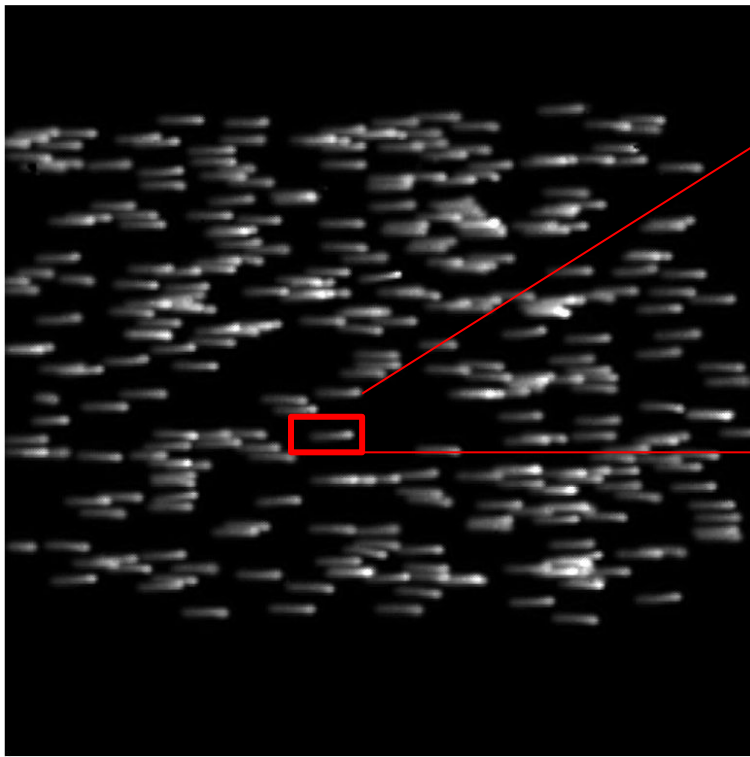


What is the spatial resolution? X- and Y-coordinates are determined with a precision of about 500nm. Determination of angle is with a precision of about 1° . It needs additional experiments.

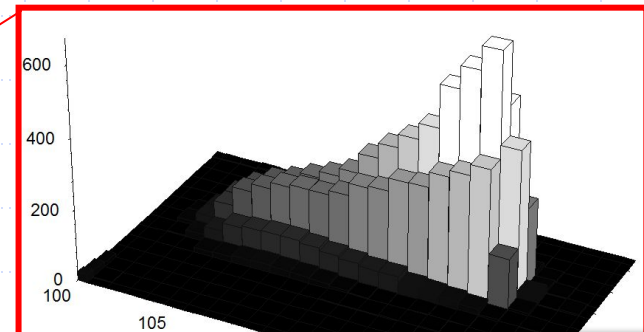


Flock of 11 MeV protons entering the silicon sensor under 85°

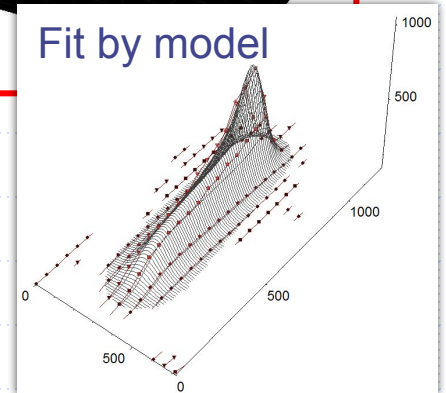
11 MeV protons, 85 degrees



$\Delta E/\Delta x$ Bragg profile nicely pronounced, proton range about 960 μm



Fit by model



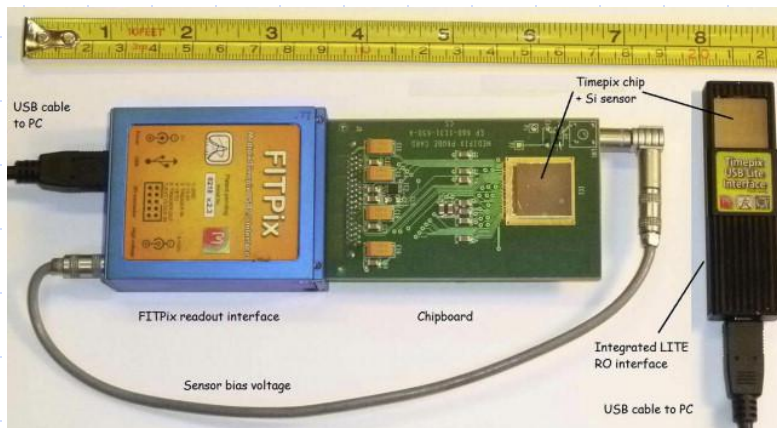


Readout interface and its role in data the acquisition process

Detector supporting electronics

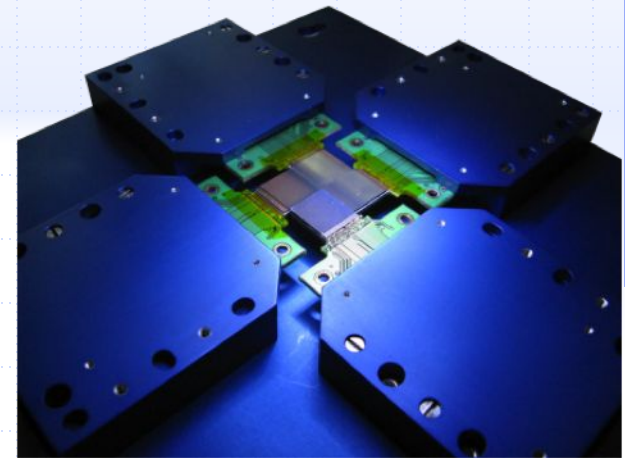
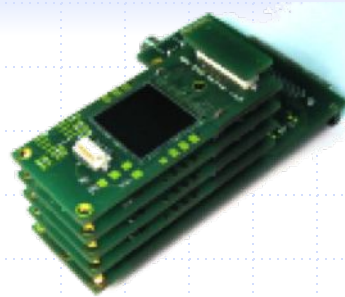
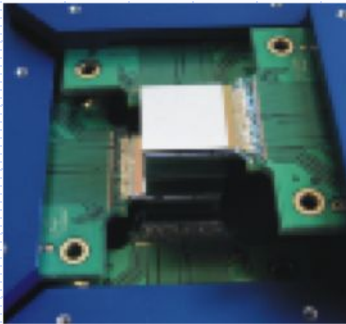
- ◆ Acquisition control
- ◆ Detector powering
- ◆ Data transfer (USB, Ethernet, ...)

Never ending need for a development of application specific variants of readout interfaces (vacuum operation, space environment, multi-detector arrays,...)





Timepix based devices developed and tested in IEAP CTU

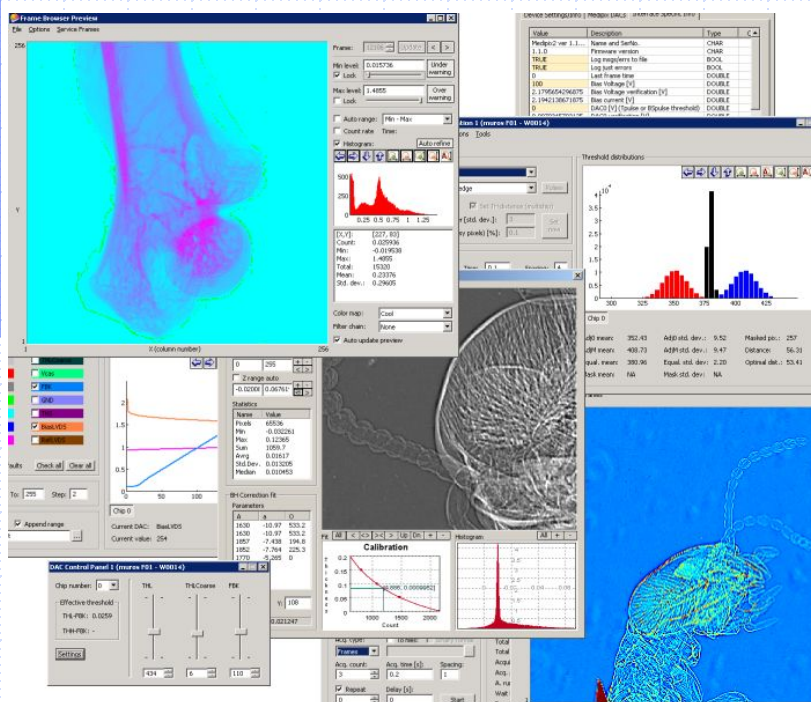




Data Acquisition Control Software

Finally, it makes Timepix detector a powerful tool.
Otherwise it is useful as a car without the steering wheel.

PIXELMAN – IEAP CTU
Acquisition and control tool for
Medipix & Timepix Detectors





SESTRA - School Education Set with Timepix for Radiation Analysis





SESTRA Kit Components



- Particle Camera MiniPixEDU (*Timepix detector, calibrated*)
- Control Software *Pixelman Simple preview & Pixet basic (acquisition, online visualisation, etc.,)*
- SZZ Alfa (*^{241}Am , α and γ source, 9.5 kBq*)
- DZZ Gamma (*^{241}Am , γ source, 300 kBq, optional*)
- Potassium Salt (*β and γ source*)
- Thoriated Tungsten Electrode (*α , β and γ source*)
- Uranium Glass (*α , β and γ source*)
- Mounting Rails
- Source Holder
- Camera Holder
- Aluminium, Stainless, Copper, Brass and Lead Shielding Plates
- Radiography Adapter Head (alpha) + *Samples with Hidden Patterns*
- Radiography Adapter Head (gamma) + *Samples with Hidden Patterns*
- Vacuum Cleaner Grate Adapter
- USB Cable
- Book of detailed guidelines *"Experiments Using Pixel Detector in Teaching Nuclear and Particle Physics"*

Set of kit accessories was designed according to practical experiences gained while using the Timepix for teaching of students...

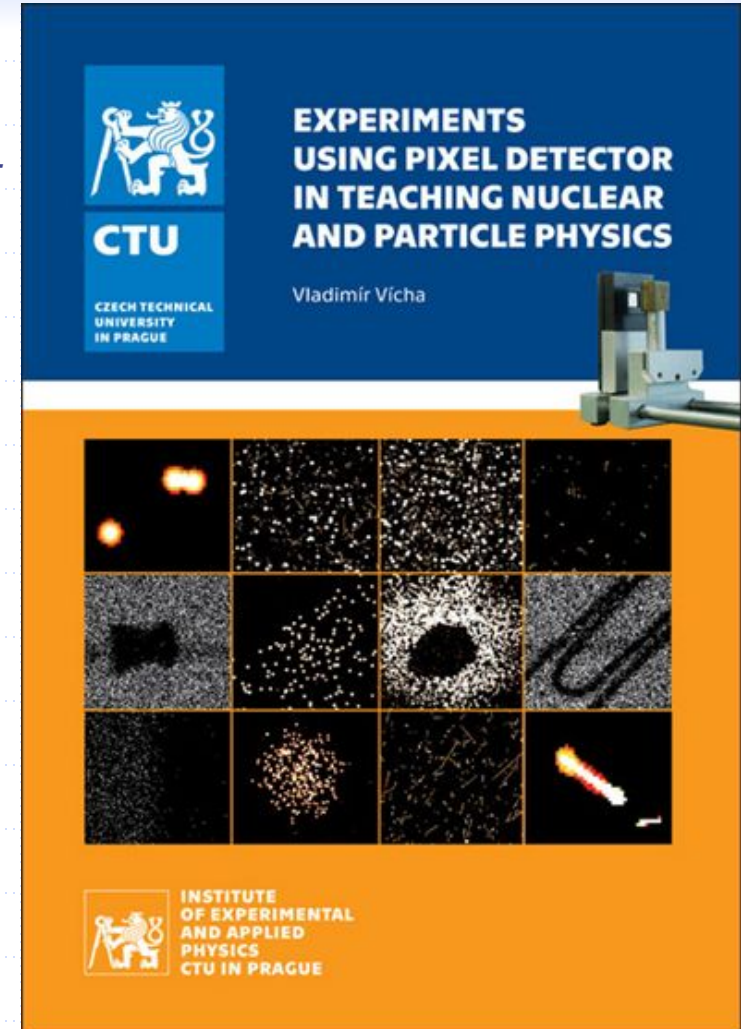


Detailed guidelines to experiments practicable with Timepix based particle camera



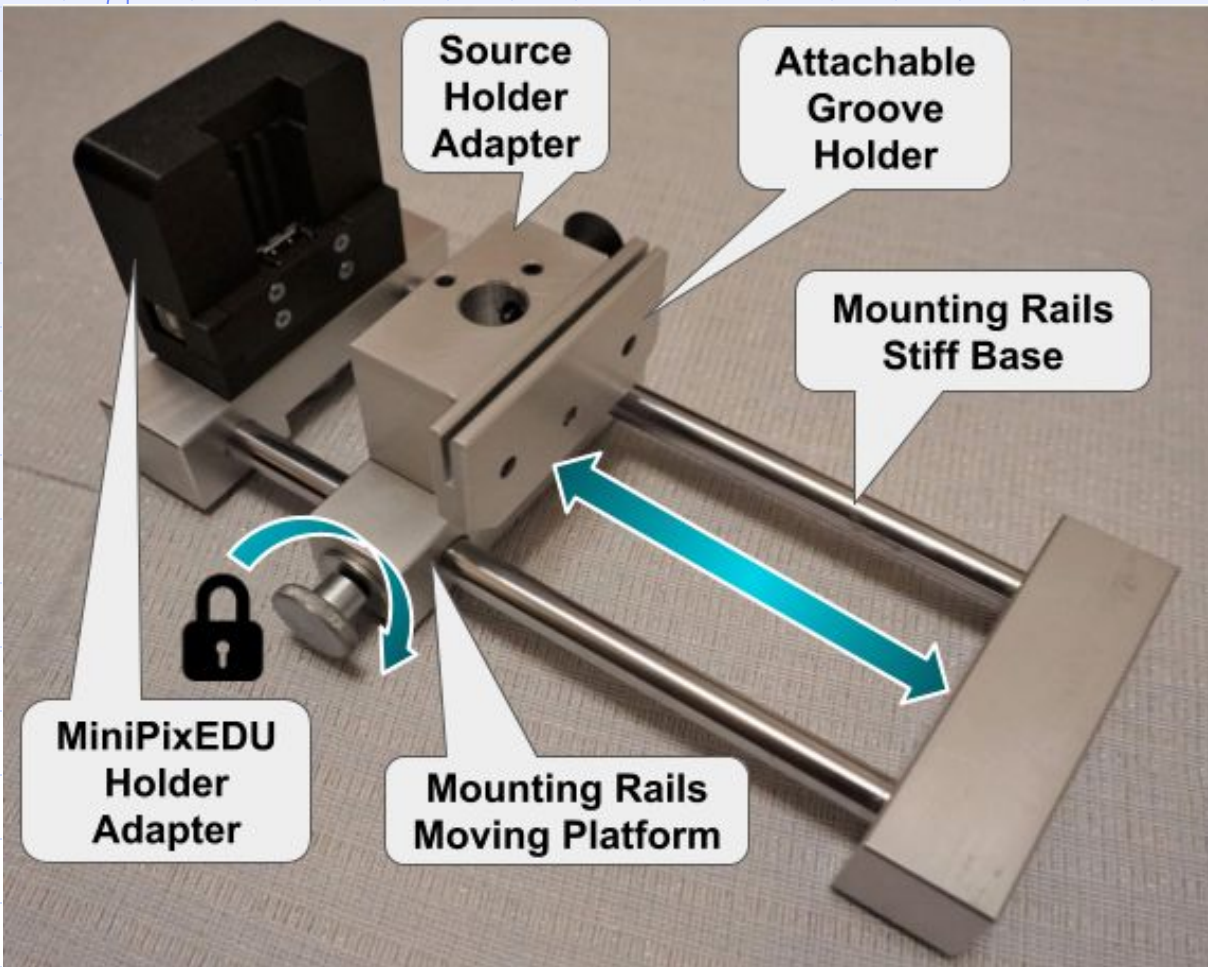
Wide set of 50 experiments
...involving detection and recognition of different kinds of ionising radiation in pixel detector, observation of natural background radiation, space originating radiation, basics of radiography....

Focused on high school students and university students by means of practical exercises to help them to understand physics and motivate them in further learning





Mechanics of the SESTRA educational kit



- Fast (re)arrangement of various experimental set-up(s)
- *Prepared for performing of real time demonstrations within a time span of the standard classes*



Set of Radiation Sources in the SESTRA kit



- Thoriated Tungsten Electrode
(α , β and γ source)
- Uranium Glass
(α , β and γ source)
- Potassium Salt
(β and γ source)
- ^{241}Am α and γ source 9.5 kBq (optional)
- ^{241}Am γ source 300 kBq (optional)



URANIUM Glass Radioactivity



Sample included in the kit



- Uranium serving as a yellowish-green glass colorant
- Freely available on the market in the form of various decorative items

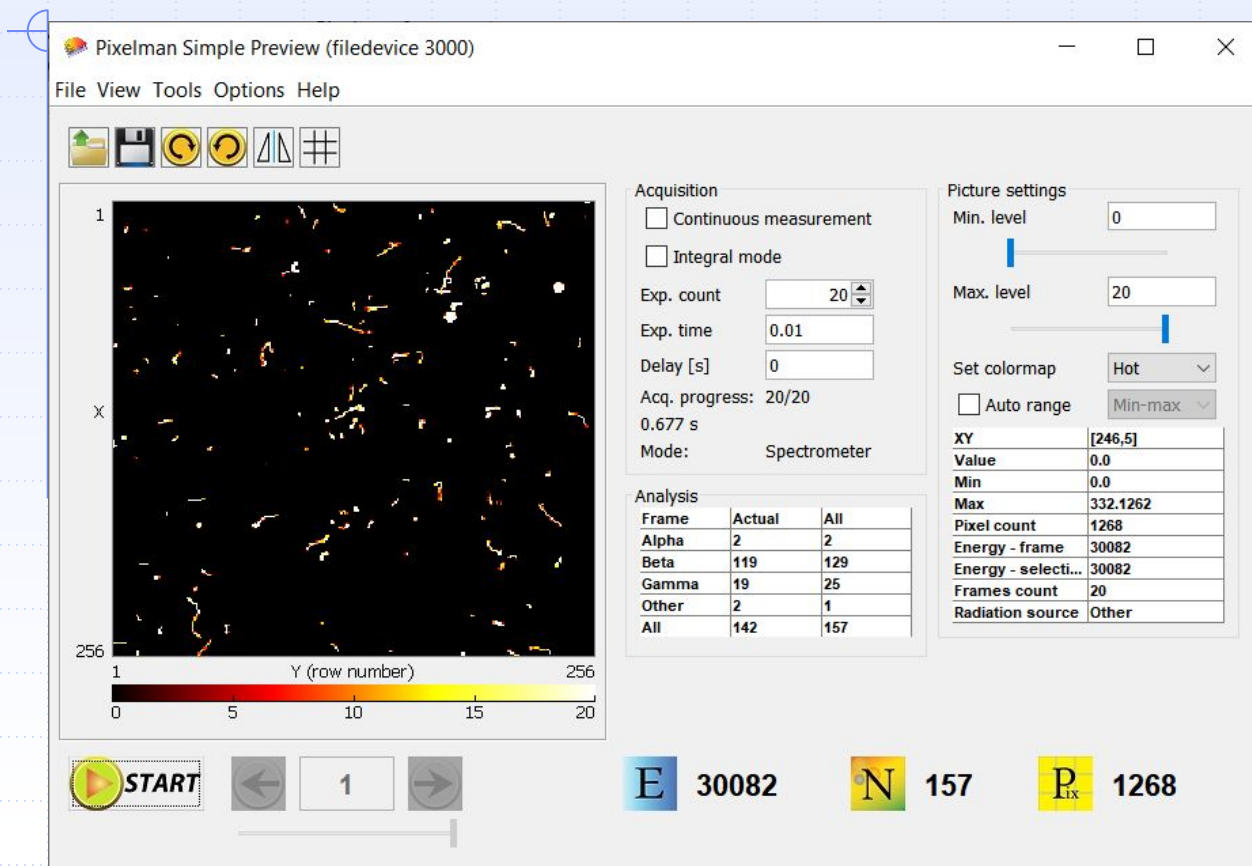
Uranium

^{235}U , 0.72%, $T_{1/2} = 7,0 \cdot 10^8$ years
 ^{238}U , 99.274%, $T_{1/2} = 4,5 \cdot 10^9$ years



Pixelman - Simple Preview

Measurement control, data acquisition and visualisation software

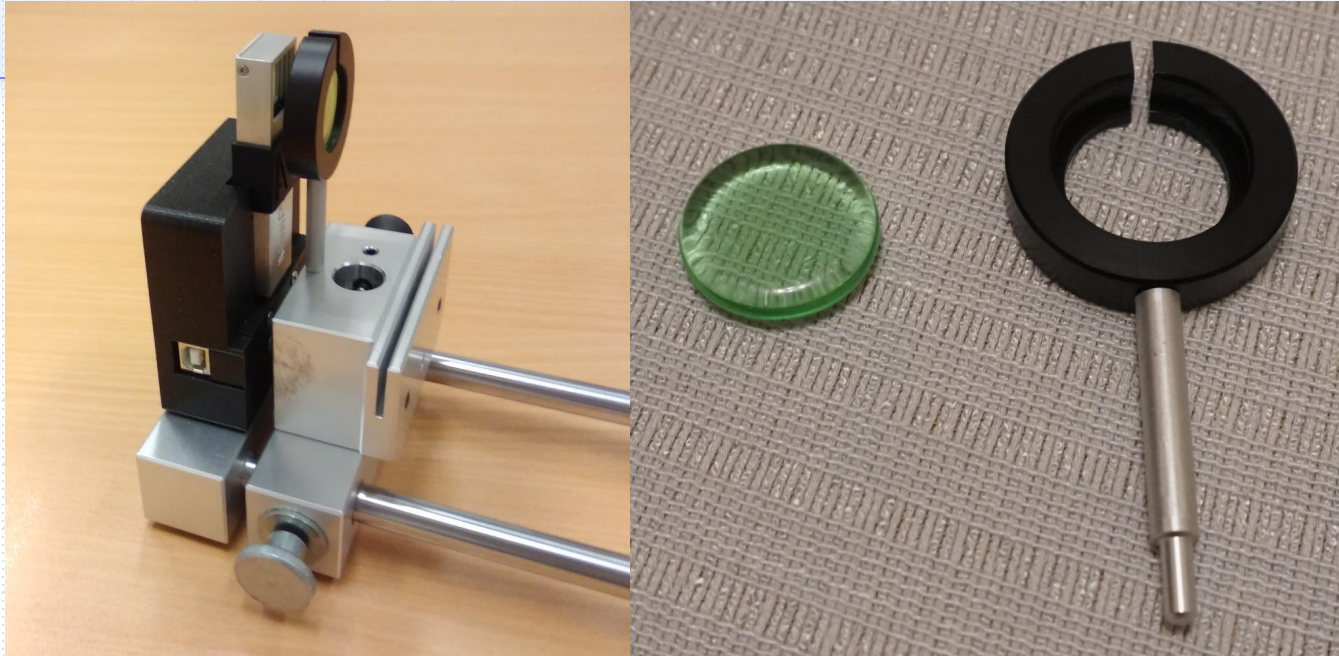


- The main control window appears after starting application "Pixelman.exe"
- One window instance for the real Particle camera and one instance for a virtual (file reader) device will appear on the screen.



Uranium Glass

preparation of measurement



Experiment set-up:

Particle camera and computer, mounting rails, uranium glass sample placed in the source holder close to the detector

Settings:

Exp. count: 60

Mode: Spectrometer

Analysis type: Basic

Bias voltage: 20V

Integral mode: No

Continuous measurement: No

Exp. time: 1 s

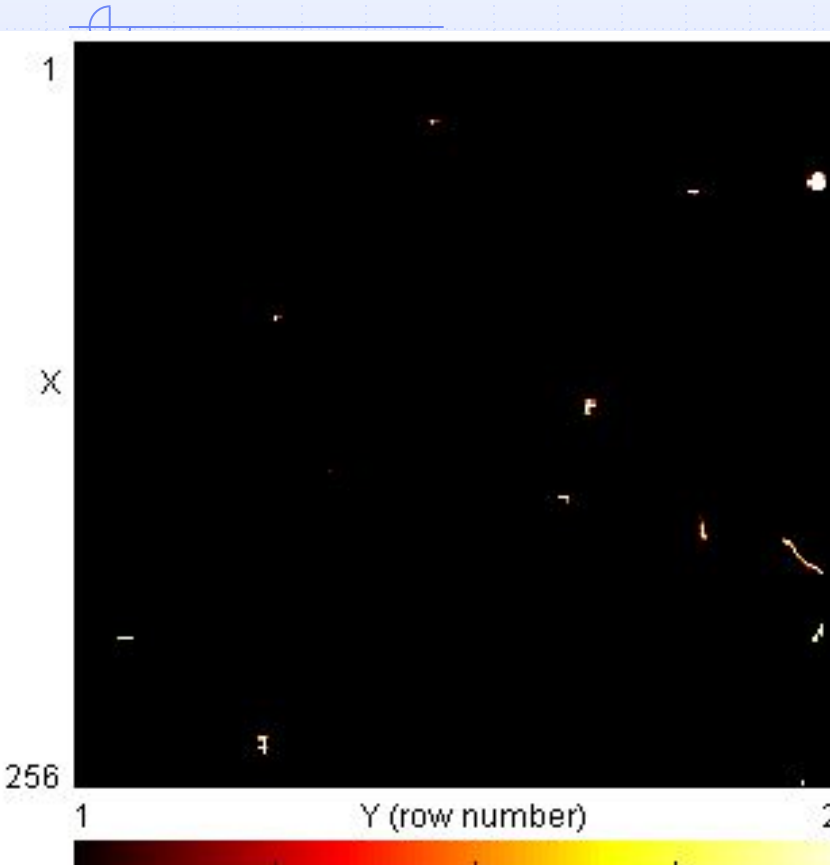
Min. level: 0

Max. level: 20

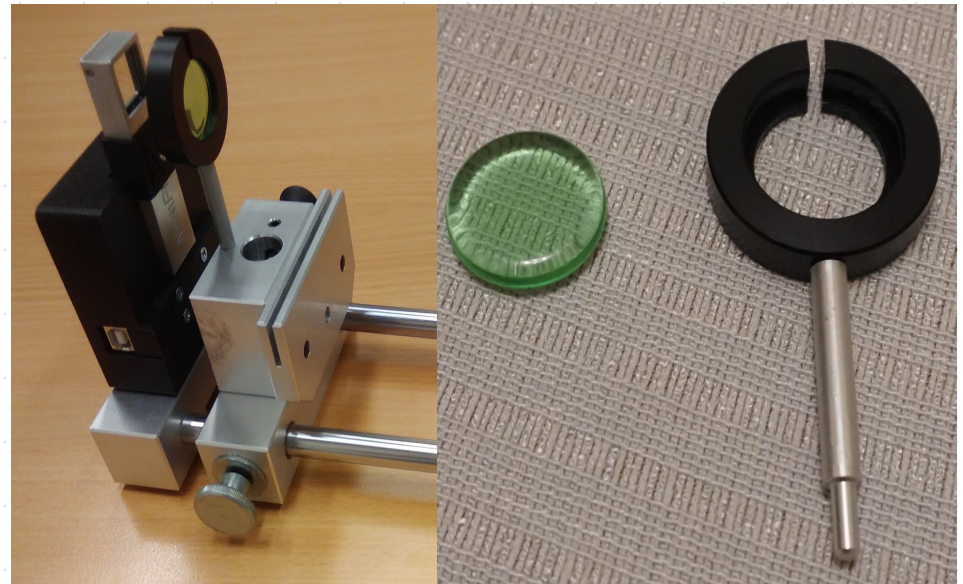
Colormap: Hot



URANIUM Glass Radioactivity Observation



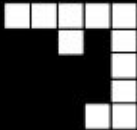
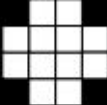
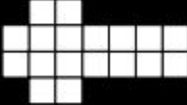
- Uranium glass as a weak source of mixed radiation field
- Evidence of alpha, beta, gamma particle detection





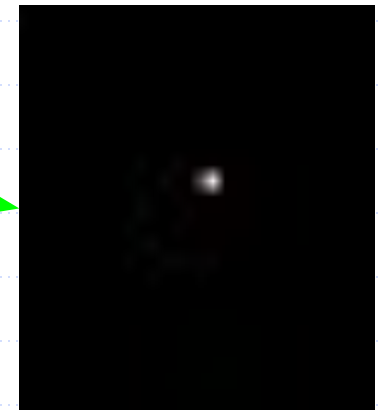
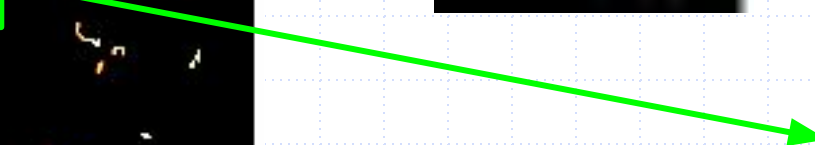
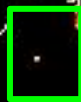
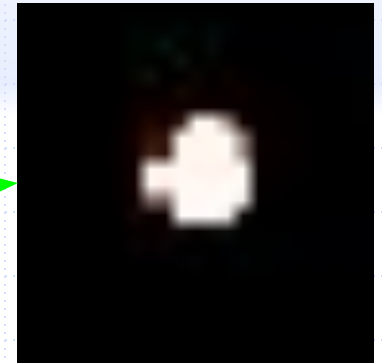
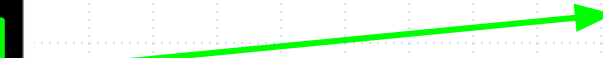
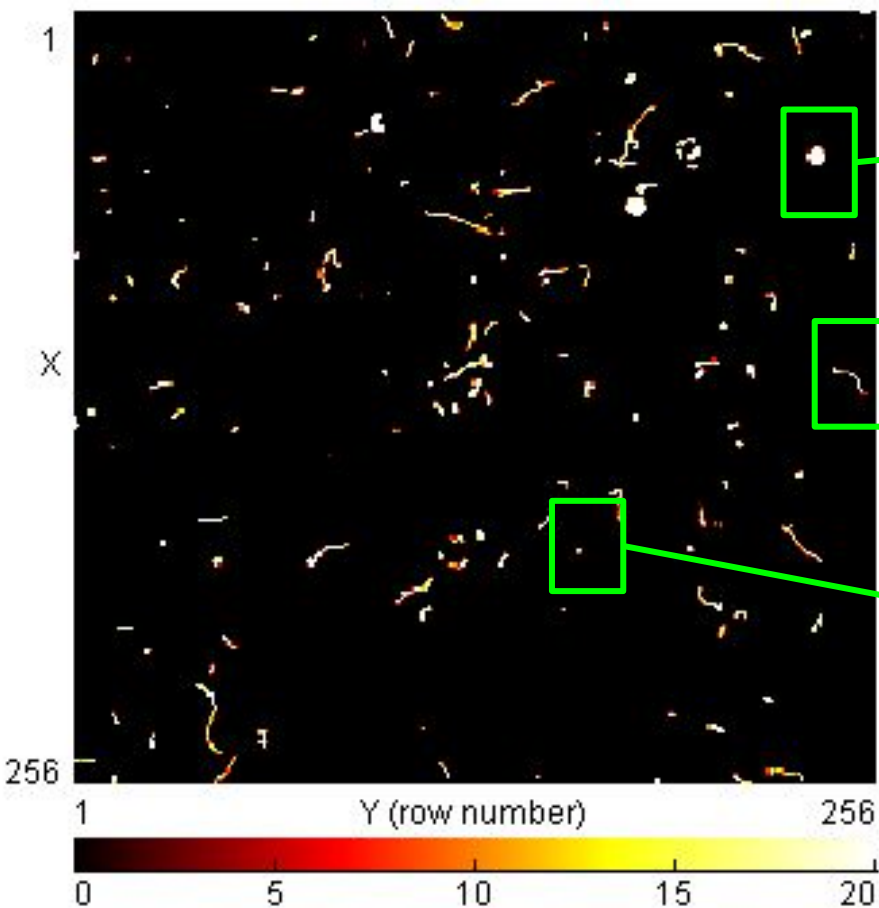
Review of the characteristic patterns



1) Dot		Photons and electrons (10keV)
2) Small blob		Photons and electrons
3) Curly track		Electrons (MeV range)
4) Heavy blob		Heavy ionizing particles with low range (alpha particles,...)
5) Heavy track		Heavy ionizing particles (protons,...)
6) Straight track		Energetic light charged particles (MIP, Muons,...)



Uranium Glass Radioactivity Close Look Observation

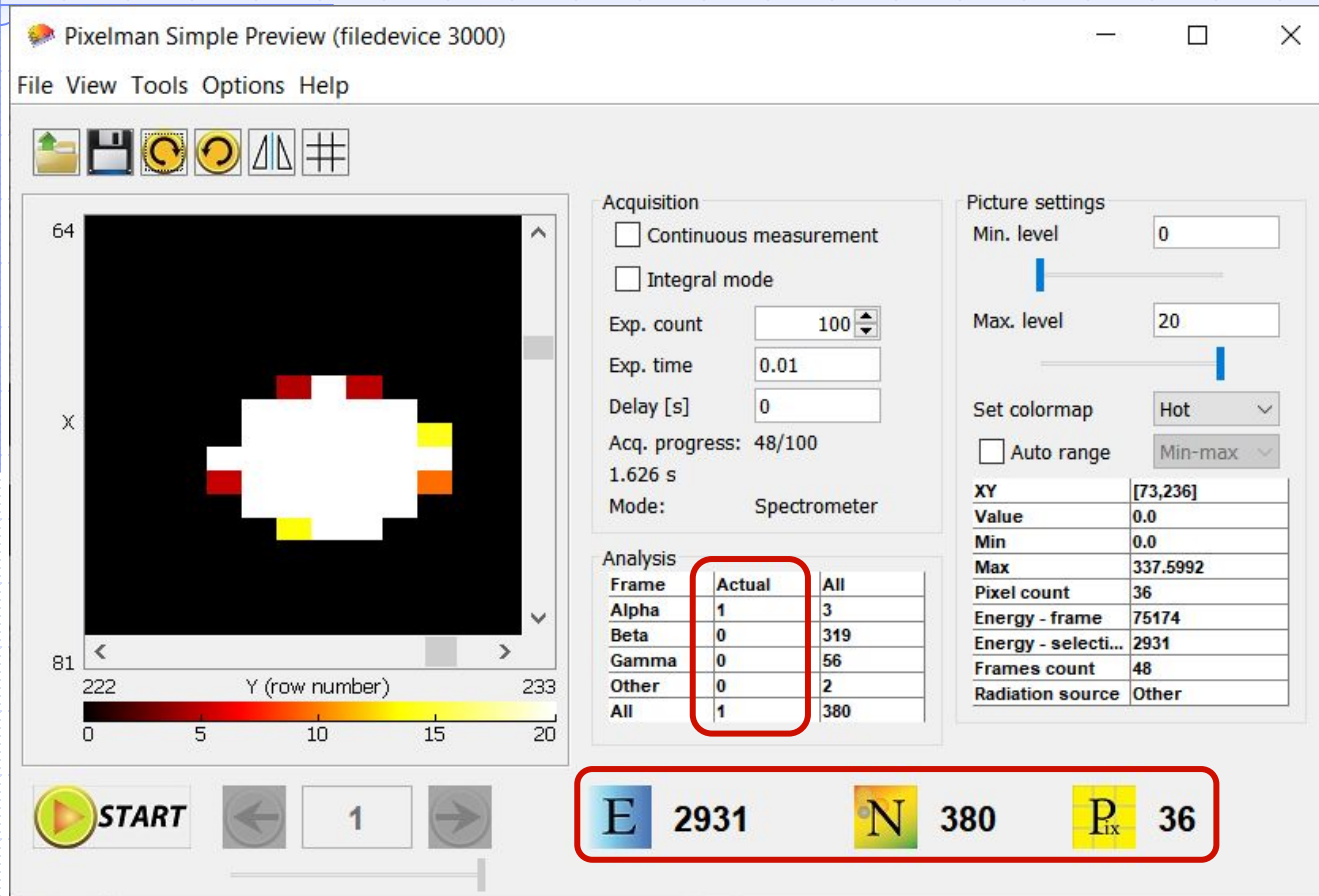


- Evidence of alpha, beta, gamma particle detection



Uranium Glass Radioactivity

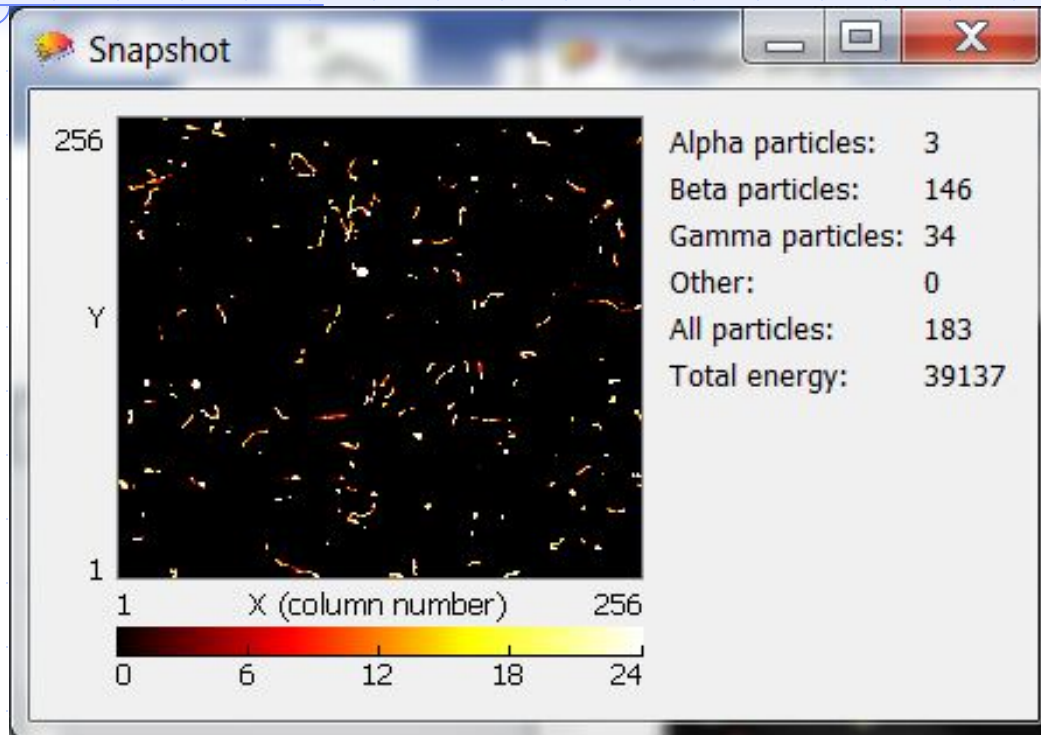
Close look on a single particle



When zoomed-in, properties of single particle track can be obtained (deposited energy in keV, cluster area, track type classification)



Uranium Glass Radioactivity *as seen by the Timepix pixel detector*



Several useful hints on software functionality

Getting a snapshot (Menu: Tools -> Snapshot)

*Displaying of integral frame composed of separate frames
(Menu: Tools -> Integral frame)*



Kinetic Energy Absorbed in the Sensor, How to find-out speed of hitting particle



The Einstein relation

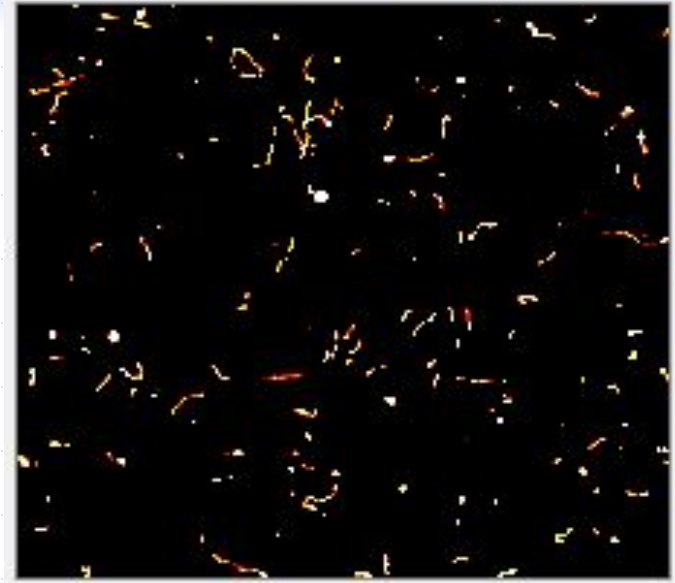
$$E = E_0 + E_k$$

$$v = c \sqrt{1 - \frac{1}{\left(1 + \frac{E_k}{E_0}\right)^2}}$$

Rest mass energy:

Alpha: $E_0 = 3,735,000$ keV ,

Beta: $E_0 = 511$ keV



Speed of alphas?

Speed of betas?

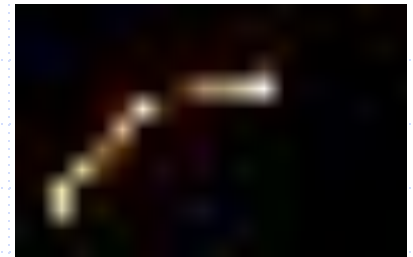
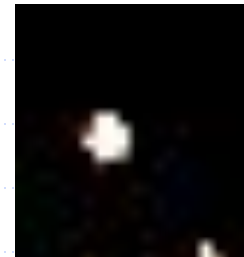
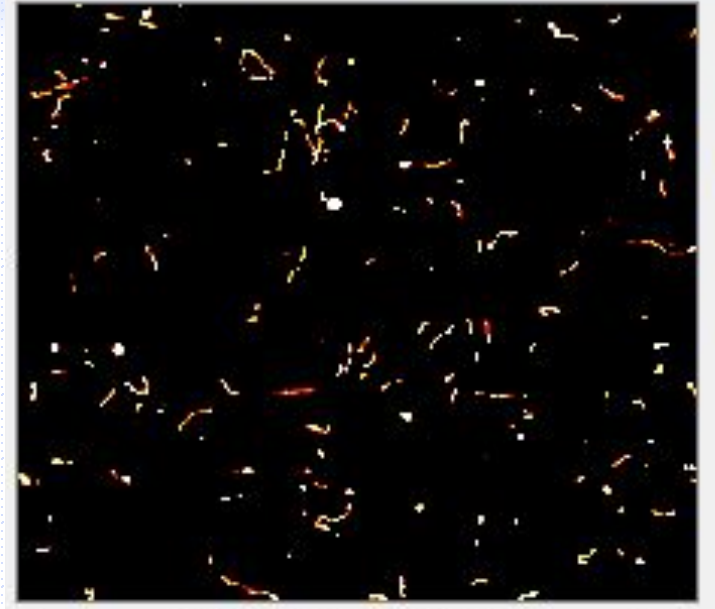
Hint:

*Use the excel table "Speeds of particles i lambda.xlsx"
to compute speed of observed particles*



Kinetic Energy Absorbed in the Sensor, How to find-out speed of hitting particle

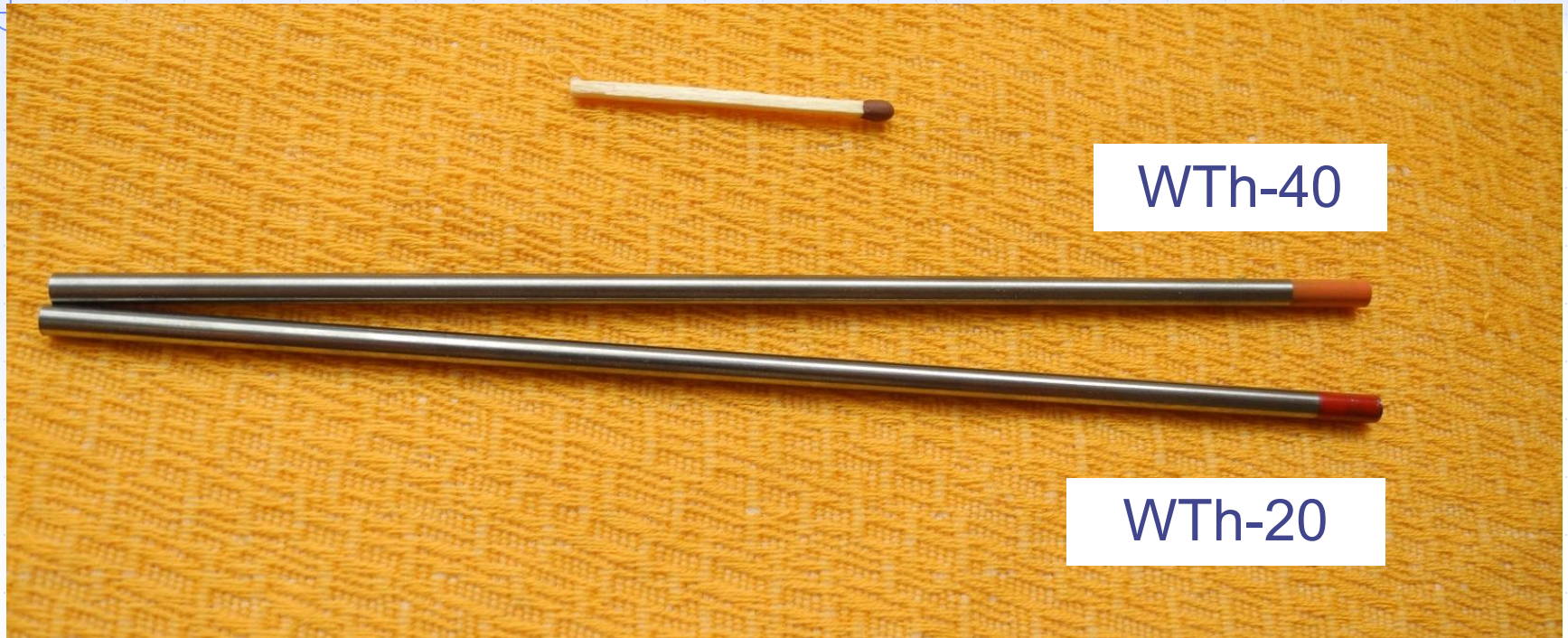
- What are the highest energies of detected alphas, bettas and gammas?
- Speed of alphas
VS
speed of bettas?
- Can speeds of alphas or betas be comparable to speed of light?



$E_{\max}$?



Tungsten Welding Electrode with THORIUM



Thorium ^{232}Th , $T_{1/2} = 1,4 \cdot 10^{10}$ years

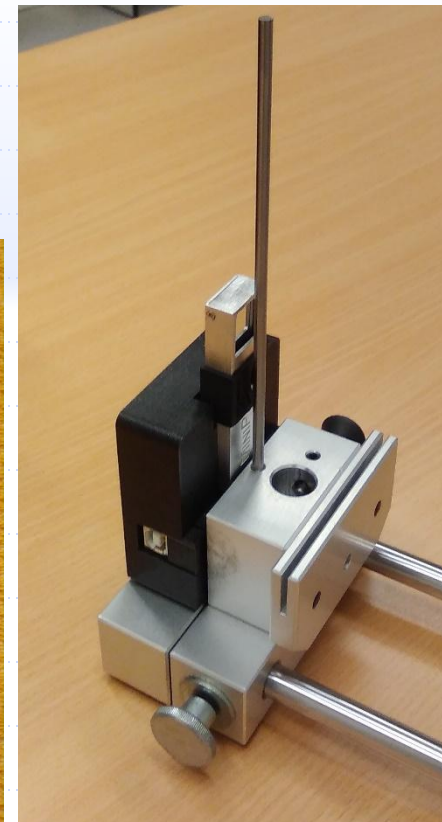


Age of the Earth: $4,5 \cdot 10^9$ years



Thoriated Welding Rod

preparation of measurement



Experiment set-up:

Particle camera, computer, mounting rails, thorium welding rod in a holder.

Settings:

Exp. count: 60

Mode: Spectrometer

Analysis type: Basic

Bias voltage: 20V

Integral mode: No

Continuous measurement: No

Exposition time: 1 s

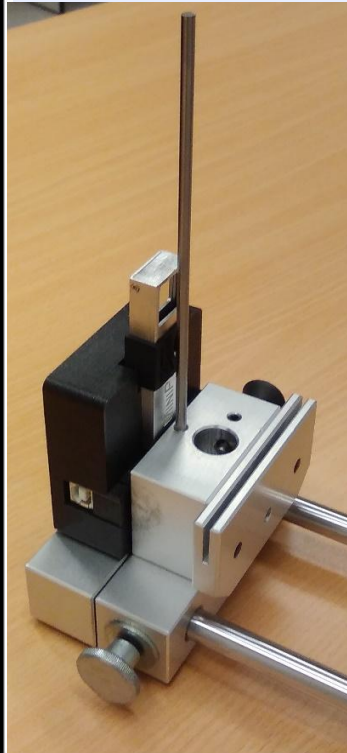
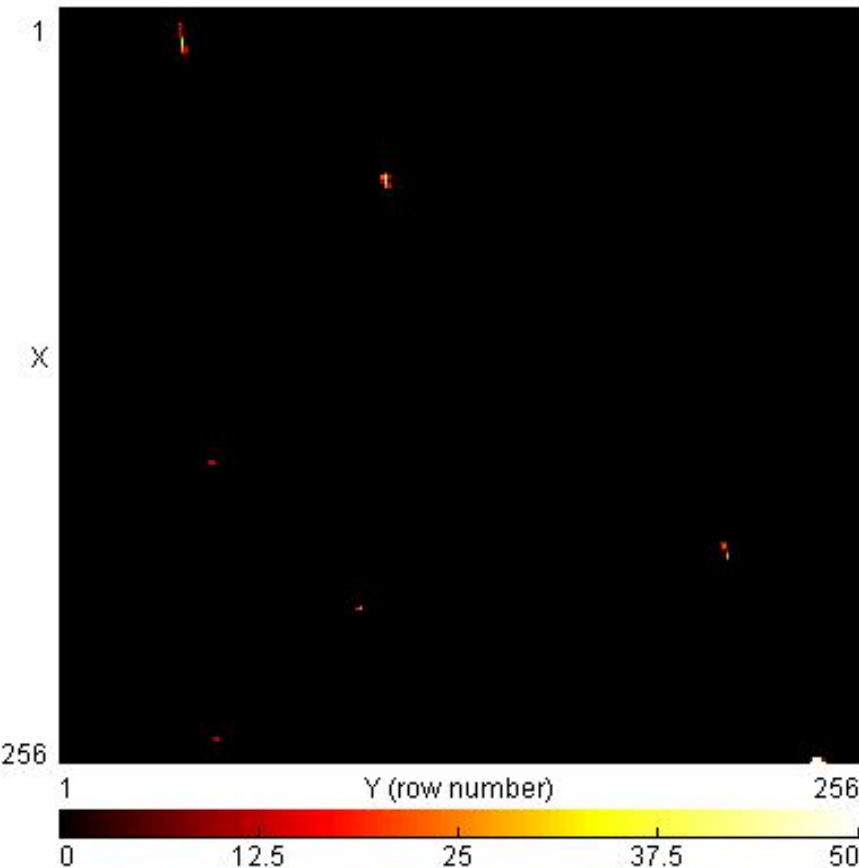
Min. level: 0

Max. level: 20

Colormap: Hot



Thoriated Welding Rod Radioactivity Observation

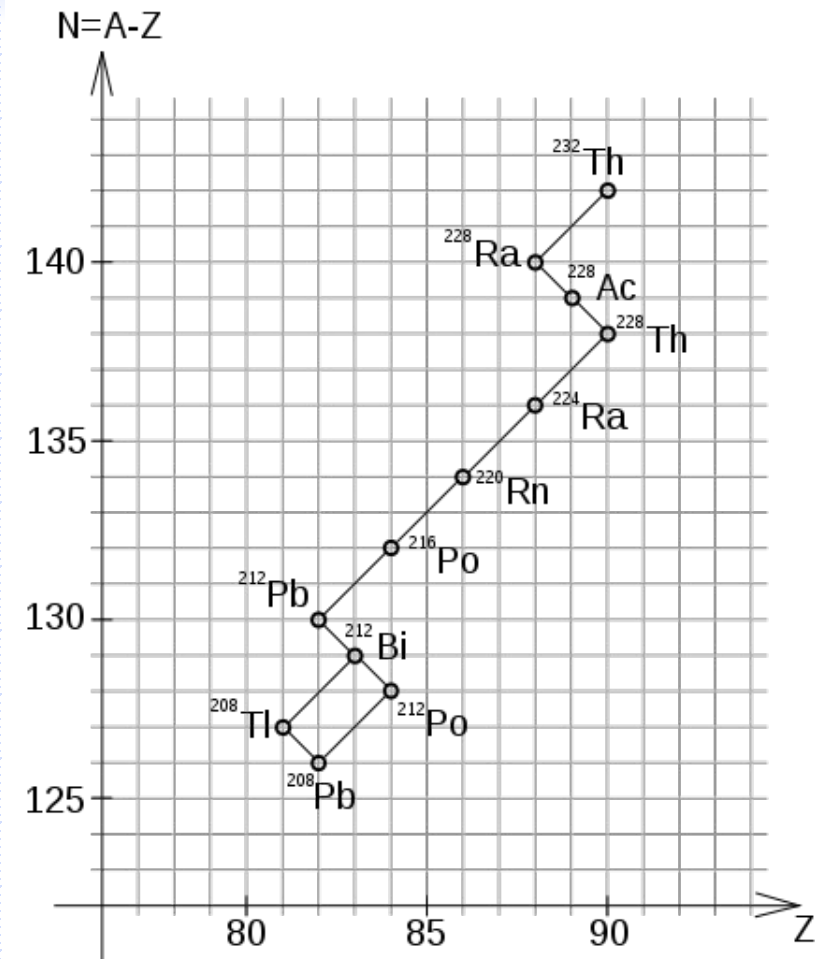
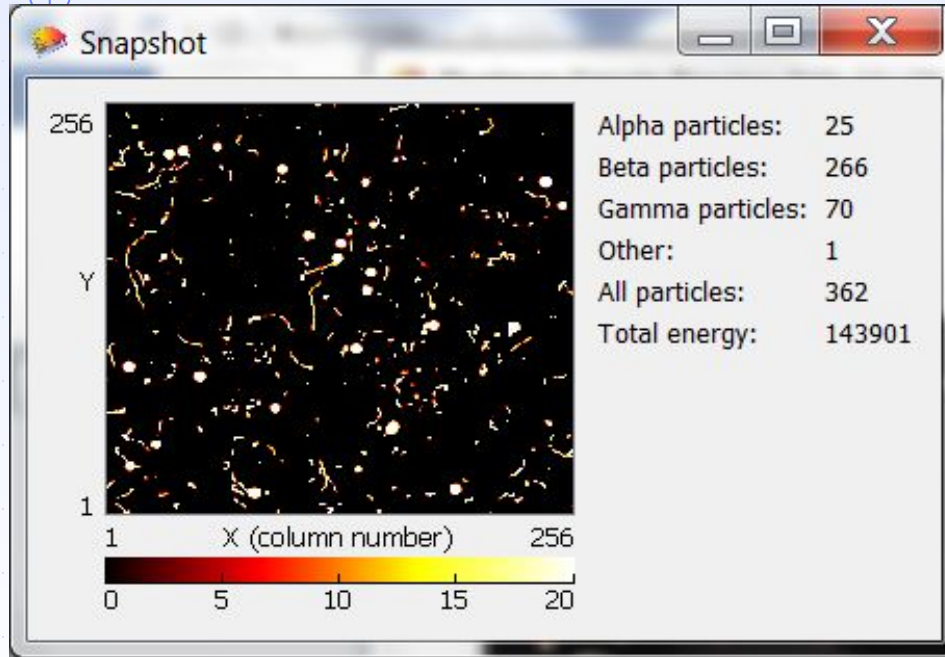


- Thoriated electrode as a weak source of mixed radiation field
- again, evidence of alpha, beta, gamma detection
- Higher portion of alpha radiation is apparent





Thorium Radioactivity *as seen by Timepix pixel detector*



Are there more energetic alpha and beta tracks in comparison to uranium glass observation?



POTASSIUM Chloride Salt



- KCl - Potassium Chloride
- Natural abundance 0.0117% of radioactive isotope ^{40}K with half life $1.277 \cdot 10^9$ years



Decay modes

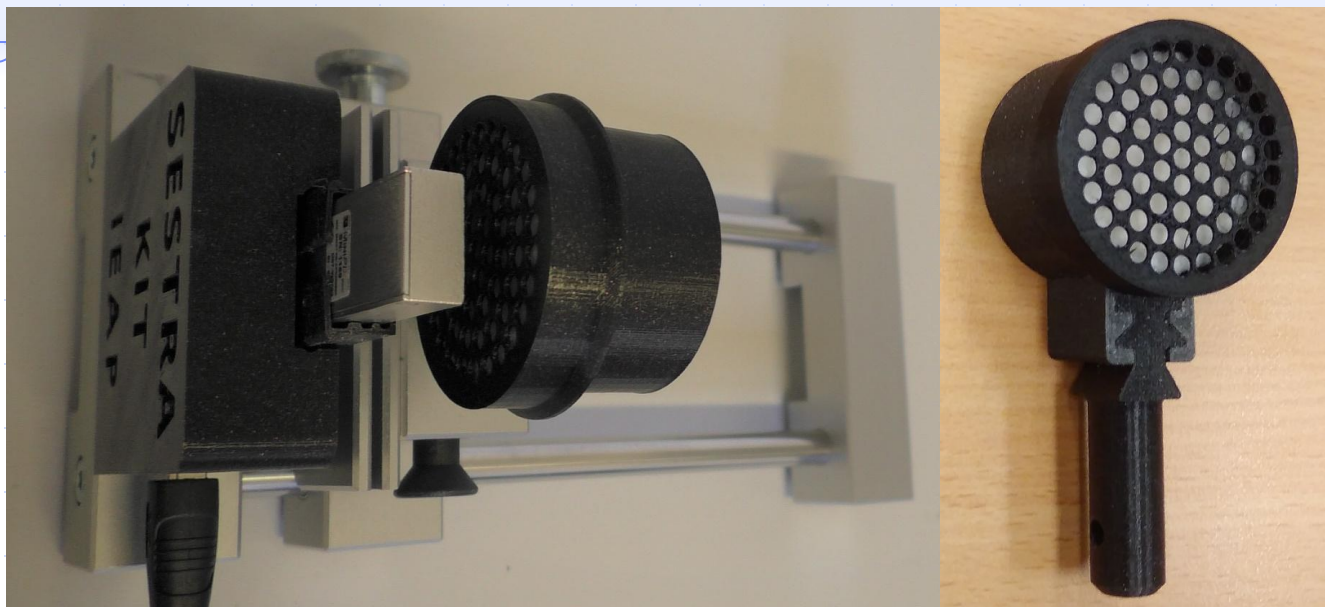
$^{40}_{19}\text{K} \rightarrow ^{40}_{20}\text{Ca} + ^{-1}_0\text{e} + \bar{\nu}_e$ 89.28 %, emission of β^- with energy 1.311 MeV

$^{40}_{19}\text{K} + ^{-1}_0\text{e} \rightarrow ^{40}_{18}\text{Ar}^* + \nu_e$ 20.72 %, emission of gamma with energy 1.5049 MeV



Potassium Chloride Salt

preparation of measurement



Experiment set-up:

Particle camera and computer, container with potassium salt inserted into the holder at close position to the detector.

Settings:

Exp. count: 60

Mode: Spectrometer

Analysis type: Basic

Bias voltage: 20V

Integral mode: No

Continuous measurement: No

Exp. time: 1 s

Min. level: 0

Max. level: 20

Colormap: Hot



...Few facts about potassium

How much potassium does a human body contain?

A human body of 70 kg weigh contains about 140 g of potassium while just (0.0117%) a small portion of 0.0164 g is the radioactive isotope ^{40}K

Aren't we living radiation sources???

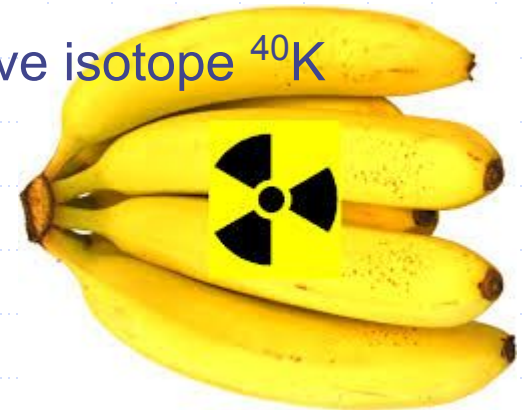
Corresponding activity of such amount is 4300 Bq (decays/sec)

Are bananas save to eat???

1kg of banana contains 0.390 g of potassium

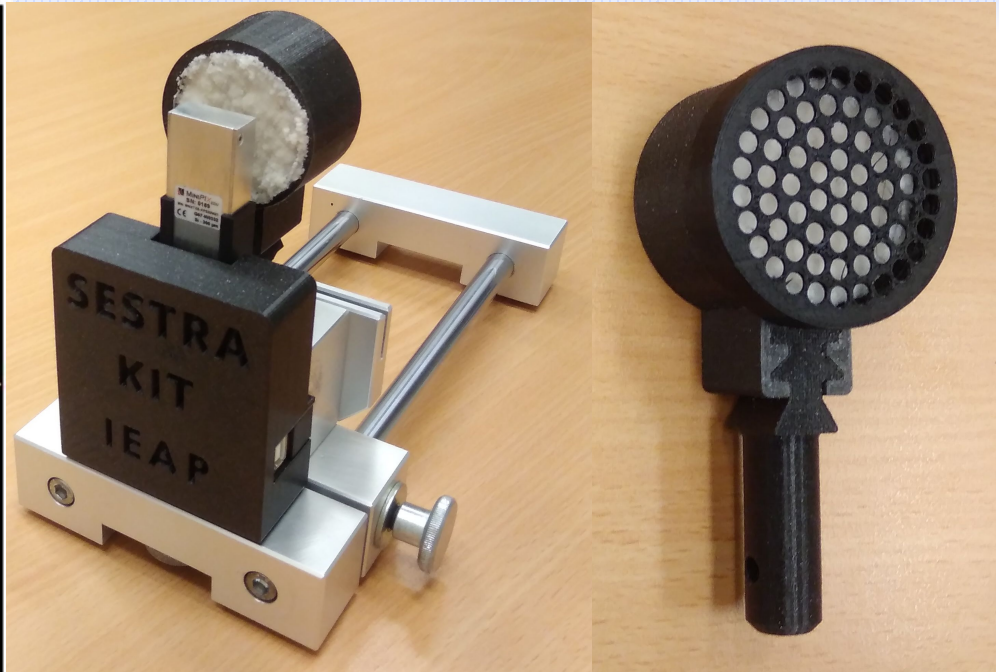
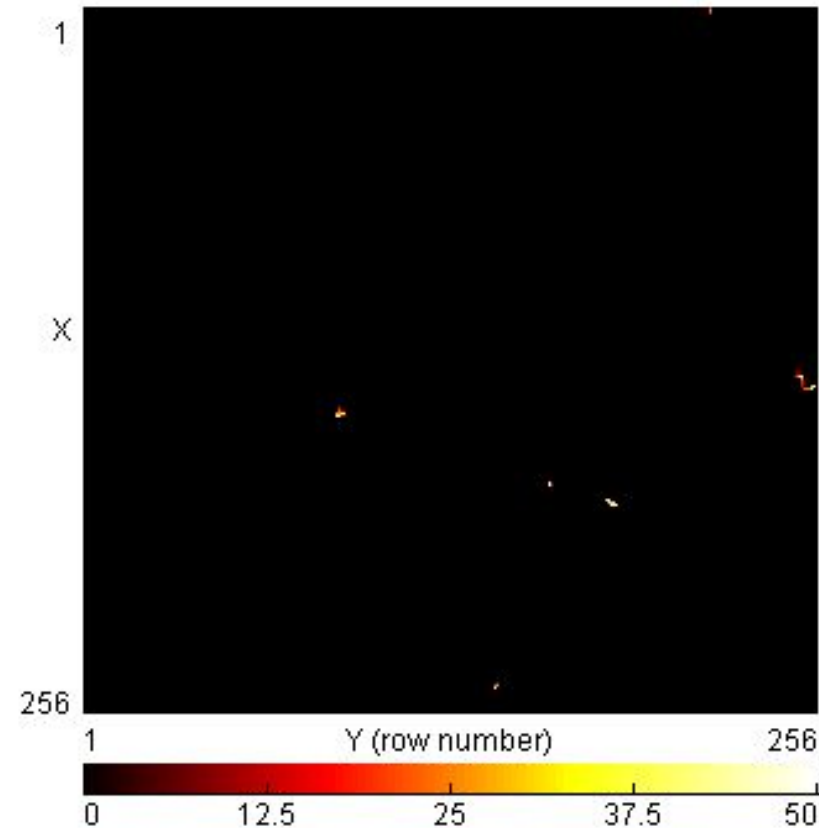
...where 0.00004563 g is a portion of the radioactive isotope ^{40}K

...with activity of 12 Bq (decays/sec)





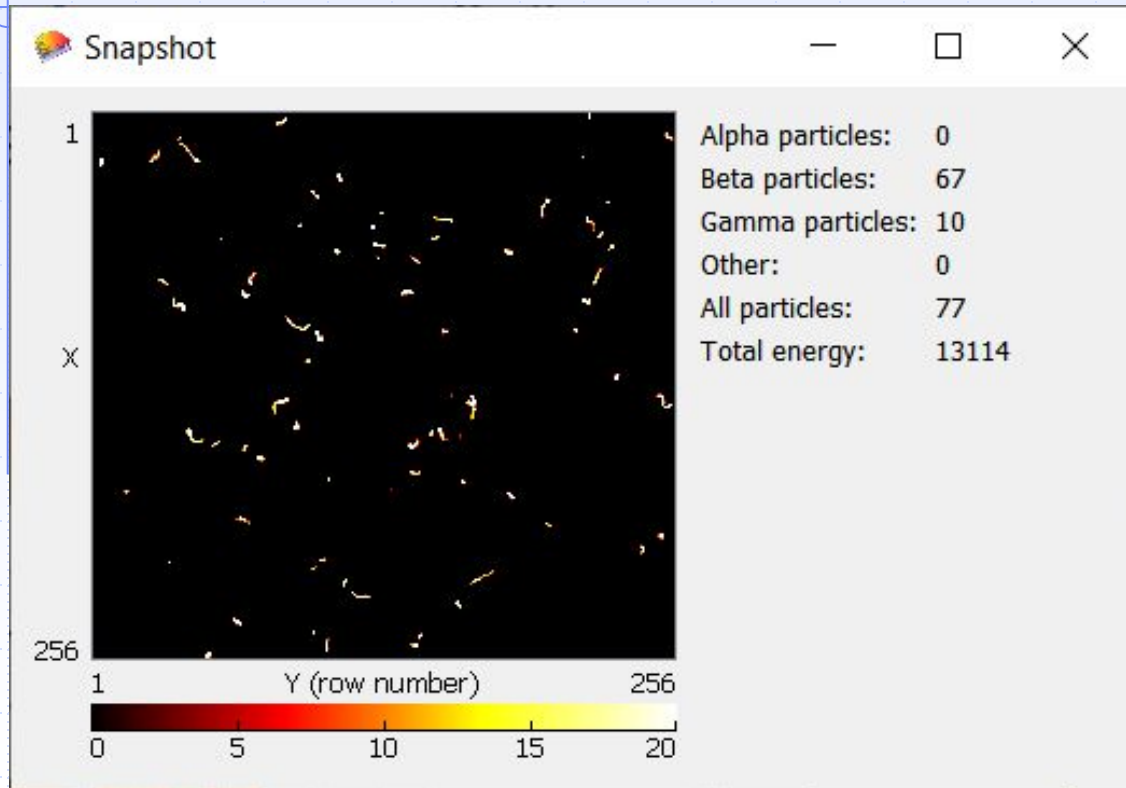
Potassium radioactivity observation



- Container filled with the potassium chloride (KCl) salt
- Weak source of gamma and beta radiation



Potassium Radioactivity *as seen by Timepix pixel detector*



Obvious difference to Uranium glass and
Thoriated electrode experiment
- no alpha particle tracks present



SZZ Alfa

^{241}Am isotope radiation source



- Activity 9.5 kBq
- Adapted for educational purposes
- Source of alpha and gamma radiation

Alfa decay - Americium:



Half-life of ^{241}Am is 432 years.

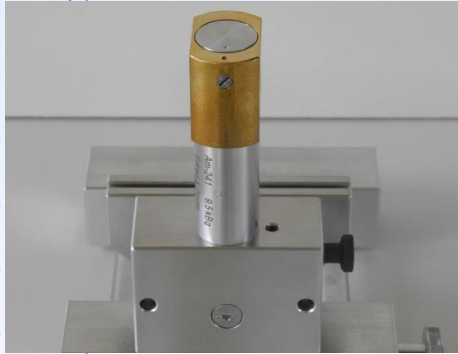
$$E(\alpha) = 5\,500\text{ keV}$$

$$E(\gamma) = 59.5\text{ keV}$$

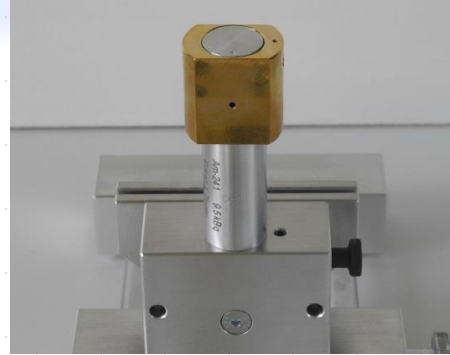


SZZ Alfa Radiation Source

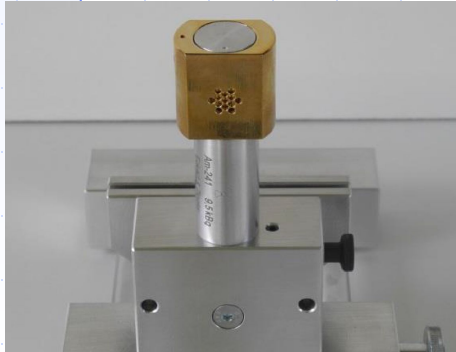
4 selectable output configurations



1 - Closed



3 –“Short collimated”



2 - „Sieve“
(multihole)

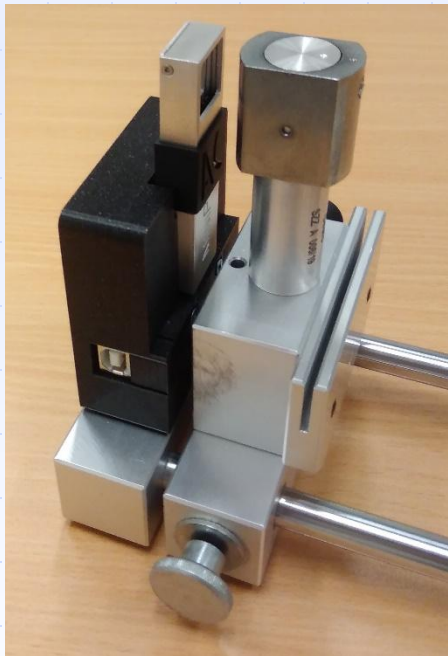


4 –“Long Collimated”

Note: Radiation in other directions is shielded out by a brass body



9. ENERGY OF ALPHA PARTICLES EMITTED FROM SZZ Alfa 241Am radiation source



Experiment set-up:

Particle camera, computer, mounting rails, SZZ Alfa placed in a holder.

Settings:

Exp. count: 60

Mode: Spectrometer

Analysis type: Basic

Bias voltage: 20V

Continuous measurement: Yes

Integral mode: No

Exposition time: 0.05 s

Min. level: 0

Max. level: 20

Colormap: Hot

Visual Tools: Histograms of particle properties - Energy all frames (tab Alpha)



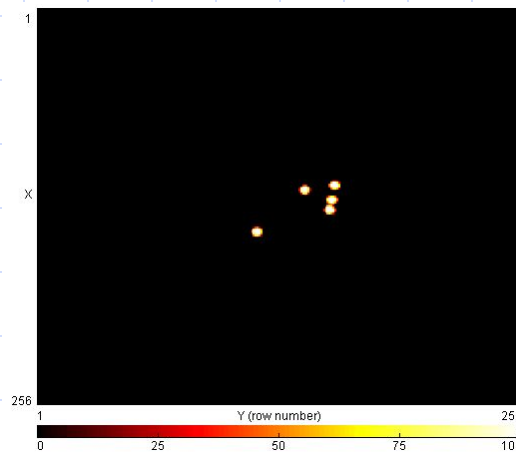
SZZ Alpha Radiation Source

Americium 241 source observation



- Source of alpha and gamma radiation
low activity, 9.6 kBq
- Suitable for demonstration of properties of heavy charged particles

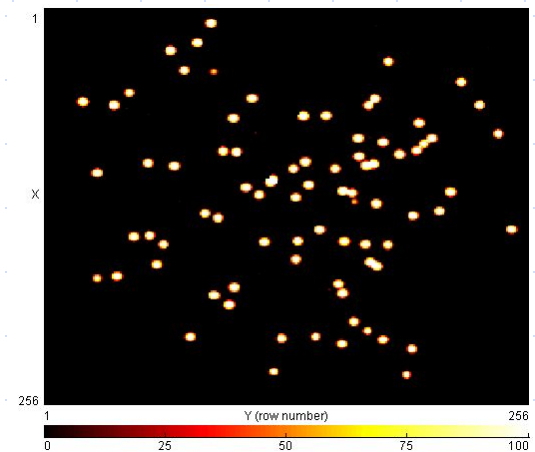
Collimated



vs

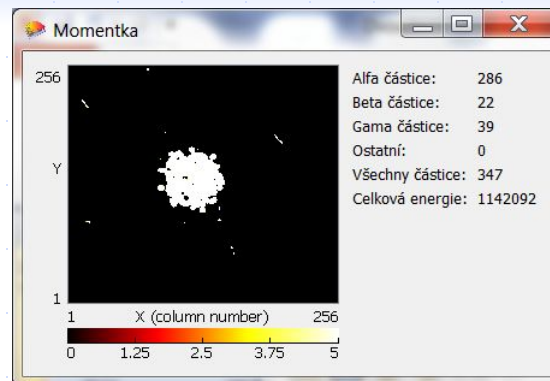
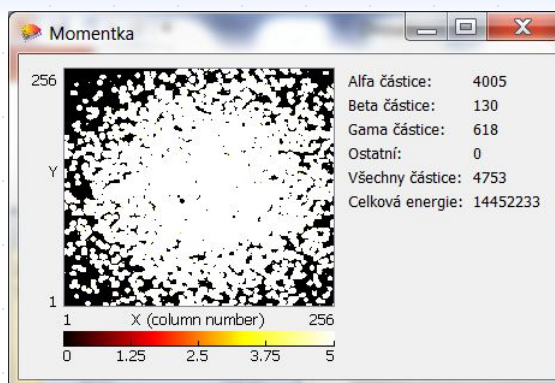
Non-collimated

Beam



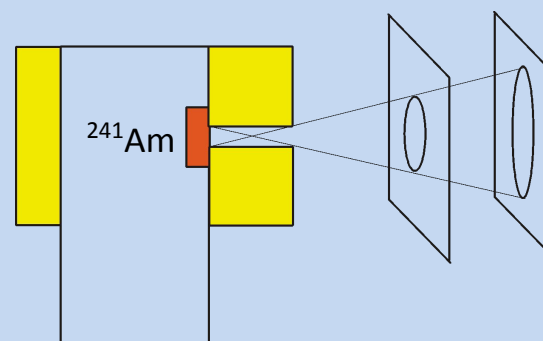
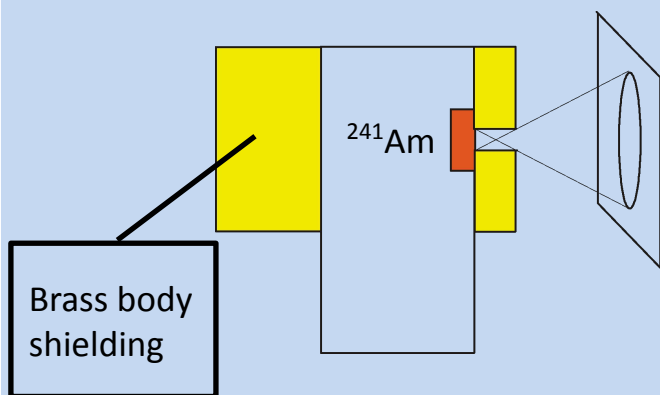


Beam Collimation



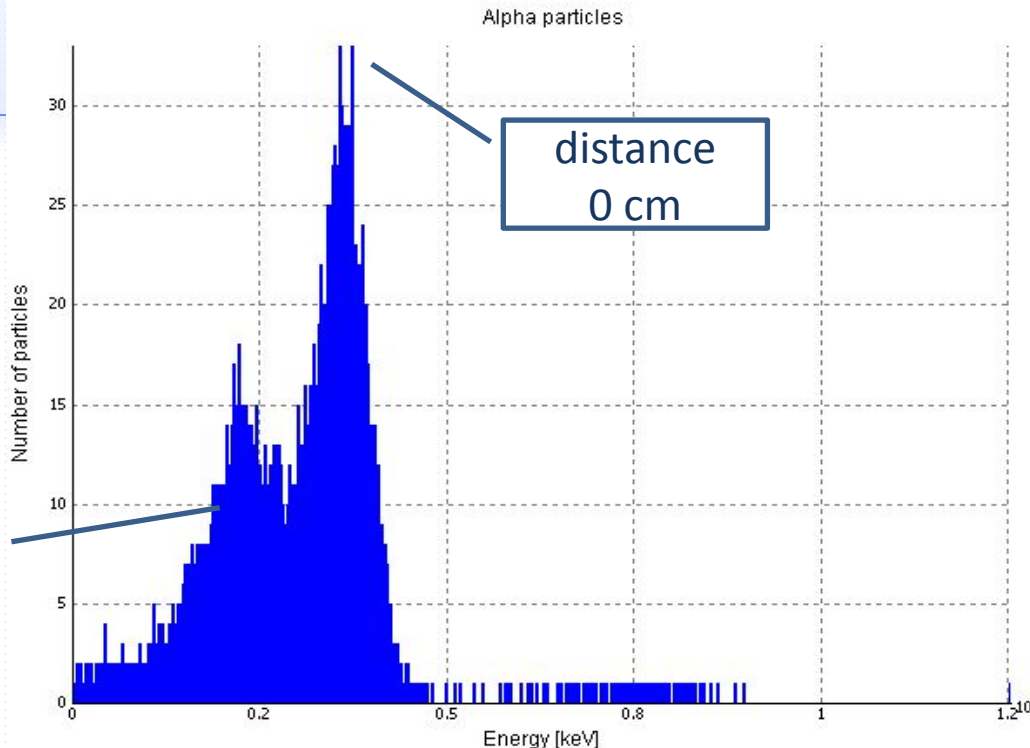
Short collimated output

Long collimated output





Loss of Alpha Particle Energy in the air



LET=1590 keV/cm
for 3.5 MeV alpha

$a \sim 10^{15} \text{ m/s}^2$

Experiment set-up:

Particle camera and computer, SZZ alfa placed in the holder, distance in between the detector and radiation source is changed

Settings:

Exp. count: 60

Mode: Spectrometer

Analysis type: Basic

Bias voltage: 20V

Integral mode: No

Continuous measurement: YES

Exp. time: 0.05 s

Min. level: 0

Max. level: 20

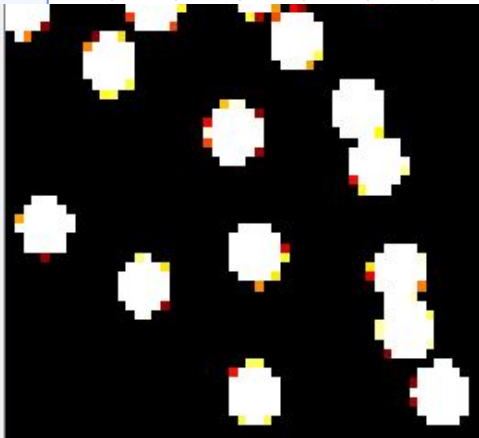
Colormap: Hot

Visual Tools: Histograms of particle properties - Energy all frames (tab Alpha)

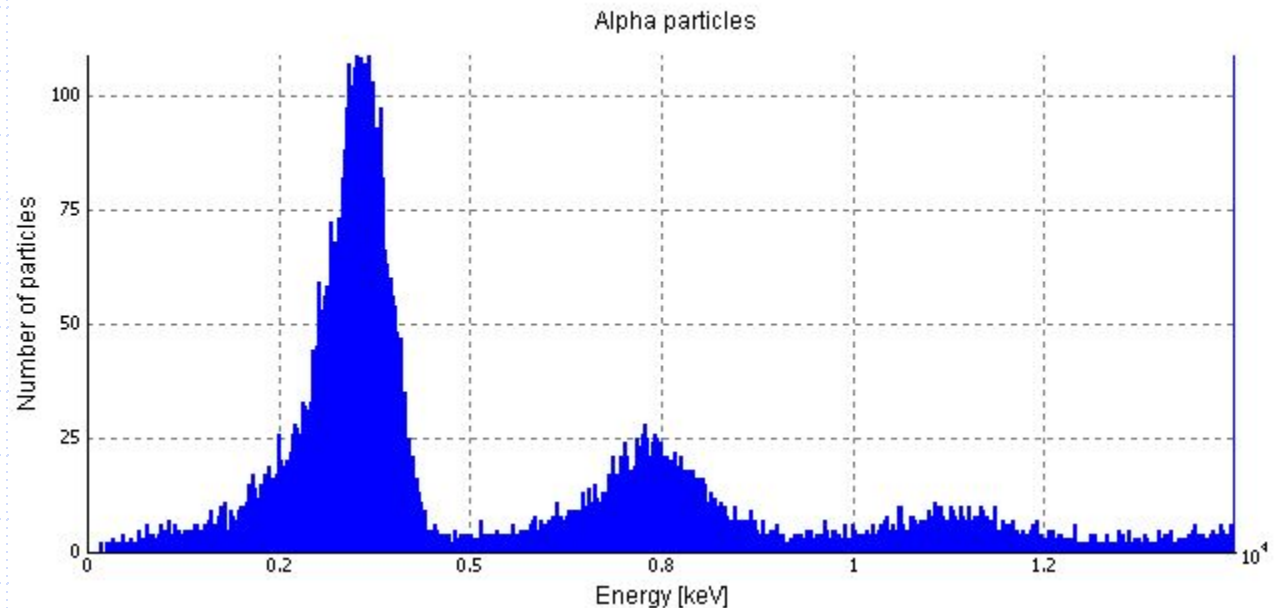


What happens if frame exposition time is significantly prolonged???

- Americium 241 represents a source of monoenergetic alpha particles but there are displayed several peaks in the spectrum
- Energy of higher peaks exceeds energy of americium alphas



- **Effect of particle track overlapping**





Loss of Alpha Particle Energy in other materials



Experiment set-up:

Particle camera and computer, SZZ alfa placed in the holder, various materials are placed in between the detector and radiation source

Visual Tools:

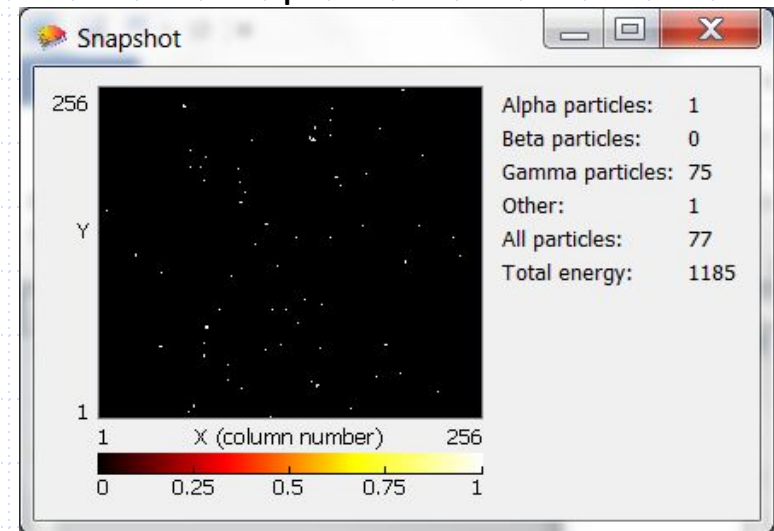
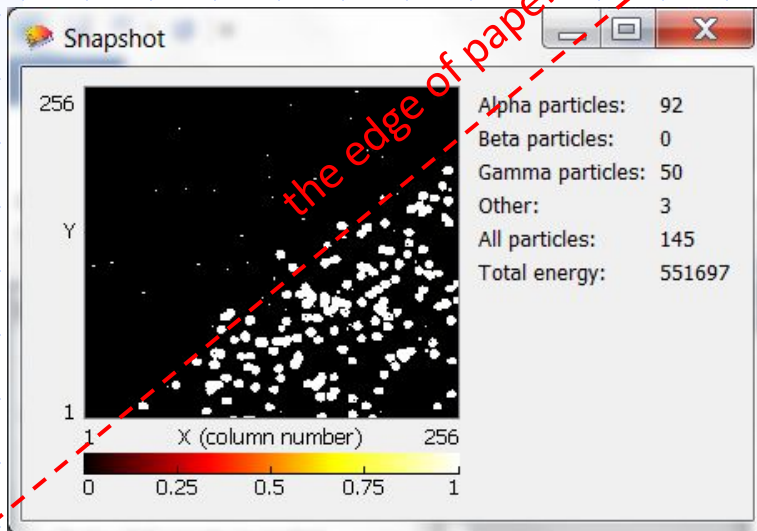
Histograms of particle properties - Energy all frames (tab Alpha)

Histograms of particle properties - Histogram of rate (tab Alpha and Gamma)

What happens if some material is placed in between the alpha source and the detector???

...Sheet of paper, aluminium wrapping foil, thin plastic foil,

The entire chip area is covered

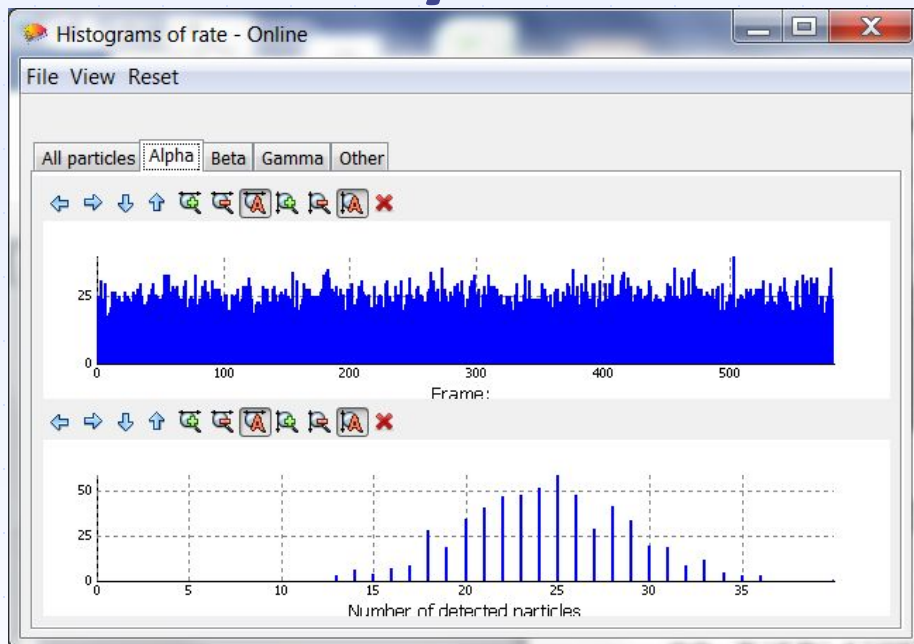


Is the covered area really empty without any tracks?



Statistical Nature of Radioactivity

Decay n - Random Variable



Experiment set-up:

Particle camera and computer, SZZ alfa placed in the holder, multihole output is selected, radiation source close to the detector

Settings:

Exp. count: 200

Mode: Spectrometer

Analysis type: Basic

Bias voltage: 20V

Integral mode: No

Visual Tools: Histograms of particle properties -

Energy all frames (tab Alpha or tab Gamma)

Continuous m.: YES

Exp. time: 0.05 s

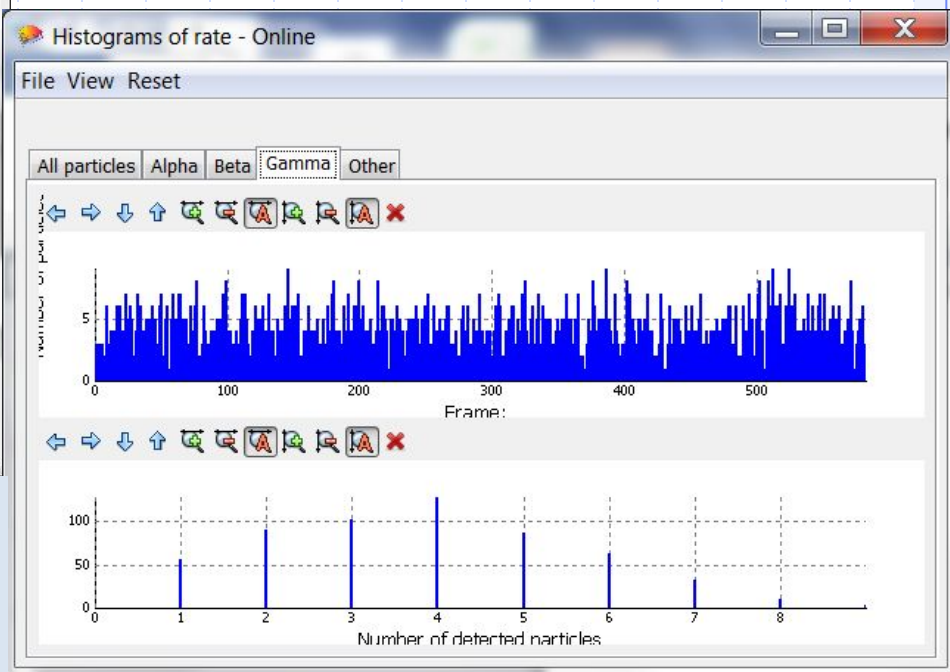
Min. level: 0

Max. level: 20

Colormap: Hot

Alpha particle flux in time

Flux histogram

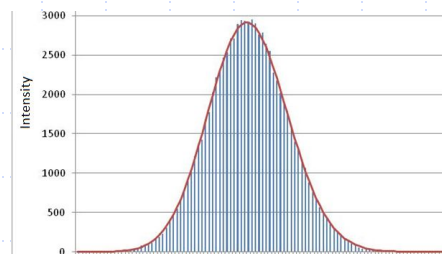


Gama particle flux in time

Flux histogram

$$p_k = \frac{\lambda^k}{k!} e^{-\lambda}$$

Poisson Distribution



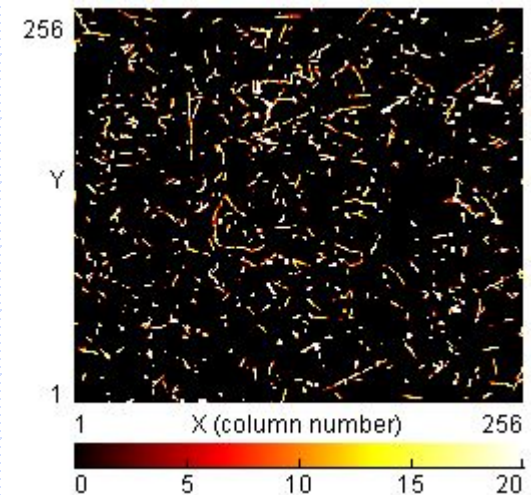
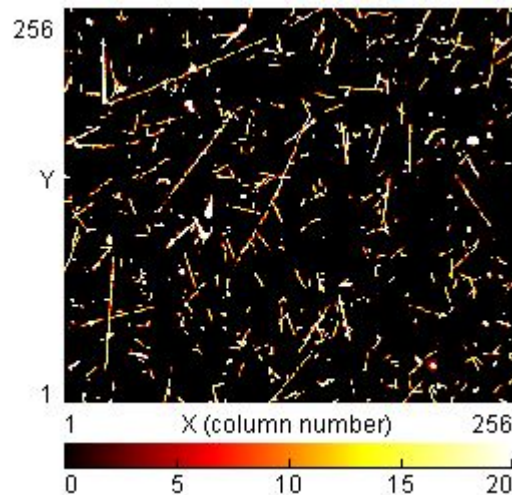
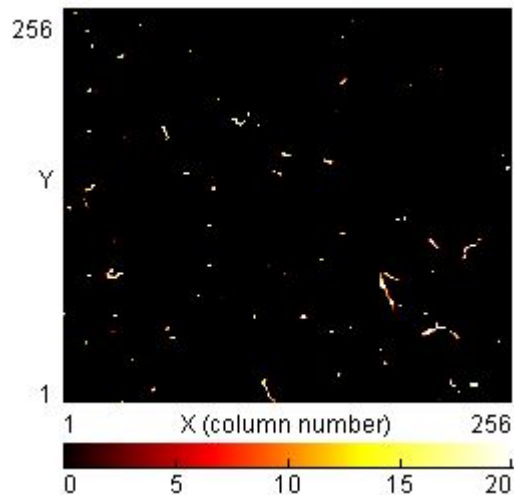


Background Radiation with Origin in Space



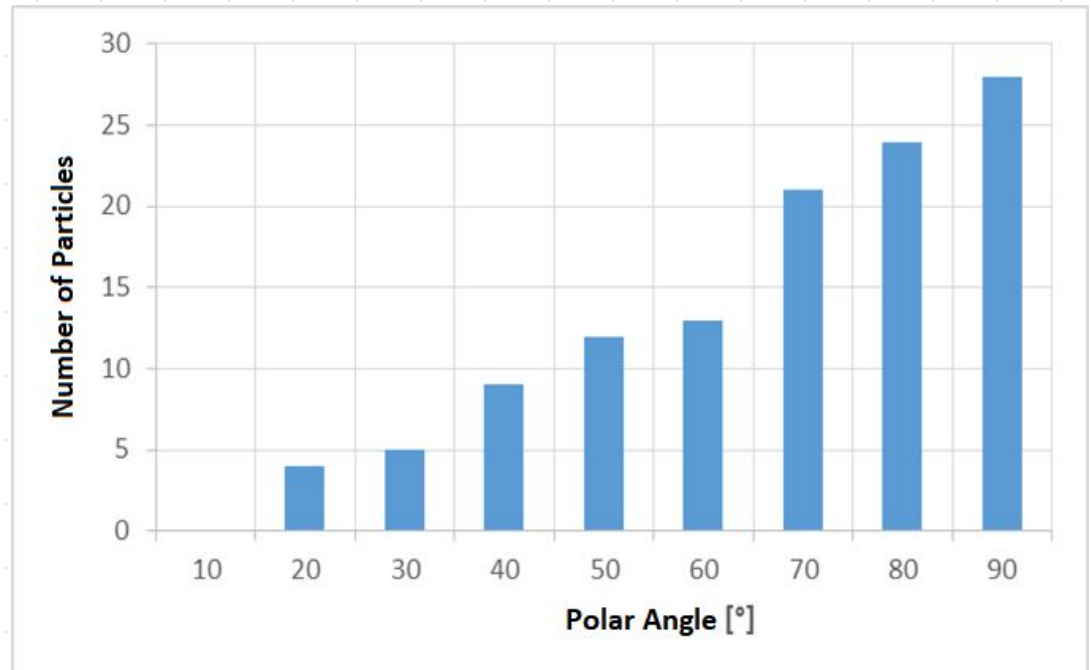
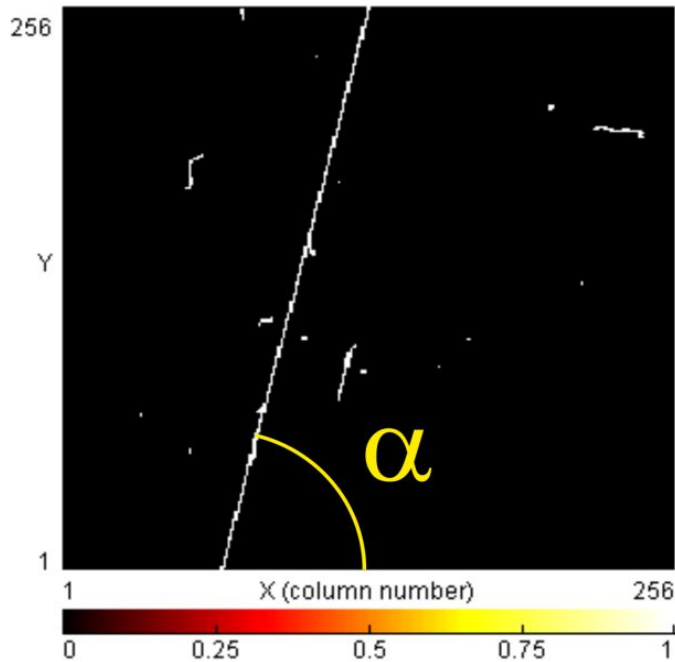
- Detection of cosmic muons and their contribution to background radiation on the ground.

Exemplar frames measured at different conditions: one frame represents response acquired at the vertically oriented sensor at the airport. The next one was acquired on board of the airplane on the standard flight level at the vertical sensor orientation and the last one at the horizontal sensor orientation.





Linear Tracks of Interacting Particles at the Polar Angle



- Significantly different results of muon observation are obtained in dependence on the detector orientation
- Note the typical dimensions of Timepix sensor 14 x 14 x 0.3 mm (w x l x h)



WHY MUONS OF LOW INCLINATION DO NOT REACH TO GROUND?



Half-life of muons is: $2,2 \mu\text{s}$.

How long distance can be reached until it decays?

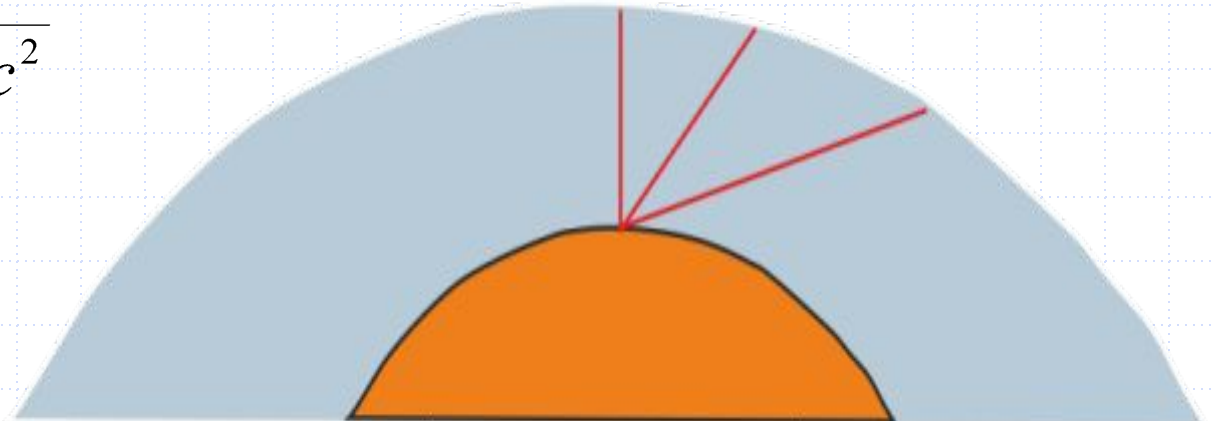
Presumption, muon was created at altitude of 10 km, its velocity is $0,999c$.

Can it reach the ground?

Classical physics: $s = v \cdot \tau_0 = 659 \text{ m}$ It can not reach the ground

STR:
$$s = v \cdot \frac{\tau_0}{\sqrt{1 - \frac{v^2}{c^2}}} = 14\,750 \text{ m}$$

It can reach the ground
(in dependence inclination on angle)





Working with Pre-measured Data Using the Virtual (filedevice 3000) Pararticle Camera



Pixelman Simple Preview (filedevice 3000)

File View Tools Options Help

Load file
Save actual frame
Set FileDev. file
Set FileDev. folder
Save measurement
Export actual frame
Export all frames
Exit

Select Folder:
Location of the
measured data

Acquisition

☐ Continuous measurement
☐ Integral mode

Exp. count: 930
Exp. time: 0.01
Delay [s]: 0
Acq. progress: 0/414
0 s
Mode: Spectrometer

Picture settings

Min. level: 0
Max. level: 20
Set colormap: Hot
☐ Auto range: Min-max

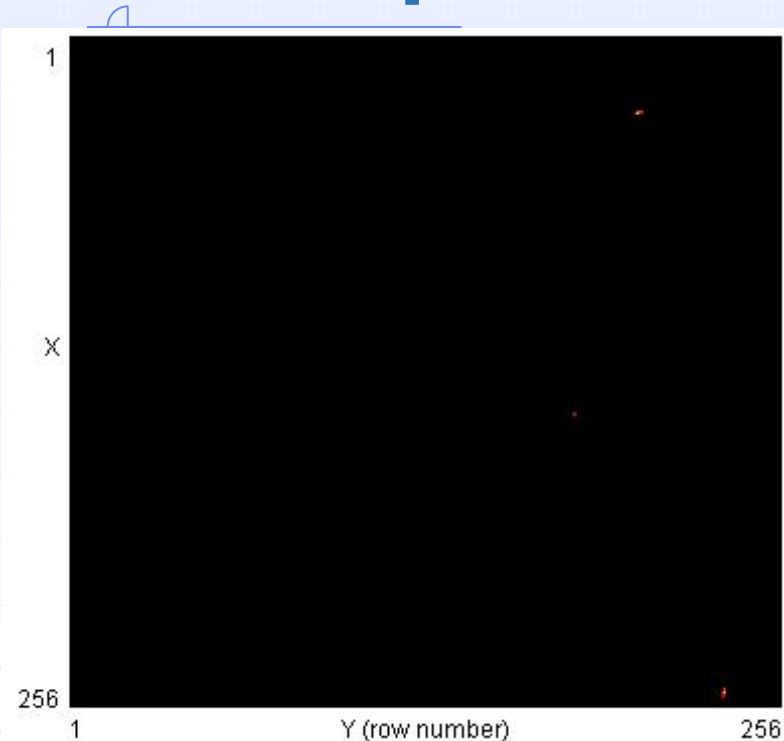
XY: [36,257]
Value: 0.0
Min: 0.0
Max: 0.0
Pixel count: 0
Energy - frame: 0
Energy - selecti...: 0
Frames count:
Radiation source: Other

Frame	Actual	All
Alpha	0	
Beta	0	
Gamma	0	
Other	0	
All	0	

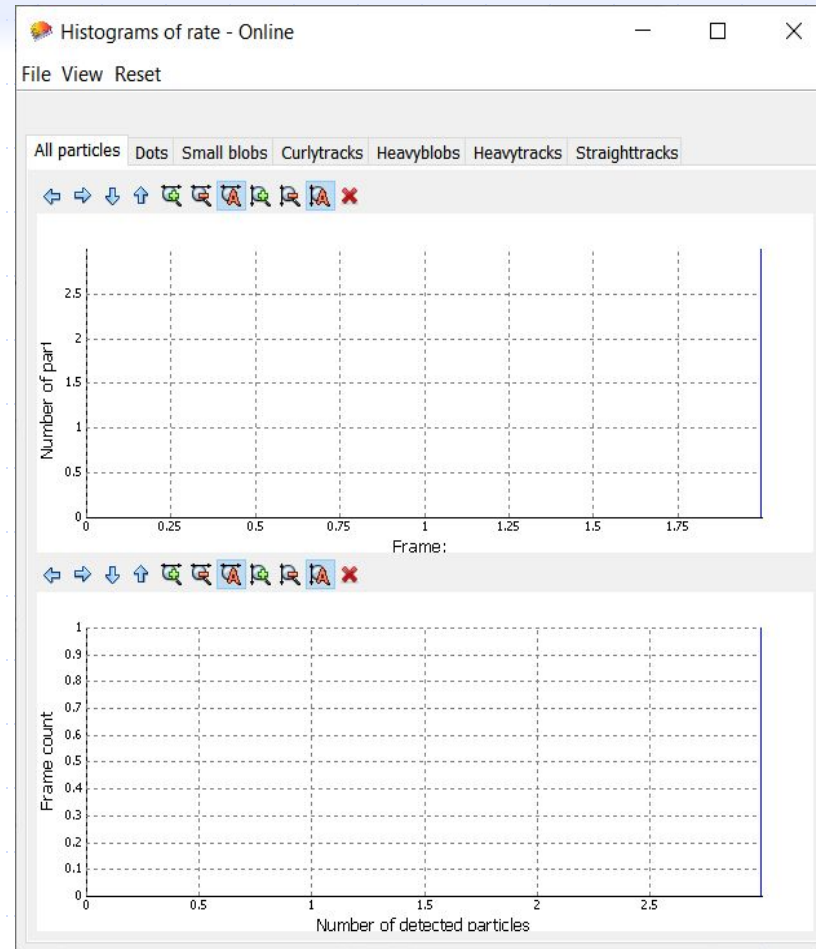
- The virtual particle camera allows to replay experimental data measured in the past
same visualisation and processing functionalities are available as if a real particle camera would be connected to a computer



Radiation field observation on board of the airplane as seen by the particle camera



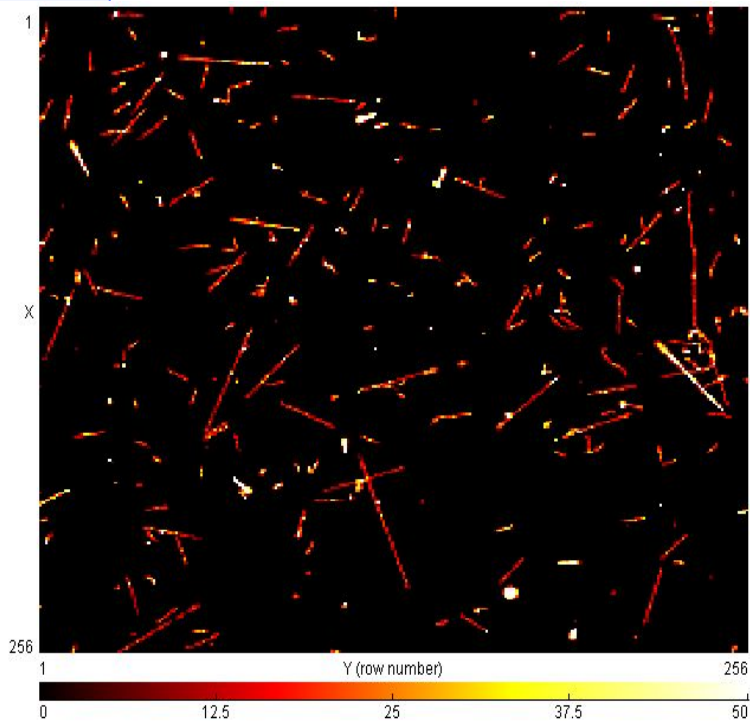
- Recording from the flight from London to Prague
- Significant difference in field composition and intensity when flying or when landed...



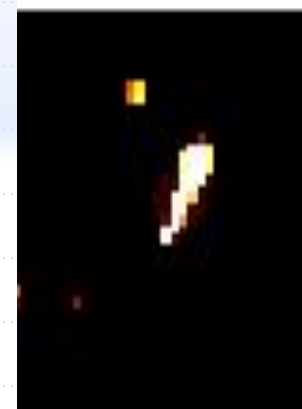
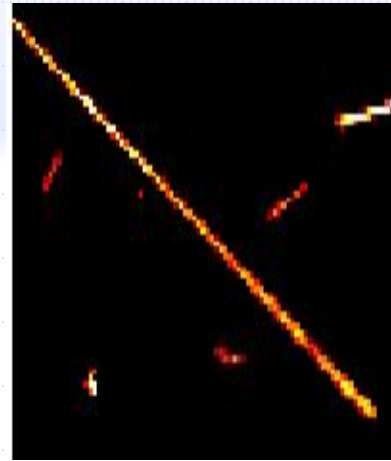


Close look on airplane flight observation recorded by the particle camera

- Typical composition of radiation field at flight level
(*high count of muons*)

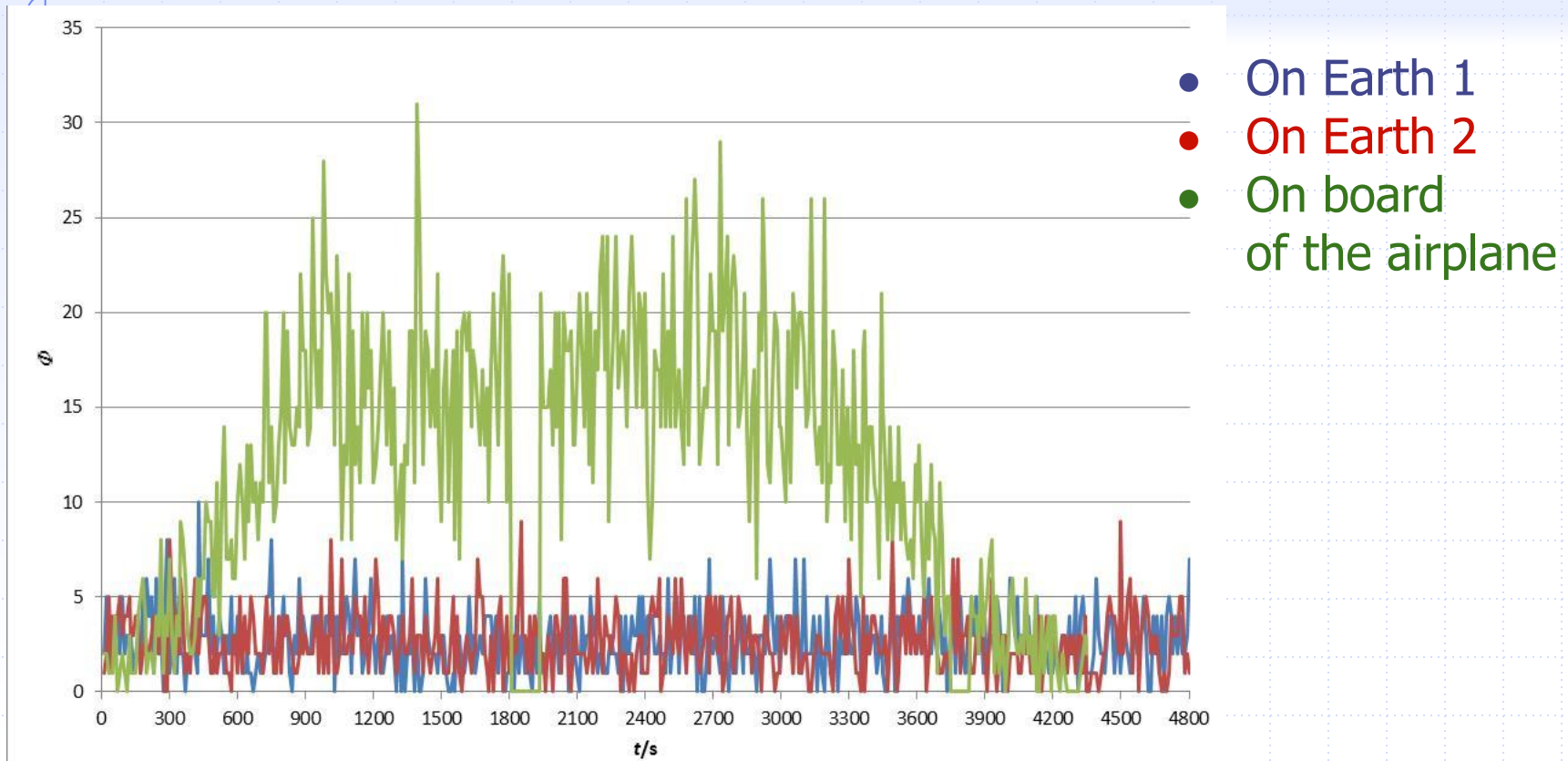


- Occasional detection of heavy energetic ions





Radiation intensity observed on Earth vs on board of the airplane





Dosimetry in space: SATRAM – ESA Proba-V satellite



Characterization of mixed radiation field on low orbit of PROBA-V satellite

- ◆ Altitude ~ 800 km
- ◆ Timepix for the first time outside in the space
- ◆ Launched in May 2013





SATRAM

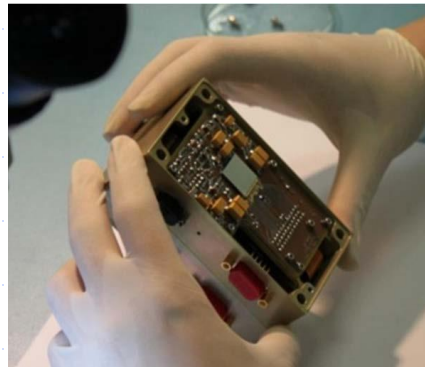
(Space Application of Timepix Radiation Monitor)



on board of Proba-V ESA satellite



- SATRAM payload (arrow) onboard ESA Proba-V satellite.
- ESA Vega-2 launcher rocket upper stage



- The same kind of pixel detector Timepix as one embedded in the particle camera MiniPixEDU of the SESTRAS educational kit



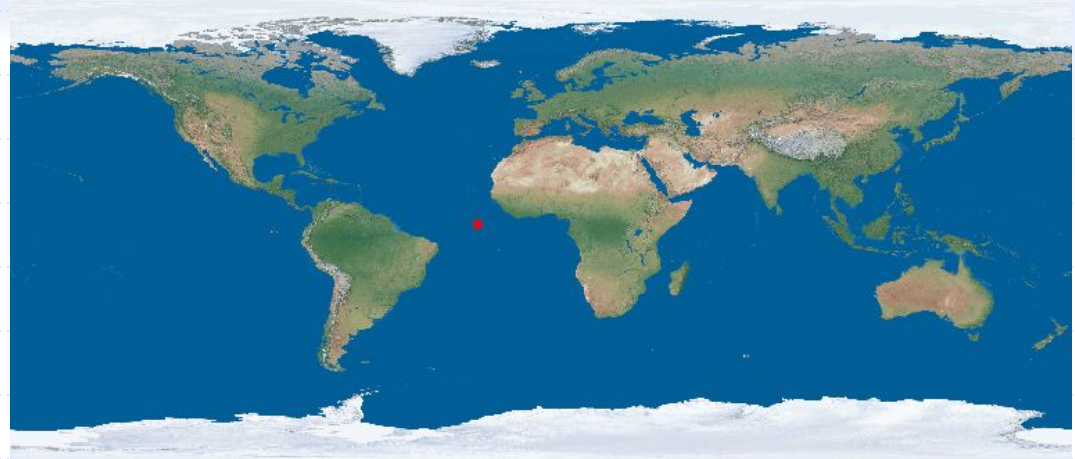
- mission on LEO - altitude 820 km
- Timepix for the first time outside in the space
- Launched in May 2013
- Radiation field recordings available for research and also for educational purposes



SATRAM Radiation Observation on the Earth Orbit



- Radiation field as seen by Timepix detector



- Variation of radiation field composition in dependence on the earth orbit position
- *Notable difference in field composition when compared to observations performed with natural radiation sources in the SESTRA educational kit*



Mission data available online at the web portal:
<https://satram.utef.cvut.cz/>

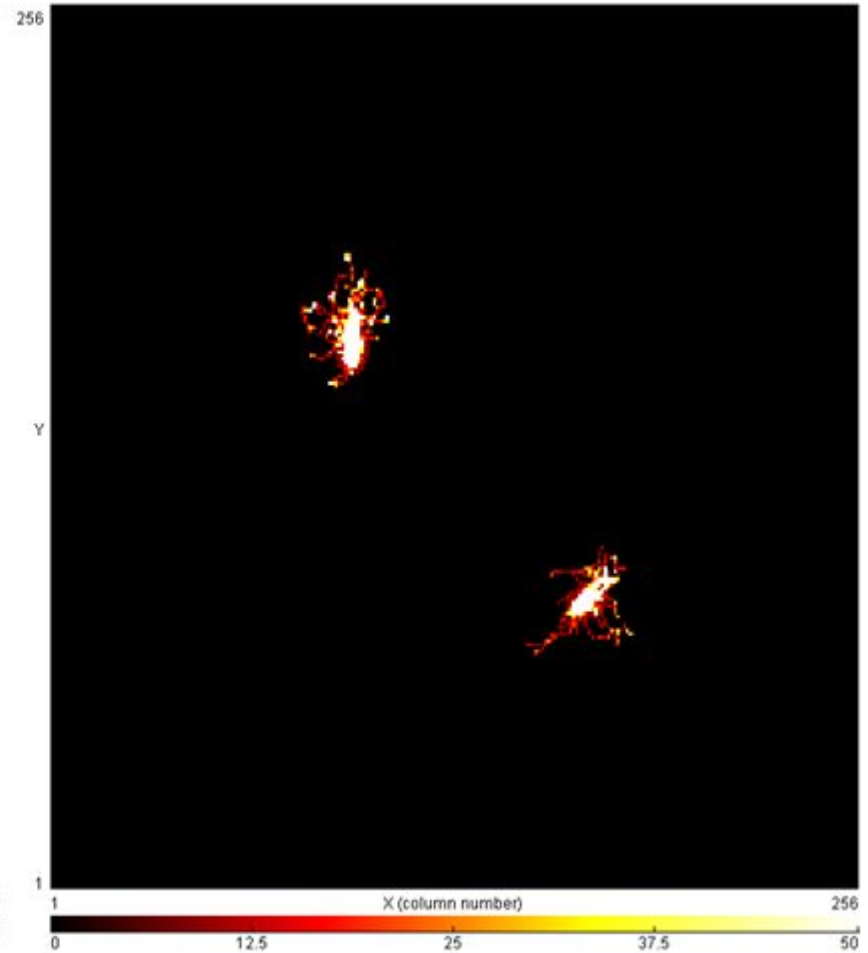
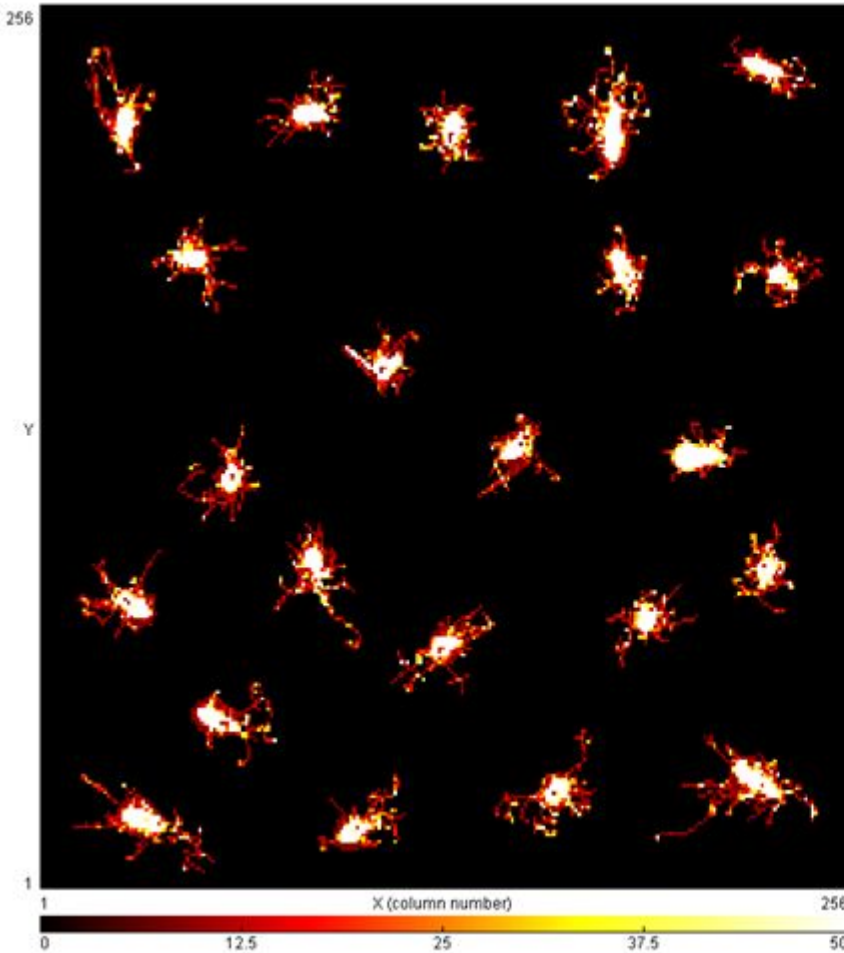


Timepix/ESA Proba-V:

LEO space radiation @ 820 km



HETPs: Highly energetic heavy charged particles (ions) □ HZE's



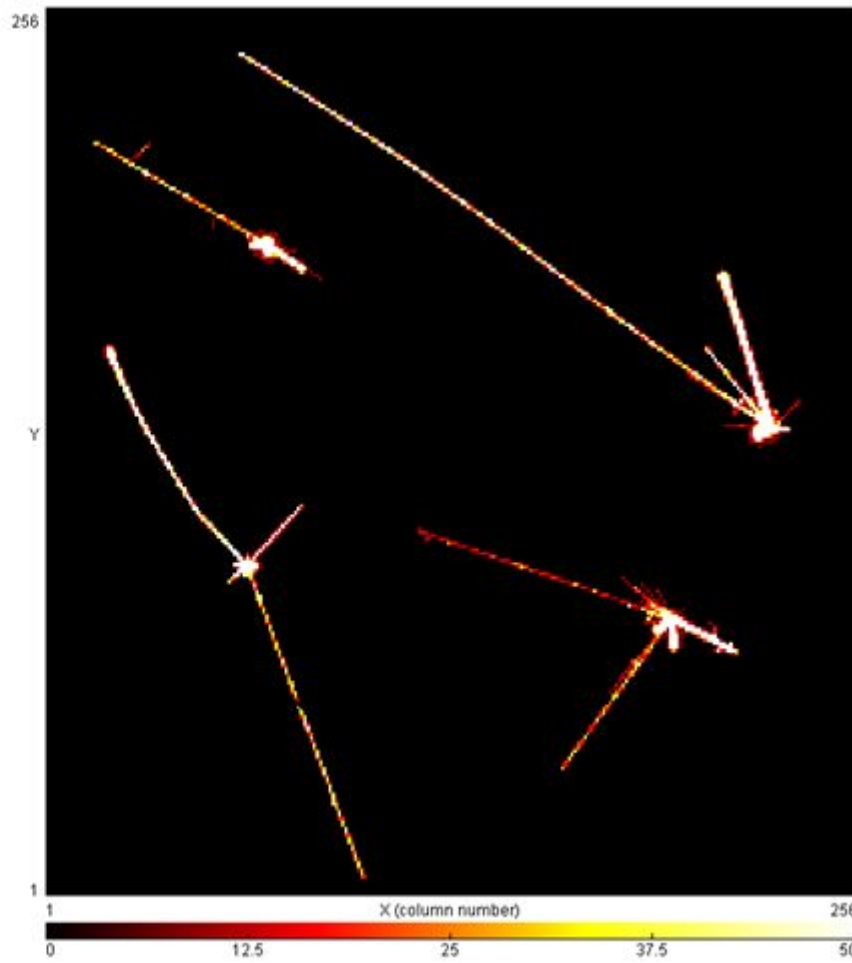
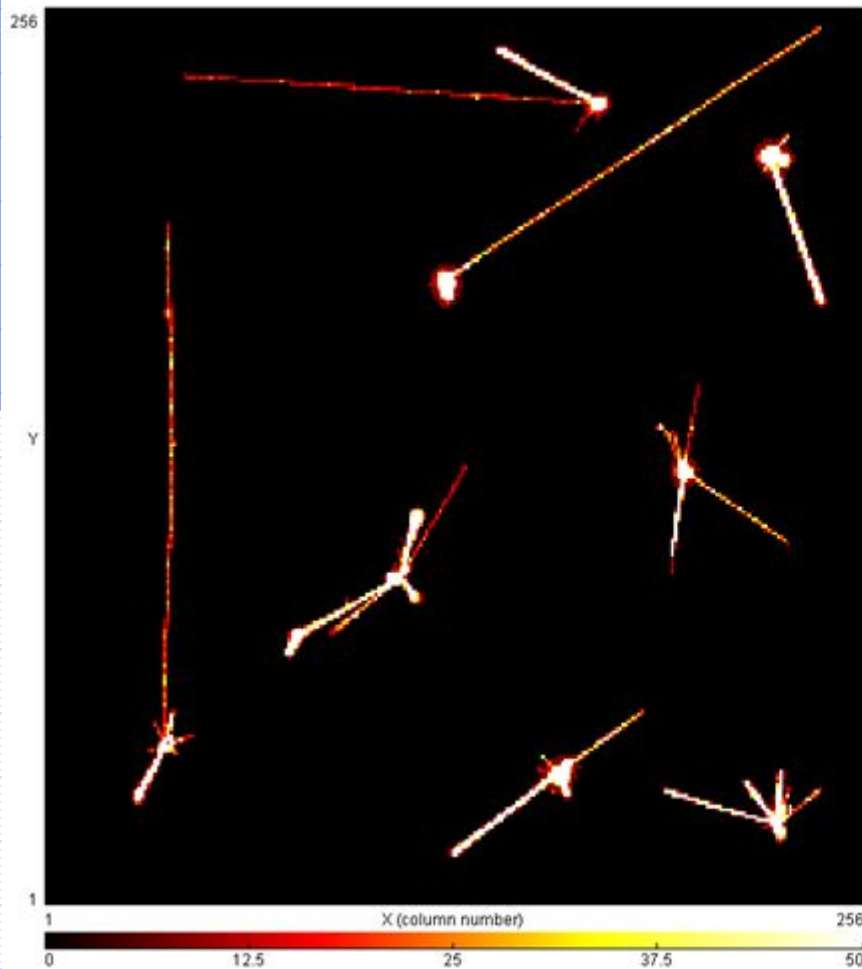


Timepix/ESA Proba-V:

LEO space radiation @ 820 km

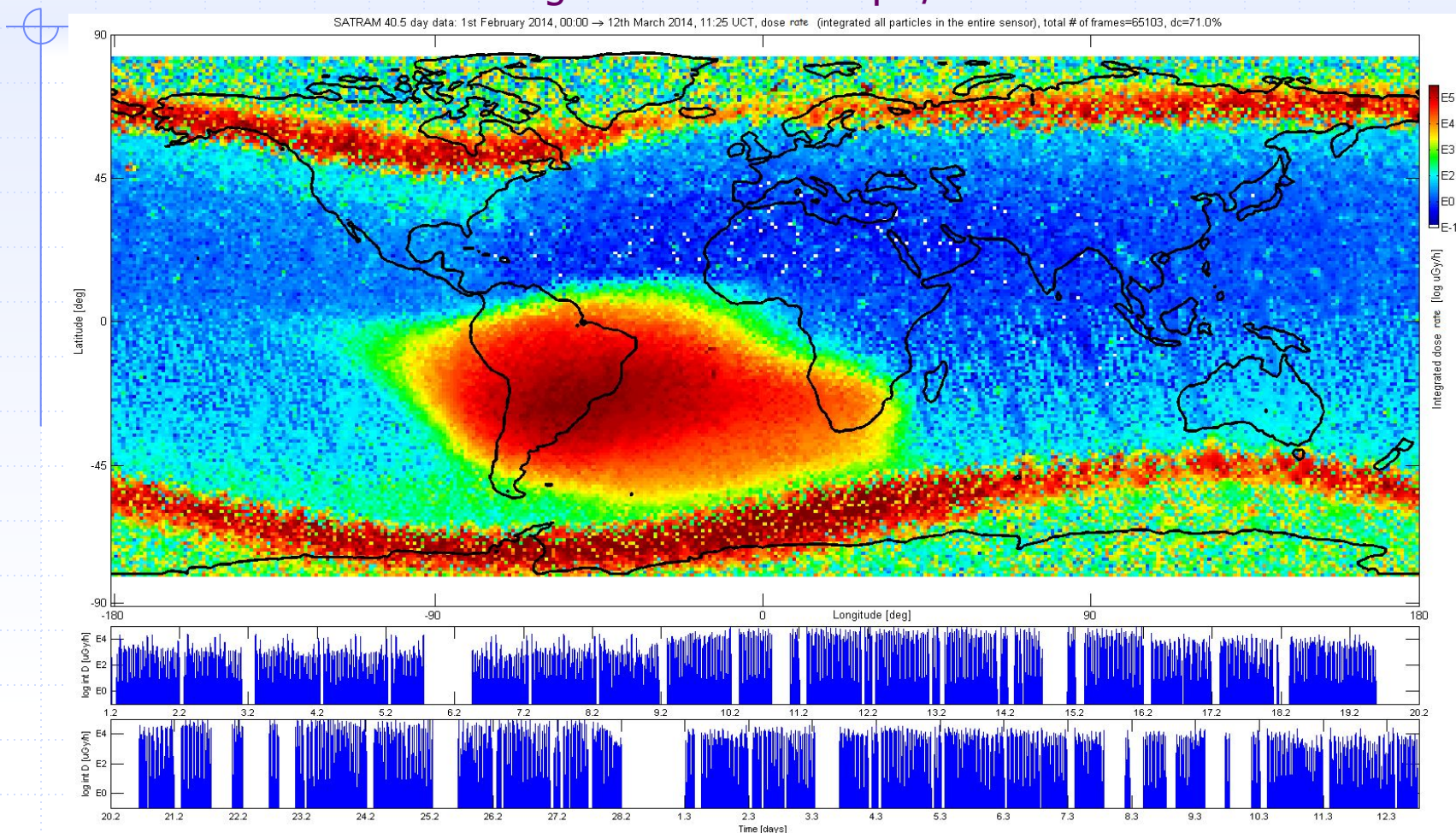


LETPs: Energetic light charged particles (l) + nuclear interactions





Measured radiation map by Satram device in orbit around the Earth at an altitude of 820 km from the earth's surface obtained within 36 days from January 1, 2014 to February 7, 2014 logarithmic scale in $\mu\text{G/hr}$.

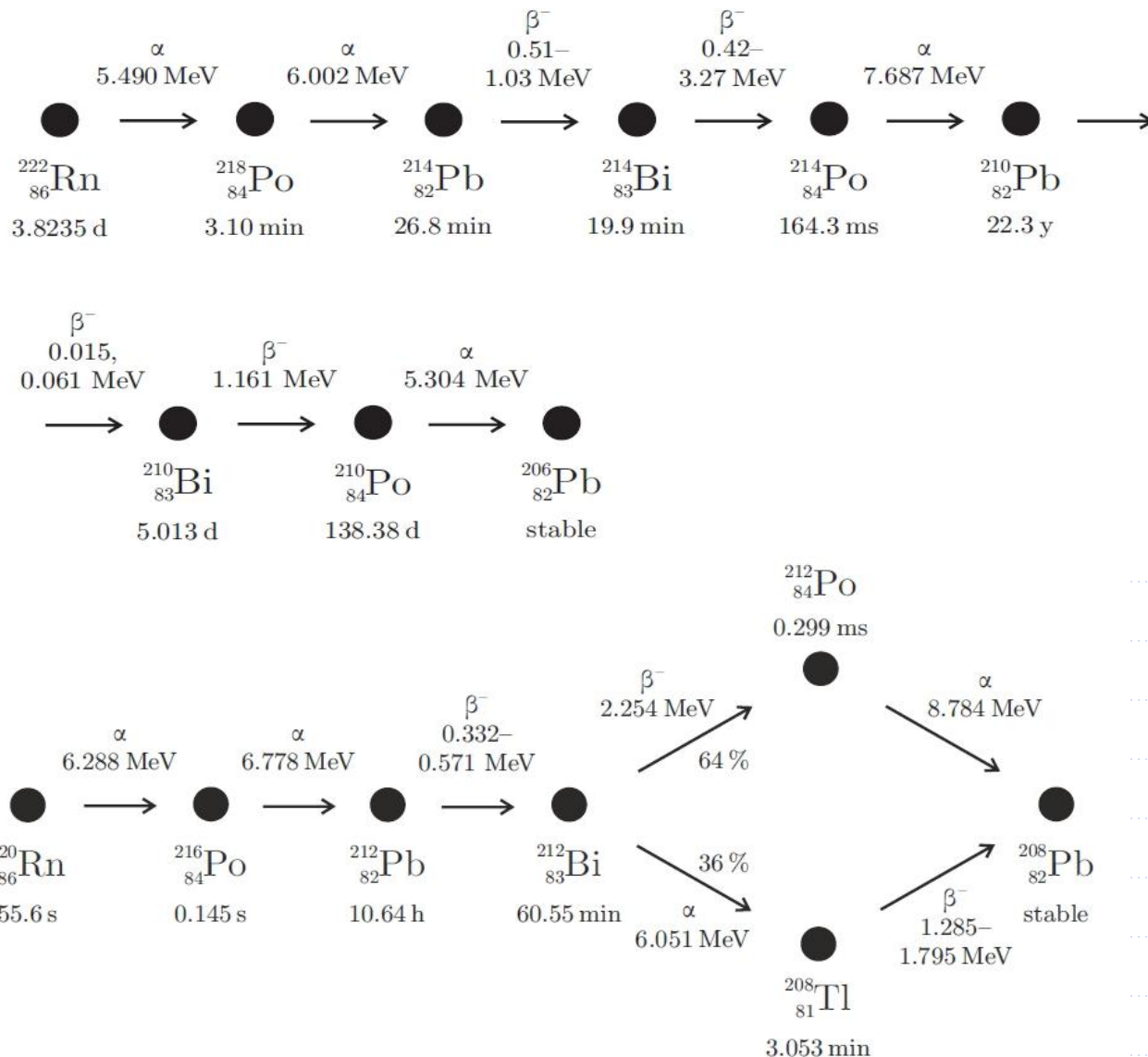




Radon as a common source of background radioactivity



- Noble gas originating from decay of radium
- It further decay through the serie of daughter products





Radon - Easy way how to prepare own radiation source

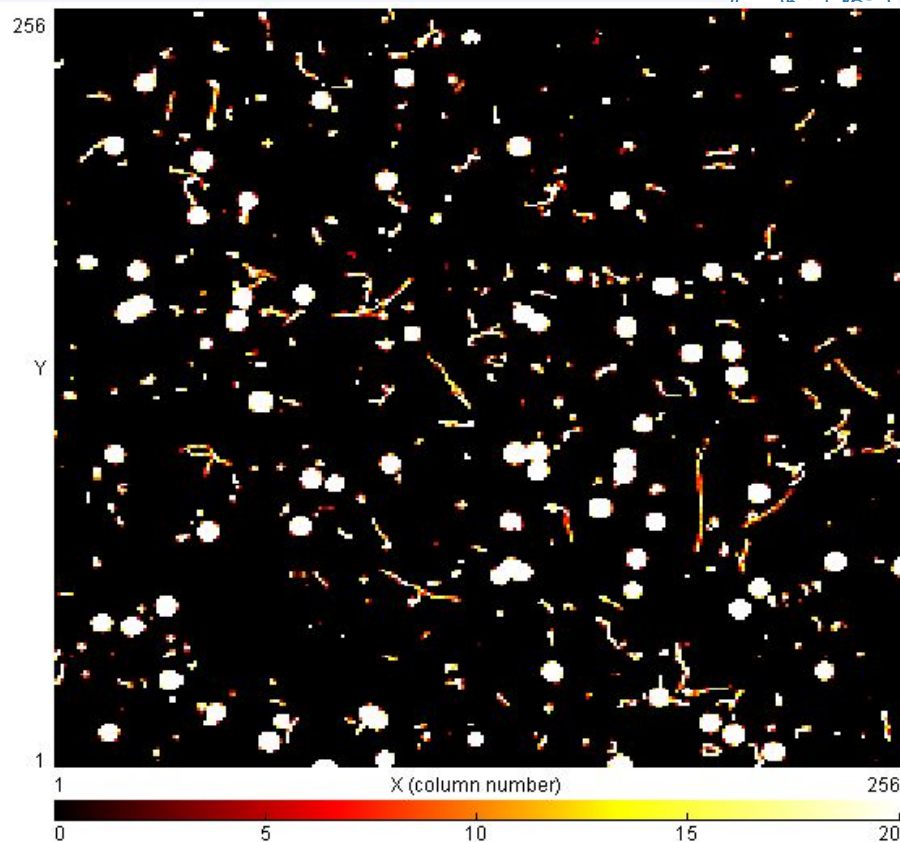
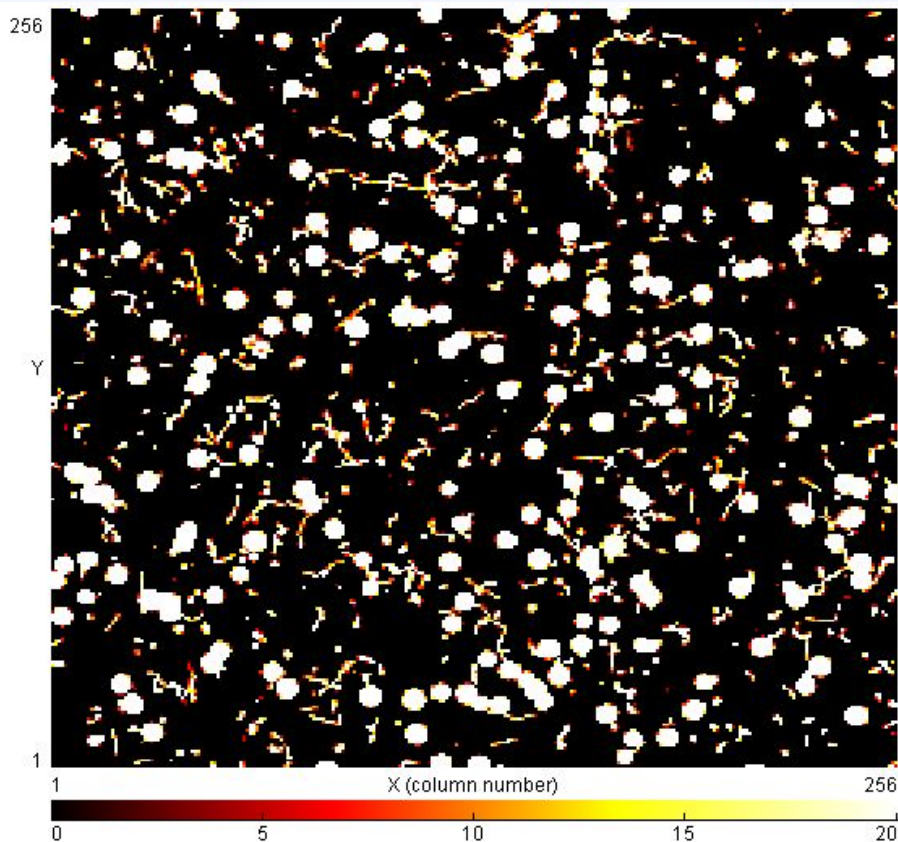
- Preparation requires just usage of a vacuum cleaner + grate adapter + paper tissue (or head mask)
(ideal location - non-ventilated basement of an old building)



- After several minutes letting air pass through the filter, radon daughter products get captured in the filter
- Then the paper tissue is placed on the top of the particle camera and radiation measurement is started



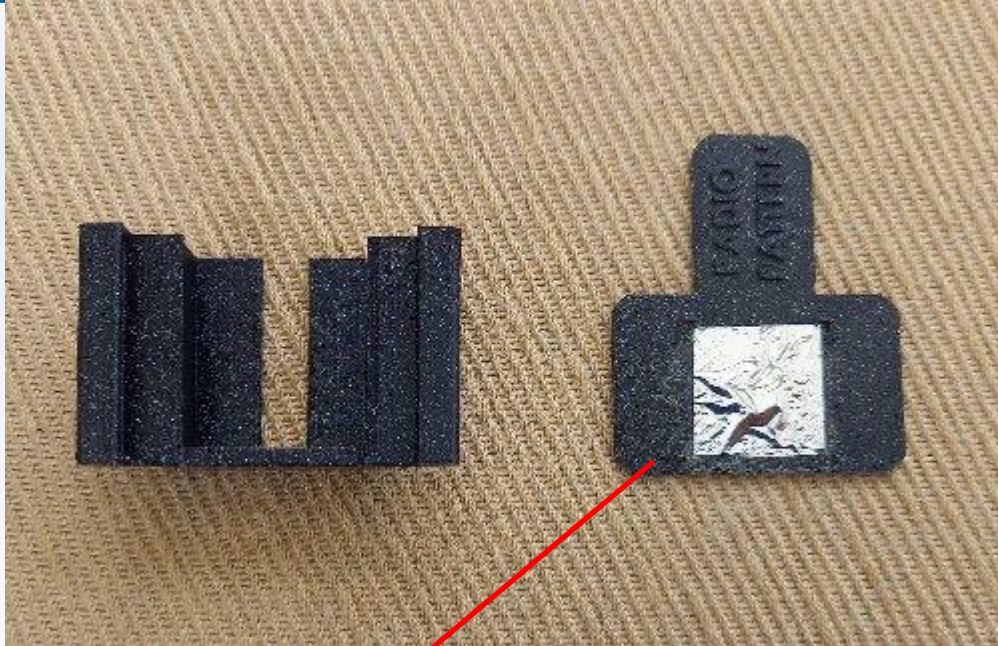
Comparison of Radioactivity observed in the Non-ventilated (left) and ventilated (right) room



- *Demonstration of the radioactivity of the paper tissue that was used as an air filter in one house in Pardubice – Czech Republic*
- *The filtering took 5 minutes and exposure time was 10 minutes in both cases*



(Alpha) Radiography



Samples with
hidden patterns

What is
Hidden
Inside?

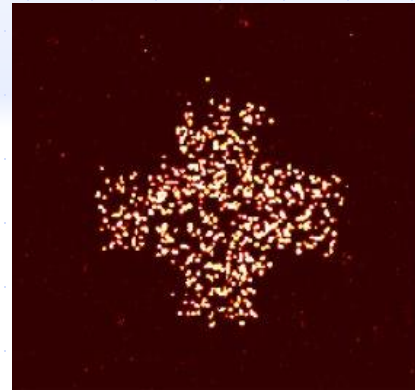


Settings of Particle camera: Radiation source other, analysis type OFF, continuous YES, integral YES, exposure count 60, exposure time 1 s, max. level 1, colormap GRAY.

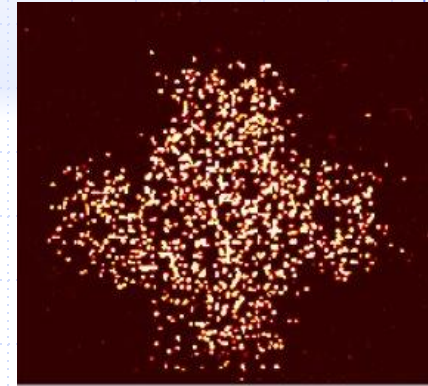
Settings of Filedevice: Radiation source other, analysis type OFF, continuous YES, integral YES, exposure count 60, exposure time 0.01 s, max. level 1, colormap GRAY.



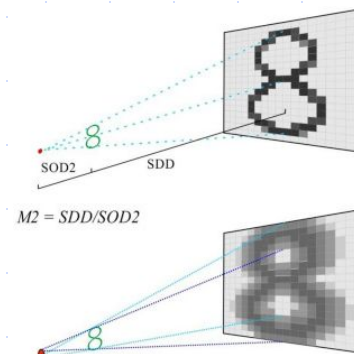
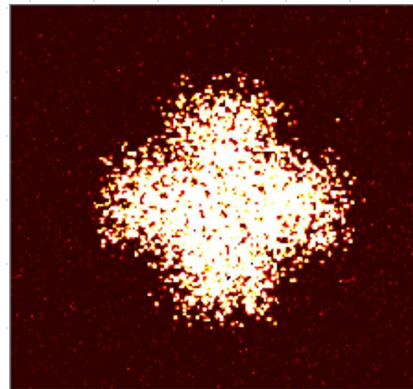
(Alfa) Radiography Demonstration



Impact of
Sample
position
(magnification)



Impact of
Beam
collimation
(sharpness)

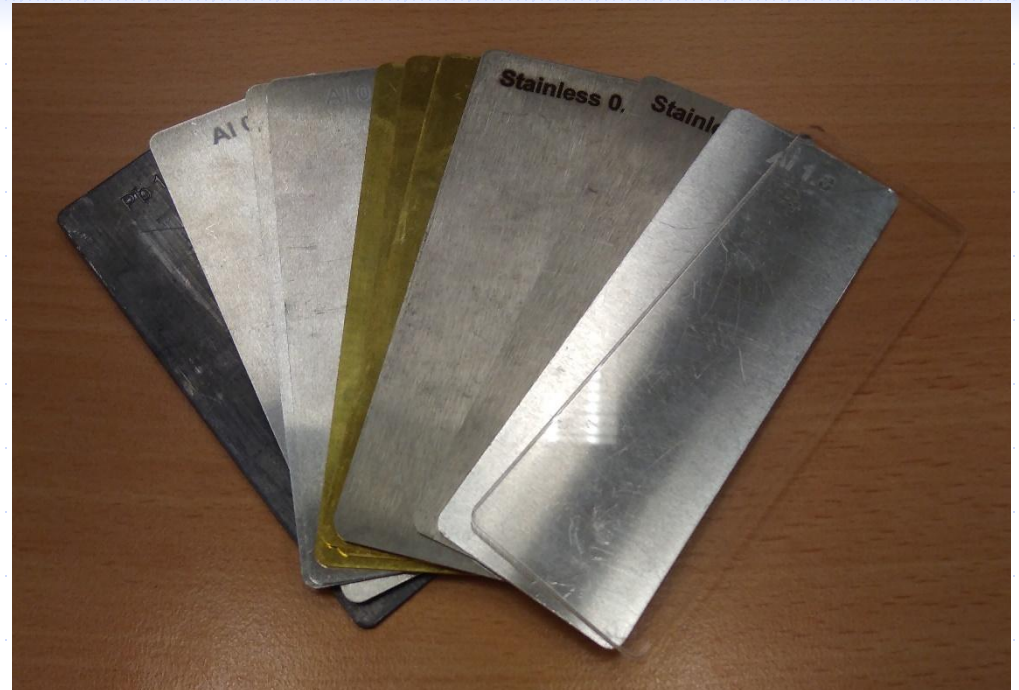
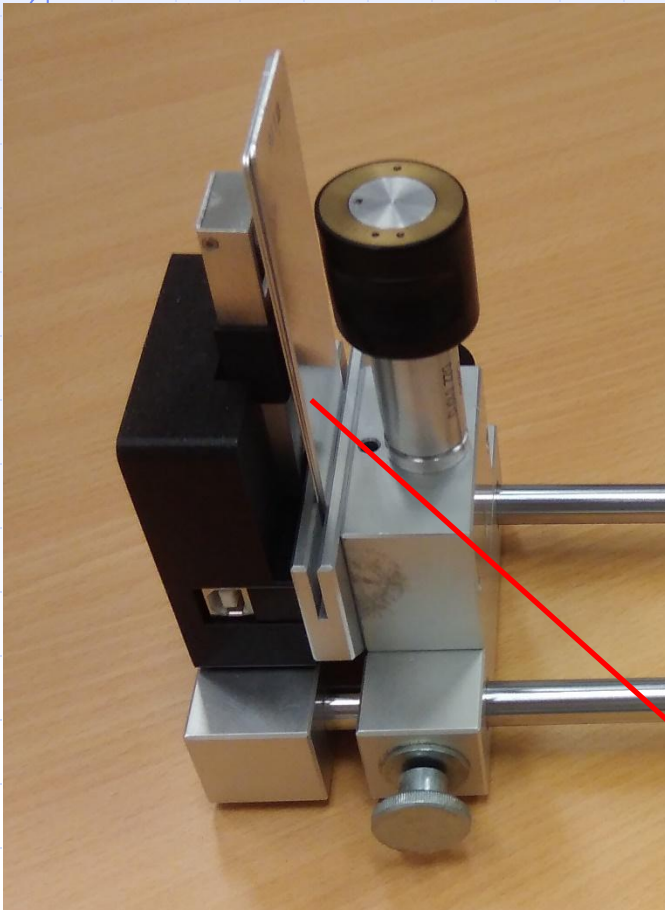


- even μm material thickness matter





ABSORPTION of gamma particles in metals

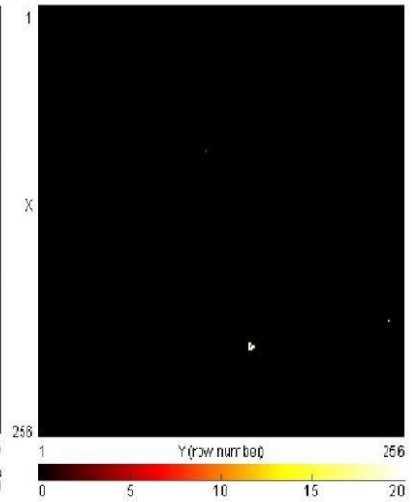
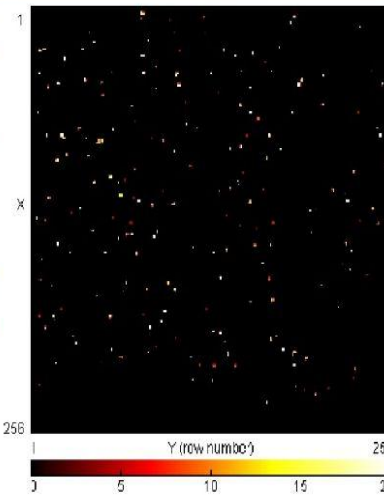
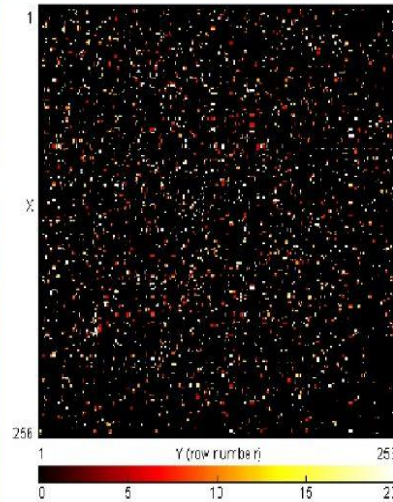
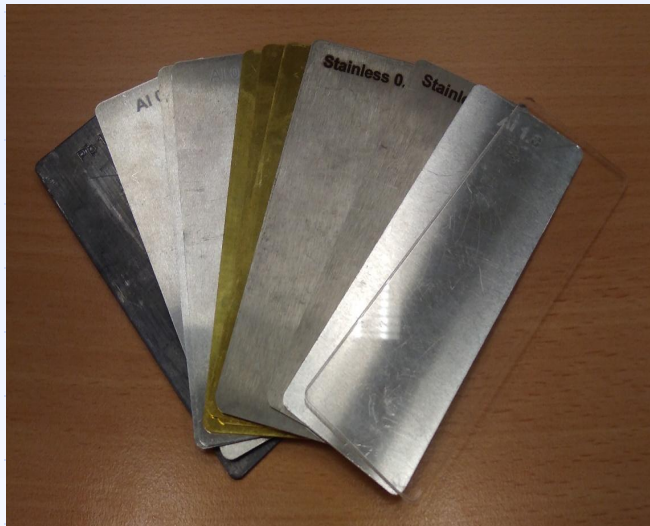


Metallic
plate

Settings of Simple preview: Radiation source americium, bias voltage 50 V, continuous NO, integral NO, exposure count 300, exposure time 0.5 s, max. level 1, colormap GRAY.



Absorption of Gamma/X-Ray in shielding plates of various materials



Aluminum vs Brass vs Lead

vs OTHER materials vs Background?

- **DZZ Gamma**
Americium 241 source
300 kBq activity
- **SZZ Alpha**
Americium 241 source
9.5 kBq activity

Settings of Filedevice: Radiation source other, analysis type OFF, continuous YES, integral YES, exposure count 60, exposure time 0.1 s, max. level 1, colormap GRAY.

Settings of Filedevice: Radiation source other, analysis type OFF, continuous YES, integral YES, exposure count 60, exposure time 1 s, max. level 1, colormap GRAY.



Radiography (Gamma/Xray)



Samples with
hidden patterns



What is hidden inside?



Settings of Particle camera: Radiation source other, analysis type OFF, continuous YES, integral YES, exposure count 60, exposure time 1 s, max. level 1, colormap GRAY.

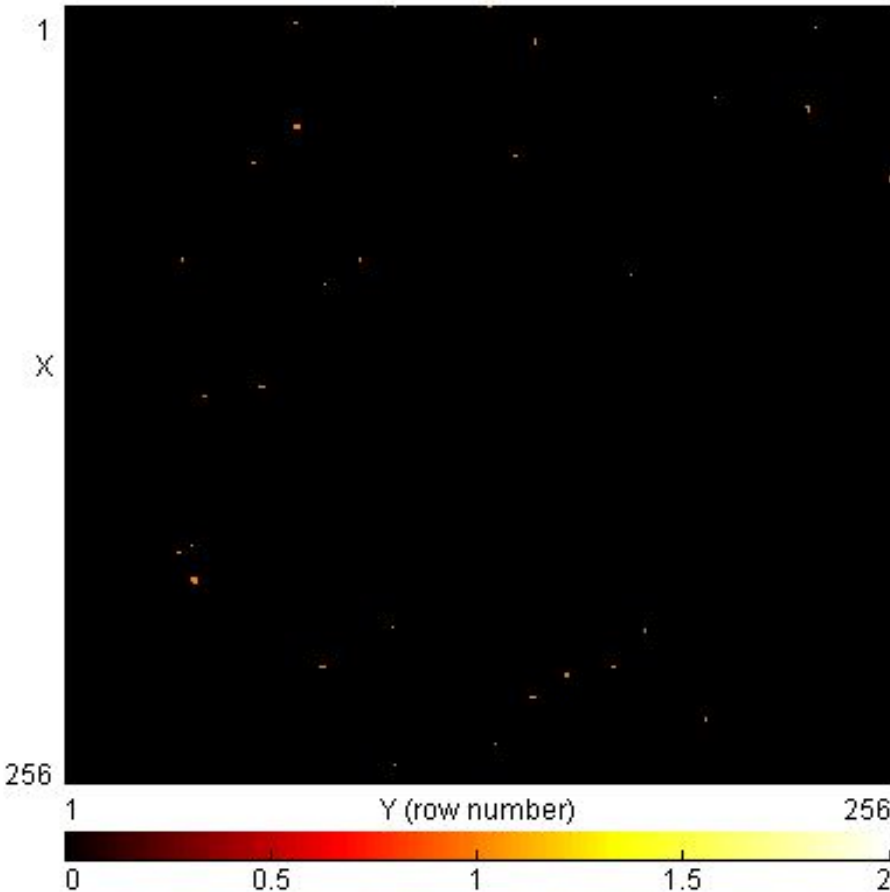
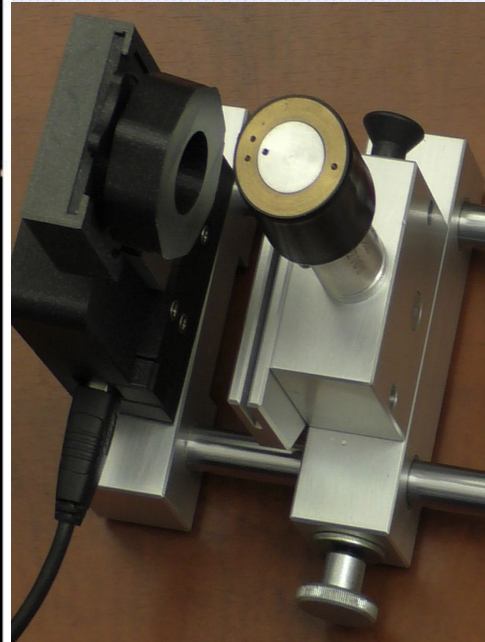
Settings of Filedevice: Radiation source other, analysis type OFF, continuous YES, integral YES, exposure count 60, exposure time 0.01 s, max. level 1, colormap GRAY.



(Xray) Radiography Demonstration



- In contrary to more advanced imaging



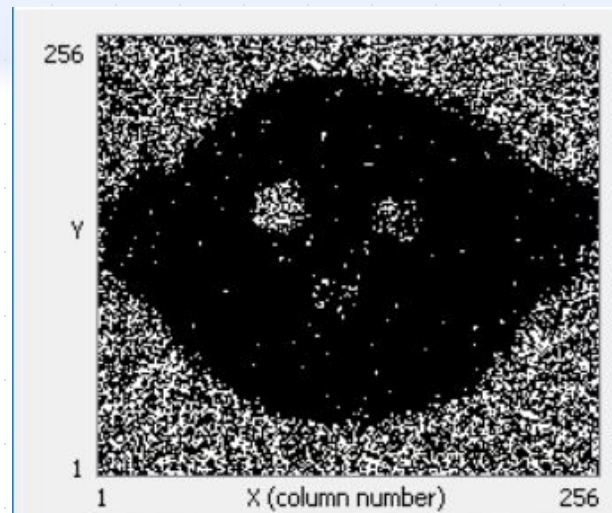
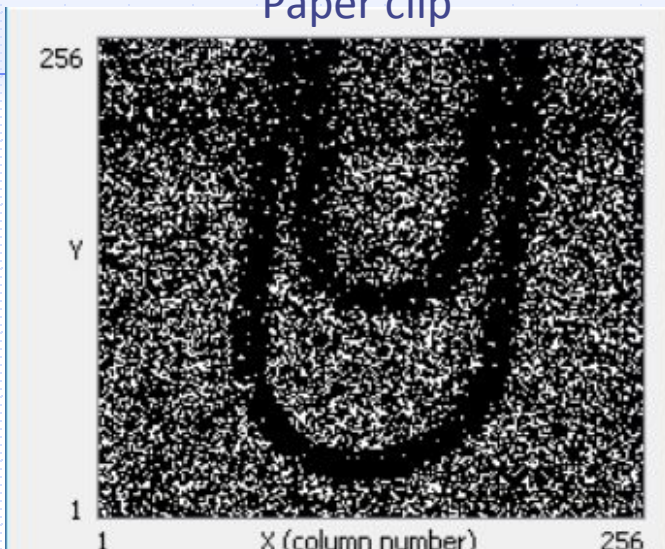
- Prove of basic working principle



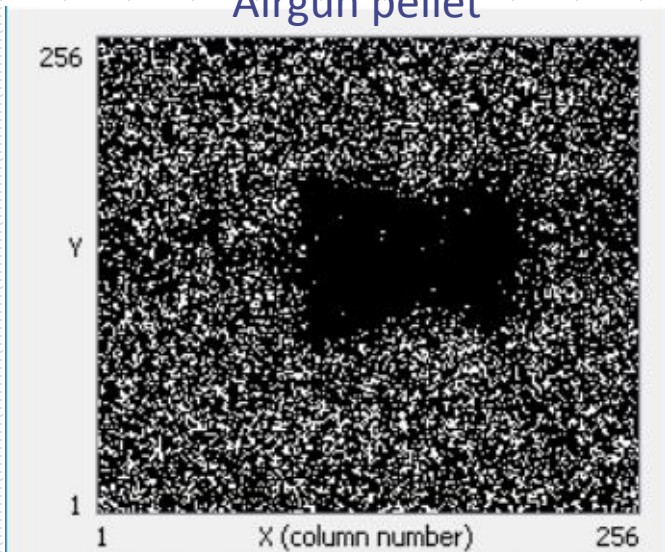
Hidden pattern examples....



Paper clip



Airgun pellet





Finally, the last item in SESTRA kit



- **Empty container dedicated for insertion and observation of own (radioactive) samples**



Using the empty container to inspect own samples



Example 1

- Potassium Sulfide (K_2SO_4) flower fertilizer



Example 2

- Gas mantle for oil lamps containing thorium dioxide

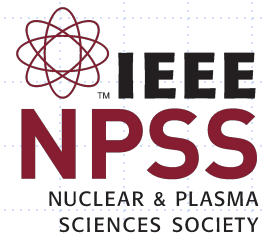
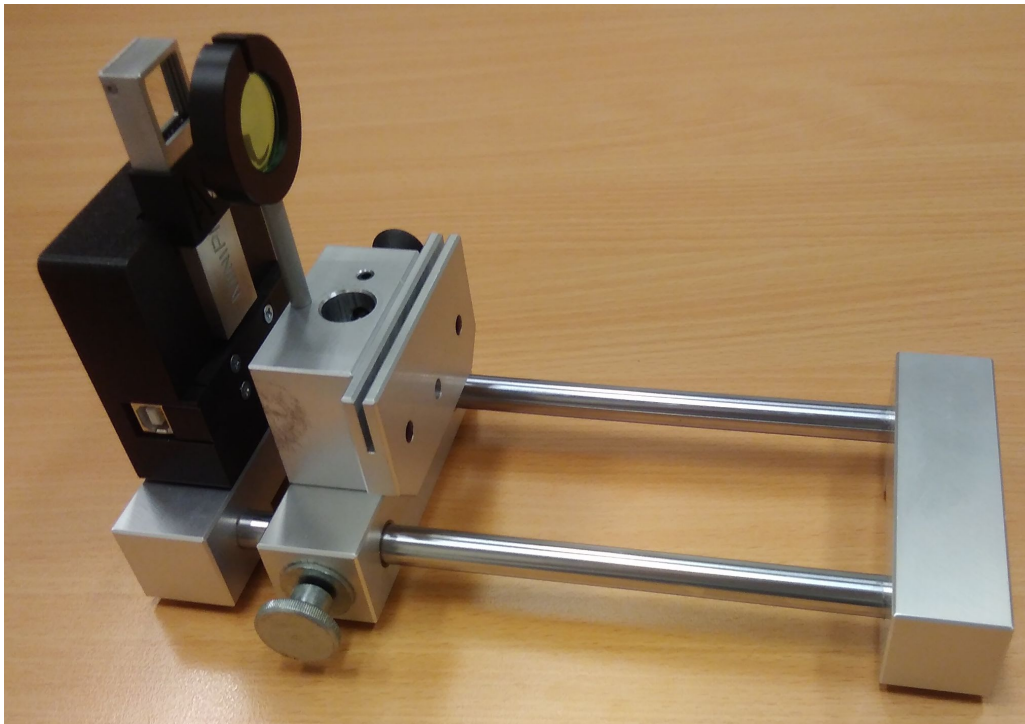




Thank you for your participation



Any initiative in a way to further cooperation is welcome!!!



outreach.and.education@utef.cvut.cz